

Description of Device Parameters

Proline Promag 300

Modbus RS485

Electromagnetic flowmeter

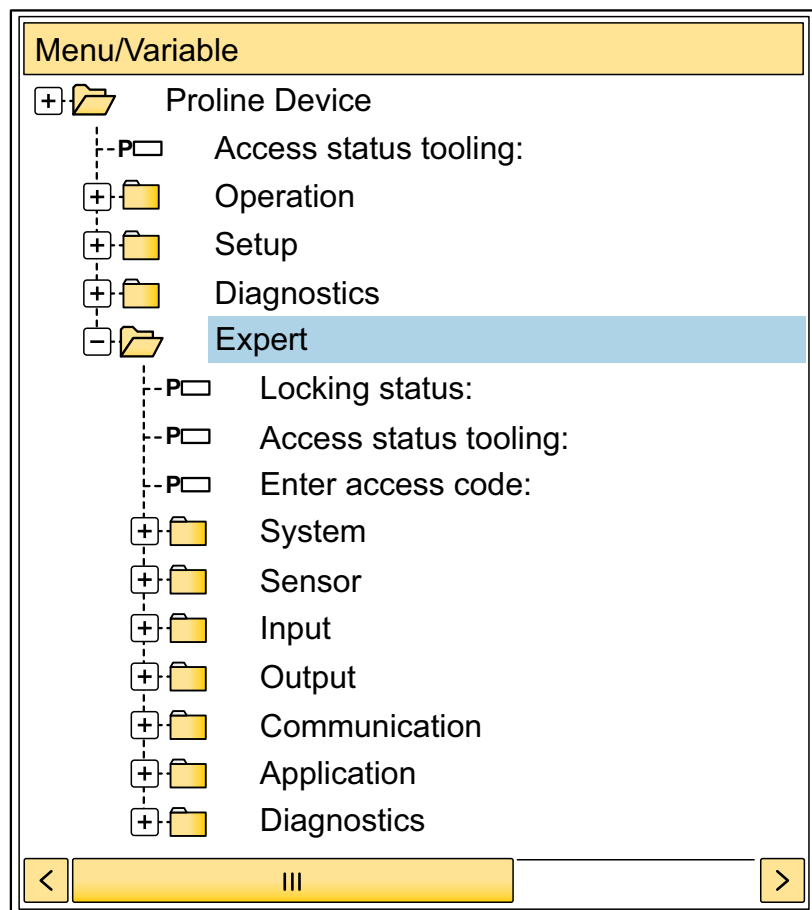


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1 Document information

1.1 Document function

The document is part of the Operating Instructions and serves as a reference for parameters, providing a detailed explanation of each individual parameter of the Expert operating menu.

It is used to perform tasks that require detailed knowledge of the function of the device:

- Commissioning measurements under difficult conditions
- Optimal adaptation of the measurement to difficult conditions
- Detailed configuration of the communication interface
- Error diagnostics in difficult cases

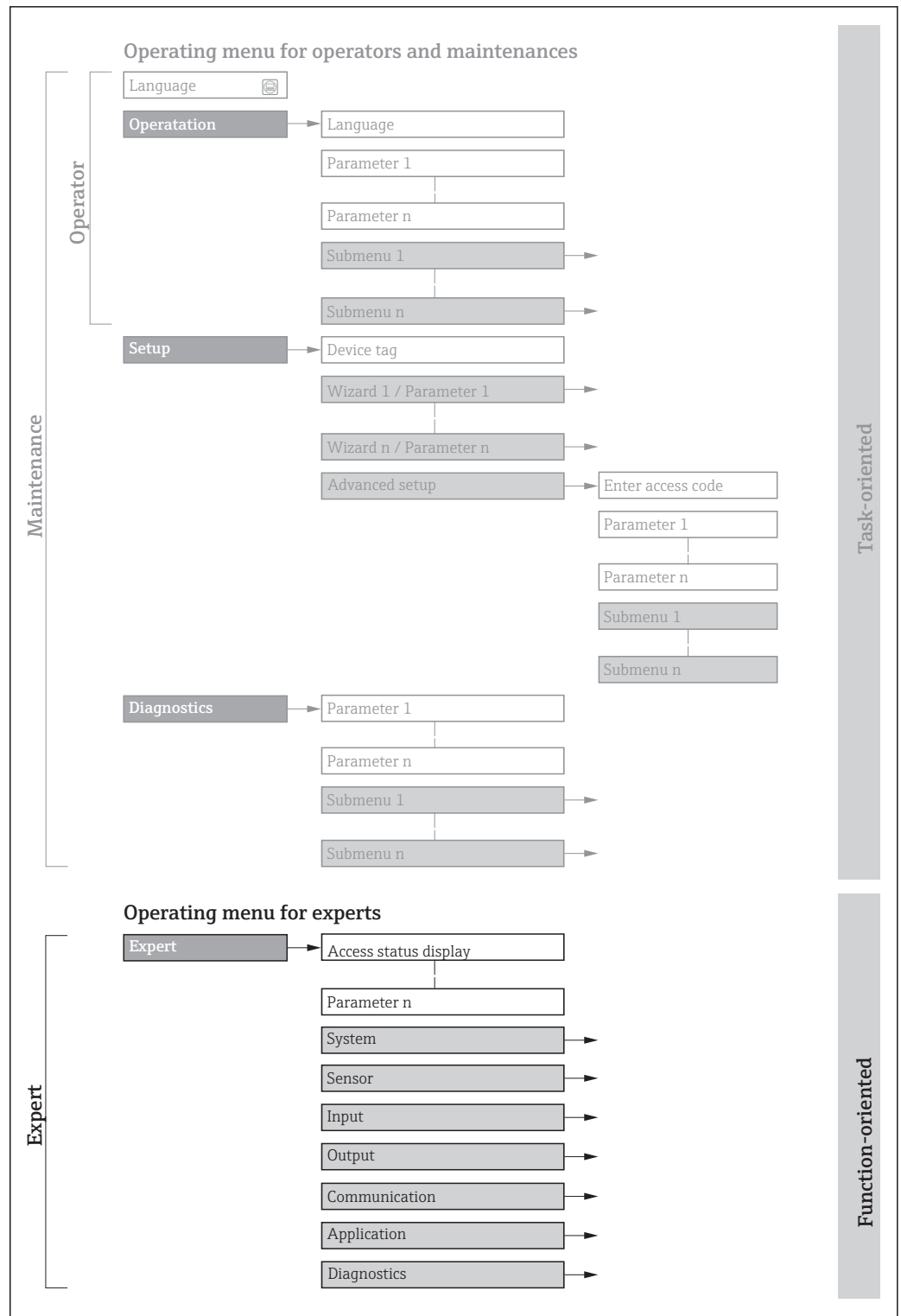
1.2 Target group

The document is aimed at specialists who work with the device over the entire life cycle and perform specific configurations.

1.3 Using this document

1.3.1 Information on the document structure

The document lists the submenus and their parameters according to the structure from the **Expert** menu (→  8), which is displayed when the **"Maintenance" user role** is enabled.



1 Sample graphic for the schematic layout of the operating menu



Detailed information concerning:

- Arrangement of the parameters according to the menu structure of the **Operation** menu, **Setup** menu, **Diagnostics** menu along with a brief description, see the Operating Instructions for the device → 7
- Operating philosophy of the operating menu: "Operating philosophy" chapter of the Operating Instructions for the device → 7

1.3.2 Structure of a parameter description

The individual parts of a parameter description are described in the following section:

Complete parameter name		Write-protected parameter = 
Navigation	 	Navigation path to the parameter via the local display (direct access code) or web browser Navigation path to the parameter via the operating tool The names of the menus, submenus and parameters are abbreviated to the form in which they appear on the display and in the operating tool.
Prerequisite		The parameter is only available under these specific conditions
Description		Description of the parameter function
Selection		List of the individual options for the parameter ■ Option 1 ■ Option 2
User entry		Input range for the parameter
User interface		Display value/data for the parameter
Factory setting		Default setting ex works
Additional information		Additional explanations (e.g. in examples): ■ On individual options ■ On display values/data ■ On the input range ■ On the factory setting ■ On the parameter function

1.4 Symbols used

1.4.1 Symbols for certain types of information

Symbol	Meaning
	Tip Indicates additional information.
	Reference to documentation
	Reference to page
	Reference to graphic
	Operation via local display
	Operation via operating tool
	Write-protected parameter

1.4.2 Symbols in graphics

Symbol	Meaning	Symbol	Meaning
1, 2, 3 ...	Item numbers	A, B, C, ...	Views
A-A, B-B, C-C, ...	Sections		

1.5 Documentation

1.5.1 Standard documentation

Operating Instructions

Measuring device	Documentation code
Promag H 300	BA01394D
Promag P 300	BA01395D

1.5.2 Supplementary device-dependent documentation

Special Documentation

Contents	Documentation code
Information on the Pressure Equipment Directive	SD01614D
Remote display and operating module DKX001	SD01763D
Web server	SD01655D

2 Overview of the Expert operating menu

The following table provides an overview of the menu structure of the expert operating menu and its parameters. The page reference indicates where the associated description of the submenu or parameter can be found.

 Expert		
Direct access (0106)	→ 	11
Locking status (0004)	→ 	12
Access status (0005)	→ 	13
Enter access code (0003)	→ 	13
▶ System	→ 	14
▶ Display	→ 	14
▶ Configuration backup	→ 	28
▶ Diagnostic handling	→ 	31
▶ Administration	→ 	37
▶ Sensor	→ 	42
▶ Measured values	→ 	42
▶ System units	→ 	52
▶ Process parameters	→ 	65
▶ External compensation	→ 	77
▶ Sensor adjustment	→ 	82
▶ Calibration	→ 	87
▶ I/O configuration	→ 	89
I/O module 1 to n terminal numbers (3902-1 to n)	→ 	89
I/O module 1 to n information (3906-1 to n)	→ 	89
I/O module 1 to n type (3901-1 to n)	→ 	90

Apply I/O configuration (3907)	→ 90
Conversion code (2762)	→ 91
► Input	→ 91
► Current input 1 to n	→ 91
► Status input 1 to n	→ 94
► Output	→ 96
► Current output 1 to n	→ 97
► Pulse/frequency/switch output 1 to n	→ 110
► Relay output 1 to n	→ 130
► Double pulse output	→ 137
► Communication	→ 141
► Modbus configuration	→ 142
► Modbus information	→ 146
► Modbus data map	→ 147
► Web server	→ 147
► WLAN settings	→ 150
► Application	→ 154
Reset all totalizers (2806)	→ 155
► Totalizer 1 to n	→ 155
► Custody transfer	→ 159
► Diagnostics	→ 159
Actual diagnostics (0691)	→ 160
Previous diagnostics (0690)	→ 161
Operating time from restart (0653)	→ 162

Operating time (0652)	→  162
▶ Diagnostic list	→  162
▶ Event logbook	→  167
▶ Custody transfer logbook	→  169
▶ Device information	→  169
▶ Mainboard module	→  173
▶ Sensor electronic module (ISEM)	→  173
▶ I/O module 1	→  174
▶ I/O module 2	→  174
▶ I/O module 3	→  175
▶ Display module	→  176
▶ Min/max values	→  184
▶ Data logging	→  176
▶ Heartbeat	→  187
▶ Simulation	→  187

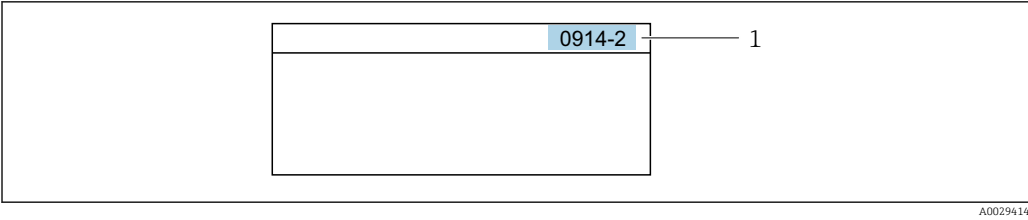
3 Description of device parameters

In the following section, the parameters are listed according to the menu structure of the local display. Specific parameters for the operating tools are included at the appropriate points in the menu structure.

Expert		
Direct access (0106)	→	11
Locking status (0004)	→	12
Access status (0005)	→	13
Enter access code (0003)	→	13
► System	→	14
► Sensor	→	42
► I/O configuration	→	89
► Input	→	91
► Output	→	96
► Communication	→	141
► Application	→	154
► Diagnostics	→	159

Direct access

Navigation	Expert → Direct access (0106)
Description	Use this function to enter the access code to enable direct access to the desired parameter via the local display. A parameter number is assigned to each parameter for this purpose.
User entry	0 to 65 535
Additional information	User entry The direct access code consists of a 4-digit number and the channel number, which identifies the channel of a process variable: e.g. 0914-1. In the navigation view, this appears on the right-hand side in the header of the selected parameter.



1 Direct access code

- Note the following when entering the direct access code:
- The leading zeros in the direct access code do not have to be entered.
Example: Input of "914" instead of "0914"
 - If no channel number is entered, channel 1 is jumped to automatically.
Example: Enter 0914 → **Assign process variable** parameter
 - If a different channel is jumped to: Enter the direct access code with the corresponding channel number.
Example: Enter 0914-2 → **Assign process variable** parameter

Locking status

Navigation Expert → Locking status (0004)

Description Displays the active write protection.

- User interface
- Hardware locked
 - CT active - all parameters
 - CT active - defined parameters
 - Temporarily locked

Additional information *Display*

If two or more types of write protection are active, the write protection with the highest priority is shown on the local display. In the operating tool all active types of write protection are displayed.

Detailed information on access authorization is provided in the "User roles and associated access authorization" and "Operating concept" sections of the Operations Instructions for the device → 7

Selection

Options	Description
None	The access status displayed in the Access status parameter (→ 13) applies . Only appears on local display.
Hardware locked (priority 1)	The DIP switch for hardware locking is activated on the PCB board. This locks write access to the parameters (e.g. via local display or operating tool).
CT active - all parameters (priority 2)	Only available for Promag H. The DIP switch for custody transfer mode is activated on the PCB board. This locks write access to all the parameters (e.g. via local display or operating tool). For detailed information on custody transfer mode, see the Special Documentation for the device → 7

Options	Description
CT active - defined parameters (priority 3)	 Only available for Promag H. The DIP switch for custody transfer mode is activated on the PCB board. This locks write access to the defined parameters (e.g. via local display or operating tool).  For detailed information on custody transfer mode, see the Special Documentation for the device →  7
Temporarily locked (Priority 4)	Write access to the parameters is temporarily locked on account of internal processes running in the device (e.g. data upload/download, reset etc.). Once the internal processing has been completed, the parameters can be changed once again.

Access status

Navigation	  Expert → Access status (0005)
Description	Displays the access authorization to the parameters via the local display, Web browser or operating tool.
User interface	▪ Operator ▪ Maintenance
Factory setting	Maintenance
Additional information	<i>Description</i>  Access authorization can be modified via the Enter access code parameter (→  13).  If additional write protection is active, this restricts the current access authorization even further. <i>Display</i>  Detailed information on access authorization is provided in the "User roles and associated access authorization" and "Operating concept" sections of the Operations Instructions for the device →  7

Enter access code

Navigation	  Expert → Ent. access code (0003)
Description	Use this function to enter the user-specific release code to remove parameter write protection.
User entry	0 to 9999

3.1 "System" submenu

Navigation  Expert → System

► System	
► Display	→  14
► Configuration backup	→  28
► Diagnostic handling	→  31
► Administration	→  37

3.1.1 "Display" submenu

Navigation  Expert → System → Display

► Display	
Display language (0104)	→  15
Format display (0098)	→  16
Value 1 display (0107)	→  18
0% bargraph value 1 (0123)	→  18
100% bargraph value 1 (0125)	→  19
Decimal places 1 (0095)	→  19
Value 2 display (0108)	→  20
Decimal places 2 (0117)	→  21
Value 3 display (0110)	→  21
0% bargraph value 3 (0124)	→  22
100% bargraph value 3 (0126)	→  22
Decimal places 3 (0118)	→  23
Value 4 display (0109)	→  23
Decimal places 4 (0119)	→  24

Display interval (0096)	→  24
Display damping (0094)	→  25
Header (0097)	→  25
Header text (0112)	→  26
Separator (0101)	→  27
Contrast display (0105)	→  27
Backlight (0111)	→  27

Display language

Navigation	 Expert → System → Display → Display language (0104)
Prerequisite	A local display is provided.
Description	Use this function to select the configured language on the local display.
Selection	■ English ■ Deutsch * ■ Français * ■ Español * ■ Italiano * ■ Nederlands * ■ Portuguesa * ■ Polski * ■ русский язык (Russian) * ■ Svenska * ■ Türkçe * ■ 中文 (Chinese) * ■ 日本語 (Japanese) * ■ 한국어 (Korean) * ■ العربية (Arabic) * ■ Bahasa Indonesia * ■ ภาษาไทย (Thai) * ■ tiếng Việt (Vietnamese) * ■ čeština (Czech) *
Factory setting	English (alternatively, the ordered language is preset in the device)

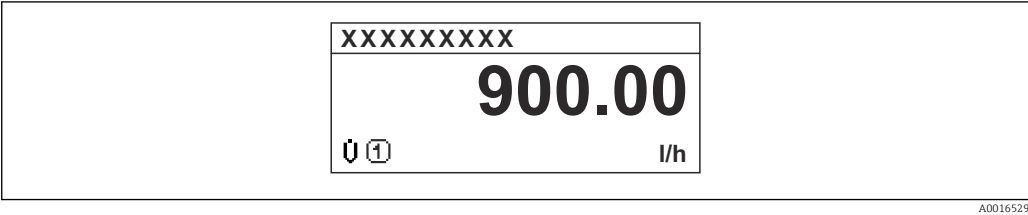
* Visibility depends on order options or device settings

Format display

Navigation	 Expert → System → Display → Format display (0098)
Prerequisite	A local display is provided.
Description	Use this function to select how the measured value is shown on the local display.
Selection	■ 1 value, max. size ■ 1 bargraph + 1 value ■ 2 values ■ 1 value large + 2 values ■ 4 values
Factory setting	1 value, max. size
Additional information	<i>Description</i> The display format (size, bar graph etc.) and number of measured values displayed simultaneously (1 to 4) can be configured. This setting only applies to normal operation.  ■ The Value 1 display parameter (→  18) to Value 4 display parameter (→  23) are used to specify which measured values are shown on the local display and in what order. ■ If more measured values are specified than the display mode selected permits, then the values alternate on the device display. The display time until the next change is configured via the Display interval parameter (→  24). <i>Custody transfer measurement</i>  Only available for Promag H. ■ Once the measuring device has been enabled for custody transfer mode, depending on the custody transfer approval selected the display can switch between showing the relevant information and the custody transfer counter. ■ In addition, a padlock symbol appears in the header of the display (.  For detailed information on custody transfer mode, see the Special Documentation for the device →  7

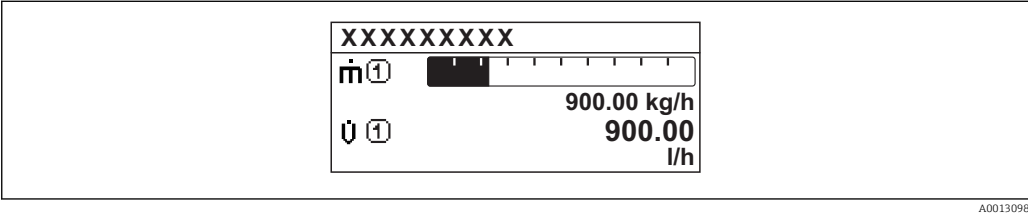
Possible measured values shown on the local display:

"1 value, max. size" option



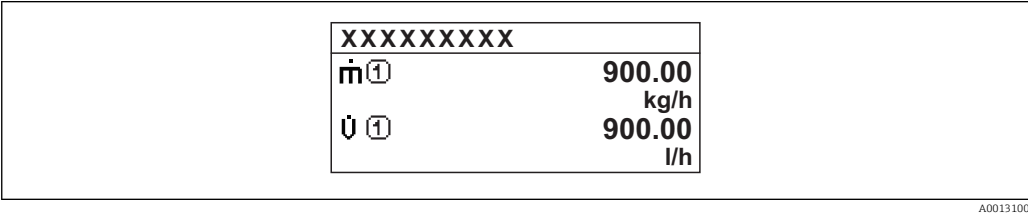
A0016529

"1 bargraph + 1 value" option



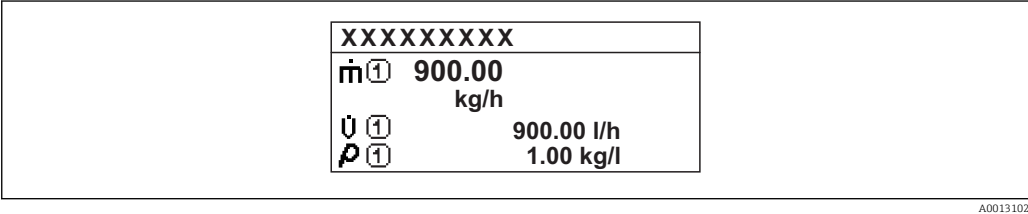
A0013098

"2 values" option



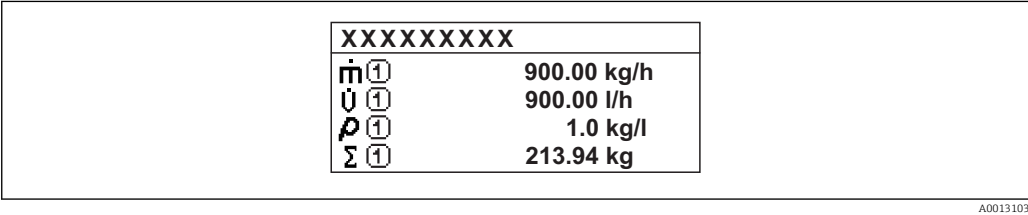
A0013100

"1 value large + 2 values" option



A0013102

"4 values" option



A0013103

Value 1 display 	
Navigation	  Expert → System → Display → Value 1 display (0107)
Prerequisite	A local display is provided.
Description	Use this function to select one of the measured values to be shown on the local display.
Selection	■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity[*] ■ Conductivity[*] ■ Corrected conductivity[*] ■ Totalizer 1 ■ Totalizer 2 ■ Totalizer 3 ■ Current output 1 ■ Current output 2[*] ■ Temperature[*] ■ Electronic temperature
Factory setting	Volume flow
Additional information	<i>Description</i> If several measured values are displayed at once, the measured value selected here will be the first value to be displayed. The value is only displayed during normal operation.  The Format display parameter (→  16) is used to specify how many measured values are displayed simultaneously and how. <i>Custody transfer measurement</i>  Only available for Promag H. Once the measuring device has been enabled for custody transfer mode, depending on the custody transfer approval selected the display can switch to show the relevant information.  For detailed information on custody transfer mode, see the Special Documentation for the device →  7 <i>Dependency</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).

0% bargraph value 1 	
Navigation	  Expert → System → Display → 0% bargraph 1 (0123)
Prerequisite	A local display is provided.

* Visibility depends on order options or device settings

Description	Use this function to enter the 0% bar graph value to be shown on the display for the measured value 1.
User entry	Signed floating-point number
Factory setting	Country-specific: ■ 0 l/h ■ 0 gal/min (us)
Additional information	<i>Description</i>  The Format display parameter (→  16) is used to specify that the measured value is to be displayed as a bar graph. <i>User entry</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).

100% bargraph value 1

Navigation	  Expert → System → Display → 100% bargraph 1 (0125)
Prerequisite	A local display is provided.
Description	Use this function to enter the 100% bar graph value to be shown on the display for the measured value 1.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter →  198
Additional information	<i>Description</i>  The Format display parameter (→  16) is used to specify that the measured value is to be displayed as a bar graph. <i>User entry</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).

Decimal places 1

Navigation	  Expert → System → Display → Decimal places 1 (0095)
Prerequisite	A measured value is specified in the Value 1 display parameter (→  18).
Description	Use this function to select the number of decimal places for measured value 1.

Selection	■ X ■ X.X ■ X.XX ■ X.XXX ■ X.XXXX
Factory setting	X.XX
Additional information	<i>Description</i>  This setting does not affect the measuring or computational accuracy of the device. The arrow displayed between the measured value and the unit indicates that the device computes with more digits than are shown on the local display.

Value 2 display



Navigation	 Expert → System → Display → Value 2 display (0108)
Prerequisite	A local display is provided.
Description	Use this function to select one of the measured values to be shown on the local display.
Selection	■ None ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature ■ Totalizer 1 ■ Totalizer 2 ■ Totalizer 3 ■ Current output 1 ■ Current output 2[*] ■ Current output 3[*] ■ Current output 4[*] ■ Custody transfer counter[*]
Factory setting	None

^{*} Visibility depends on order options or device settings

Additional information*Description*

If several measured values are displayed at once, the measured value selected here will be the second value to be displayed. The value is only displayed during normal operation.

 The **Format display** parameter (→  16) is used to specify how many measured values are displayed simultaneously and how.

Custody transfer measurement

 Only available for Promag H.

Once the measuring device has been enabled for custody transfer mode, depending on the custody transfer approval selected the display can switch to showing the custody transfer counter.

 For detailed information on custody transfer mode, see the Special Documentation for the device →  7

Dependency

 The unit of the displayed measured value is taken from the **System units** submenu (→  52).

Decimal places 2**Navigation**

  Expert → System → Display → Decimal places 2 (0117)

Prerequisite

A measured value is specified in the **Value 2 display** parameter (→  20).

Description

Use this function to select the number of decimal places for measured value 2.

Selection

- x
- x.x
- x.xx
- x.xxx
- x.xxxx

Factory setting

x.xx

Additional information*Description*

 This setting does not affect the measuring or computational accuracy of the device. The arrow displayed between the measured value and the unit indicates that the device computes with more digits than are shown on the local display.

Value 3 display**Navigation**

  Expert → System → Display → Value 3 display (0110)

Prerequisite

A local display is provided.

Description

Use this function to select one of the measured values to be shown on the local display.

Selection	For the picklist, see the Value 2 display parameter (→  20)
Factory setting	None
Additional information	<i>Description</i> If several measured values are displayed at once, the measured value selected here will be the third value to be displayed. The value is only displayed during normal operation.  The Format display parameter (→  16) is used to specify how many measured values are displayed simultaneously and how. <i>Selection</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).

0% bargraph value 3 	
Navigation	  Expert → System → Display → 0% bargraph 3 (0124)
Prerequisite	A selection was made in the Value 3 display parameter (→  21).
Description	Use this function to enter the 0% bar graph value to be shown on the display for the measured value 3.
User entry	Signed floating-point number
Factory setting	Country-specific: ■ 0 l/h ■ 0 gal/min (us)
Additional information	<i>Description</i>  The Format display parameter (→  16) is used to specify that the measured value is to be displayed as a bar graph. <i>User entry</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).

100% bargraph value 3 	
Navigation	  Expert → System → Display → 100% bargraph 3 (0126)
Prerequisite	A selection was made in the Value 3 display parameter (→  21).
Description	Use this function to enter the 100% bar graph value to be shown on the display for the measured value 3.
User entry	Signed floating-point number

Factory setting 0

Additional information *Description*



The **Format display** parameter (→ 16) is used to specify that the measured value is to be displayed as a bar graph.

User entry



The unit of the displayed measured value is taken from the **System units** submenu (→ 52).

Decimal places 3

Navigation Expert → System → Display → Decimal places 3 (0118)

Prerequisite A measured value is specified in the **Value 3 display** parameter (→ 21).

Description Use this function to select the number of decimal places for measured value 3.

Selection

- x
- x.x
- x.xx
- x.xxx
- x.xxxx

Factory setting x.xx

Additional information *Description*



This setting does not affect the measuring or computational accuracy of the device. The arrow displayed between the measured value and the unit indicates that the device computes with more digits than are shown on the local display.

Value 4 display

Navigation Expert → System → Display → Value 4 display (0109)

Prerequisite A local display is provided.

Description Use this function to select one of the measured values to be shown on the local display.

Selection For the picklist, see the **Value 2 display** parameter (→ 20)

Factory setting None

Additional information	<i>Description</i> If several measured values are displayed at once, the measured value selected here will be the fourth value to be displayed. The value is only displayed during normal operation.  The Format display parameter (→  16) is used to specify how many measured values are displayed simultaneously and how. <i>Selection</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).
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Decimal places 4 	
Navigation	  Expert → System → Display → Decimal places 4 (0119)
Prerequisite	A measured value is specified in the Value 4 display parameter (→  23).
Description	Use this function to select the number of decimal places for measured value 4.
Selection	■ x ■ x.x ■ x.xx ■ x.xxx ■ x.xxxx
Factory setting	x.xx
Additional information	<i>Description</i>  This setting does not affect the measuring or computational accuracy of the device. The arrow displayed between the measured value and the unit indicates that the device computes with more digits than are shown on the local display.

Display interval	
Navigation	  Expert → System → Display → Display interval (0096)
Prerequisite	A local display is provided.
Description	Use this function to enter the length of time the measured values are displayed if the values alternate on the display.
User entry	1 to 10 s
Factory setting	5 s

Additional information*Description*

This type of alternating display only occurs automatically if the number of measured values defined exceeds the number of values the selected display format can display simultaneously.

-  The **Value 1 display** parameter (→  18) to **Value 4 display** parameter (→  23) are used to specify which measured values are shown on the local display.
- The display format of the displayed measured values is specified using the **Format display** parameter (→  16).

Custody transfer measurement

-  Only available for Promag H.

Once the measuring device has been enabled for custody transfer mode, depending on the custody transfer approval selected the display can switch between showing the relevant information and the custody transfer counter.

-  For detailed information on custody transfer mode, see the Special Documentation for the device →  7

Display damping**Navigation**

  Expert → System → Display → Display damping (0094)

Prerequisite

A local display is provided.

Description

Use this function to enter a time constant for the reaction time of the local display to fluctuations in the measured value caused by process conditions.

User entry

0.0 to 999.9 s

Factory setting

0.0 s

Additional information*User entry*

Use this function to enter a time constant (PT1 element ¹⁾) for display damping:

- If a low time constant is entered, the display reacts particularly quickly to fluctuating measured variables.
- On the other hand, the display reacts more slowly if a high time constant is entered.

-  Damping is switched off if **0** is entered (factory setting).

Header**Navigation**

  Expert → System → Display → Header (0097)

Prerequisite

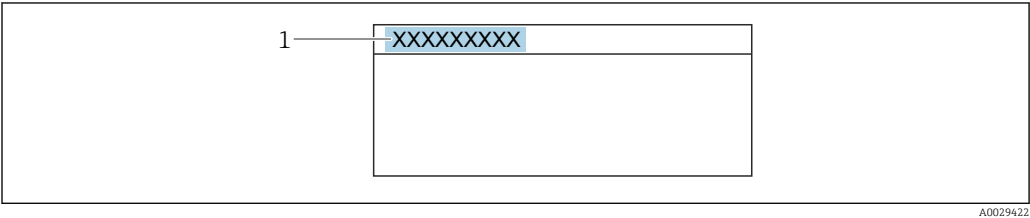
A local display is provided.

Description

Use this function to select the contents of the header of the local display.

1) proportional transmission behavior with first order delay

Selection	■ Device tag ■ Free text
Factory setting	Device tag
Additional information	Description The header text only appears during normal operation.



1 Position of the header text on the display

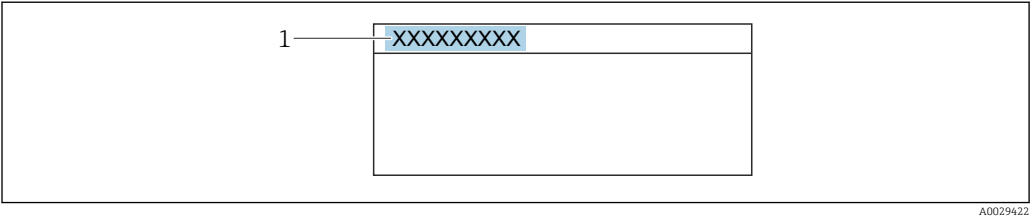
A0029422

- Selection
- Device tag
Is defined in the **Device tag** parameter (→  170).
 - Free text
Is defined in the **Header text** parameter (→  26).

Header text

Navigation	  Expert → System → Display → Header text (0112)
Prerequisite	In the Header parameter (→  25), the Free text option is selected.
Description	Use this function to enter a customer-specific text for the header of the local display.
User entry	Max. 12 characters such as letters, numbers or special characters (e.g. @, %, /)
Factory setting	-----

Additional information	Description The header text only appears during normal operation.
------------------------	---



1 Position of the header text on the display

A0029422

User entry

The number of characters displayed depends on the characters used.

Separator		
Navigation	  Expert → System → Display → Separator (0101)	
Prerequisite	A local display is provided.	
Description	Use this function to select the decimal separator.	
Selection	■ . (point) ■ , (comma)	
Factory setting	. (point)	
Contrast display		
Navigation	  Expert → System → Display → Contrast display (0105)	
Prerequisite	A local display is provided.	
Description	Use this function to enter a value to adapt the display contrast to the ambient conditions (e.g. the lighting or viewing angle).	
User entry	20 to 80 %	
Factory setting	Depends on the display	
Backlight		
Navigation	  Expert → System → Display → Backlight (0111)	
Prerequisite	One of the following conditions is met: ■ Order code for "Display; operation", option F "4-line, illum.; touch control" ■ Order code for "Display; operation", option G "4-line, illum.; touch control +WLAN" ■ Order code for "Display; operation", option O "remote 4-line display, illum; 10m/30ft cable; touch control"	
Description	Use this function to switch the backlight of the local display on and off.	
Selection	■ Disable ■ Enable	
Factory setting	Enable	

3.1.2 "Configuration backup" submenu

Navigation  Expert → System → Config. backup

► Configuration backup		
Operating time	→ 	28
Last backup	→ 	28
Configuration management	→ 	28
Backup state	→ 	29
Comparison result	→ 	30

Operating time

Navigation	 Expert → System → Config. backup → Operating time (0652)
Description	Use this function to display the length of time the device has been in operation.
User interface	Days (d), hours (h), minutes (m) and seconds (s)
Additional information	<i>User interface</i> The maximum number of days is 9999, which is equivalent to 27 years.

Last backup

Navigation	 Expert → System → Config. backup → Last backup (2757)
Description	Displays the time since a backup copy of the data was last saved to the device memory.
User interface	Days (d), hours (h), minutes (m) and seconds (s)

Configuration management



Navigation	 Expert → System → Config. backup → Config. managem. (2758)
Description	Use this function to select an action to save the data to the device memory.

- Selection**
- Cancel
 - Execute backup
 - Restore
 - Compare
 - Clear backup data

Factory setting Cancel

Additional information *Selection*

Options	Description
Cancel	No action is executed and the user exits the parameter.
Execute backup	A backup copy of the current device configuration is saved from the integrated HistoROM to the memory of the device. The backup copy includes the transmitter data of the device. The following message appears on local display: Backup active, please wait!
Restore	The last backup copy of the device configuration is restored from the device memory to the device's integrated HistoROM. The backup copy includes the transmitter data of the device. The following message appears on local display: Restore active! Do not interrupt power supply!
Compare	The device configuration saved in the device memory is compared with the current device configuration of the integrated HistoROM. The following message appears on local display: Comparing files The result can be viewed in Comparison result parameter.
Clear backup data	The backup copy of the device configuration is deleted from the memory of the device. The following message appears on local display: Deleting file

HistoROM

A HistoROM is a "non-volatile" device memory in the form of an EEPROM.

Backup state

Navigation  Expert → System → Config. backup → Backup state (2759)

Description Displays the status of the data backup process.

- User interface**
- None
 - Backup in progress
 - Restoring in progress
 - Delete in progress
 - Compare in progress
 - Restoring failed
 - Backup failed

Factory setting None

Comparison result

Navigation

 Expert → System → Config. backup → Compar. result (2760)

Description

Displays the last result of the comparison of the data records in the device memory and in the HistoROM.

User interface

- Settings identical
- Settings not identical
- No backup available
- Backup settings corrupt
- Check not done
- Dataset incompatible

Factory setting

Check not done

Additional information

Description

 The comparison is started via the **Compare** option in the **Configuration management** parameter (→  28).

Selection

Options	Description
Settings identical	The current device configuration of the HistoROM is identical to the backup copy in the device memory. If the transmitter configuration of another device has been transmitted to the device via HistoROM in the Configuration management parameter, the current device configuration of the HistoROM is only partially identical to the backup copy in the device memory: The settings for the transmitter are not identical.
Settings not identical	The current device configuration of the HistoROM is not identical to the backup copy in the device memory.
No backup available	There is no backup copy of the device configuration of the HistoROM in the device memory.
Backup settings corrupt	The current device configuration of the HistoROM is corrupt or not compatible with the backup copy in the device memory.
Check not done	The device configuration of the HistoROM has not yet been compared to the backup copy in the device memory.
Dataset incompatible	The backup copy in the device memory is not compatible with the device.

HistoROM

A HistoROM is a "non-volatile" device memory in the form of an EEPROM.

3.1.3 "Diagnostic handling" submenu

Navigation  Expert → System → Diagn. handling

► Diagnostic handling	
Alarm delay (0651)	→  31
► Diagnostic behavior	→  31

Alarm delay

Navigation  Expert → System → Diagn. handling → Alarm delay (0651)

Description Use this function to enter the time interval until the device generates a diagnostic message.



The diagnostic message is reset without a time delay.

User entry 0 to 60 s

Factory setting 0 s

Additional information *Result*

This setting affects the following diagnostic messages:

- 170 coil resistance
- 832 Electronic temperature too high
- 833 Electronic temperature too low
- 834 Process temperature too high
- 835 Process temperature too low
- 962 empty pipe

"Diagnostic behavior" submenu

Each item of diagnostic information is assigned a specific diagnostic behavior at the factory. The user can change this assignment for specific diagnostic information in the **Diagnostic behavior** submenu (→  31).

The following options are available in the **Assign behavior of diagnostic no. xxx** parameters:

Options	Description
Alarm	The device stops measurement. The measured value output via Modbus RS485 and the totalizers assume the defined alarm condition. A diagnostic message is generated. The background lighting changes to red.
Warning	The device continues to measure. The measured value output via Modbus RS485 and the totalizers are not affected. A diagnostic message is generated.

Options	Description
Logbook entry only	The device continues to measure. The diagnostic message is displayed only in the Event logbook submenu (→  167) (Event list submenu (→  168)) and is not displayed in alternation with the operational display.
Off	The diagnostic event is ignored, and no diagnostic message is generated or entered.



For a list of all the diagnostic events, see the Operating Instructions for the device

Navigation



Expert → System → Diagn. handling → Diagn. behavior

► Diagnostic behavior		
Assign behavior of diagnostic no. 302 (0739)		→  33
Assign behavior of diagnostic no. 531 (0741)		→  33
Assign behavior of diagnostic no. 441 (0657)		→  33
Assign behavior of diagnostic no. 442 (0658)		→  34
Assign behavior of diagnostic no. 443 (0659)		→  34
Assign behavior of diagnostic no. 444 (0740)		→  34
Assign behavior of diagnostic no. 543 (0643)		→  35
Assign behavior of diagnostic no. 832 (0681)		→  35
Assign behavior of diagnostic no. 833 (0682)		→  36
Assign behavior of diagnostic no. 834 (0700)		→  36
Assign behavior of diagnostic no. 835 (0702)		→  36
Assign behavior of diagnostic no. 862 (0745)		→  37
Assign behavior of diagnostic no. 937 (0743)		→  37

Assign behavior of diagnostic no. 302 (Device verification active)


Navigation	Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 302 (0739)
Description	Use this function to change the diagnostic behavior of the diagnostic message 302 Device verification active .
Selection	■ Alarm ■ Warning
Factory setting	Warning
Additional information	For a detailed description of the options available, see → 31

Assign behavior of diagnostic no. 531 (Empty pipe detection)


Navigation	Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 531 (0741)
Description	Use this function to change the diagnostic behavior of the diagnostic message 531 Empty pipe detection .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	For a detailed description of the options available, see → 31

Assign behavior of diagnostic no. 441 (Current output 1 to n)


Navigation	Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 441 (0657)
Description	Use this function to change the diagnostic behavior of the diagnostic message 441 Current output 1 to n .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	For a detailed description of the options available, see → 31

Assign behavior of diagnostic no. 442 (Frequency output 1 to n)



Navigation	 Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 442 (0658)
Prerequisite	The measuring device has a pulse/frequency/switch output.
Description	Use this function to change the diagnostic behavior of the diagnostic message 442 Frequency output 1 to n .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 443 (Pulse output 1 to n)



Navigation	 Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 443 (0659)
Prerequisite	The measuring device has a pulse/frequency/switch output.
Description	Use this function to change the diagnostic behavior of the diagnostic message 443 Pulse output 1 to n .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 444 (Current input 1 to n)



Navigation	 Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 444 (0740)
Prerequisite	The device has one current input.
Description	Use this function to change the diagnostic behavior of the diagnostic message 444 Current input 1 to n .

Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 543 (Double pulse output)



Navigation	  Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 543 (0643)
Description	Use this function to change the diagnostic behavior of the diagnostic message 543 Double pulse output .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 832 (Electronic temperature too high)



Navigation	  Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 832 (0681)
Description	Use this function to change the diagnostic behavior of the diagnostic message 832 Electronic temperature too high .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Logbook entry only
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 833 (Electronic temperature too low)



Navigation	 Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 833 (0682)
Description	Use this function to change the diagnostic behavior of the diagnostic message 833 Electronic temperature too low .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Logbook entry only
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 834 (Process temperature too high)



Navigation	 Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 834 (0700)
Description	Use this function to change the diagnostic behavior of the diagnostic message 834 Process temperature too high .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 835 (Process temperature too low)



Navigation	 Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 835 (0702)
Description	Use this function to change the diagnostic behavior of the diagnostic message 835 Process temperature too low .
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only
Factory setting	Warning
Additional information	 For a detailed description of the options available, see →  31

Assign behavior of diagnostic no. 862 (Empty pipe)		
Navigation	  Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 862 (0745)	
Description	Use this function to change the diagnostic behavior of the diagnostic message 862 Empty pipe .	
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only	
Factory setting	Warning	
Additional information	 For a detailed description of the options available, see →  31	

Assign behavior of diagnostic no. 937 (EMC interference)		
Navigation	  Expert → System → Diagn. handling → Diagn. behavior → Diagnostic no. 937 (0743)	
Description	Use this function to change the diagnostic behavior of the diagnostic message 937 EMC interference .	
Selection	■ Off ■ Alarm ■ Warning ■ Logbook entry only	
Factory setting	Warning	
Additional information	 For a detailed description of the options available, see →  31	

3.1.4 "Administration" submenu

Navigation   Expert → System → Administration

► Administration	
► Define access code	→  38
► Reset access code	→  39
Device reset (0000)	→  40

Activate SW option (0029)	→ 40
Software option overview (0015)	→ 41

"Define access code" wizard

 The **Define access code** wizard (→ 38) is only available when operating via the local display or Web browser.

If operating via the operating tool, the **Define access code** parameter can be found directly in the **Administration** submenu. There is no **Confirm access code** parameter if the device is operated via the operating tool.

Navigation  Expert → System → Administration → Def. access code

► Define access code	
Define access code	→ 38
Confirm access code	→ 39

Define access code

Navigation   Expert → System → Administration → Def. access code → Def. access code

Description Use this function to enter a user-specific release code to restrict write-access to the parameters. This protects the device configuration against any inadvertent modifications via the local display, Web browser, FieldCare or DeviceCare (via CDI-RJ45 service interface).

User entry Max. 16-digit character string comprising numbers, letters and special characters

Additional information *Description*

The write protection affects all parameters in the document marked with the  symbol.

On the local display, the  symbol in front of a parameter indicates that the parameter is write-protected.

The parameters that cannot be write-accessed are grayed out in the Web browser.

 Once the access code has been defined, write-protected parameters can only be modified if the access code is entered in the **Enter access code** parameter (→ 13).

 If you lose the access code, please contact your Endress+Hauser sales organization.

User entry

A message is displayed if the access code is not in the input range.

Factory setting

If the factory setting is not changed or 0 is defined as the access code, the parameters are not write-protected and the device configuration data can be modified. The user is logged on in the **"Maintenance"** role.

Confirm access code	
Navigation	 Expert → System → Administration → Def. access code → Confirm code
Description	Enter the defined release code a second time to confirm the release code.
User entry	Max. 16-digit character string comprising numbers, letters and special characters

"Reset access code" submenu

Navigation  Expert → System → Administration → Reset acc. code

► Reset access code

Operating time (0652)

→  39

Reset access code (0024)

→  39

Operating time	
Navigation	 Expert → System → Administration → Reset acc. code → Operating time (0652)
Description	Use this function to display the length of time the device has been in operation.
User interface	Days (d), hours (h), minutes (m) and seconds (s)
Additional information	<i>User interface</i> The maximum number of days is 9999, which is equivalent to 27 years.

Reset access code	
Navigation	 Expert → System → Administration → Reset acc. code → Reset acc. code (0024)
Description	Use this function to enter a reset code to reset the user-specific release code to the factory setting.
User entry	Character string comprising numbers, letters and special characters
Factory setting	0x00

Additional information*Description*

For a reset code, contact your Endress+Hauser service organization.

User entry

The reset code can only be entered via:

- Web browser
- DeviceCare, FieldCare (via interface CDI RJ45)
- Fieldbus

Additional parameters in the "Administration" submenu**Device reset****Navigation**

Expert → System → Administration → Device reset (0000)

Description

Use this function to choose whether to reset the device configuration - either entirely or in part - to a defined state.

Selection

- Cancel
- To delivery settings
- Restart device
- Restore S-DAT backup

Factory setting

Cancel

Additional information*Selection*

Options	Description
Cancel	No action is executed and the user exits the parameter.
To delivery settings	Every parameter for which a customer-specific default setting was ordered is reset to this customer-specific value. All other parameters are reset to the factory setting.
Restart device	The restart resets every parameter whose data are in the volatile memory (RAM) to the factory setting (e.g. measured value data). The device configuration remains unchanged.
Restore S-DAT backup	Restore the data that are saved on the S-DAT. The data record is restored from the electronics memory to the S-DAT.

Activate SW option**Navigation**

Expert → System → Administration → Activate SW opt. (0029)

Description

Use this function to enter an activation code to enable an additional, ordered software option.

User entry	Max. 10-digit string consisting of numbers.
Factory setting	Depends on the software option ordered
Additional information	<i>Description</i> If a measuring device was ordered with an additional software option, the activation code is programmed in the device at the factory. <i>User entry</i>  To activate a software option subsequently, please contact your Endress+Hauser sales organization. NOTE! The activation code is linked to the serial number of the measuring device and varies according to the device and software option. If an incorrect or invalid code is entered, this results in the loss of software options that have already been activated. ▶ Before you enter a new activation code, make a note of the current activation code . ▶ Enter the new activation code provided by Endress+Hauser when the new software option was ordered. ▶ Once the activation code has been entered, check if the new software option is displayed in the Software option overview parameter (→  41). ↳ The new software option is active if it is displayed. ↳ If the new software option is not displayed or all software options have been deleted, the code entered was either incorrect or invalid. ▶ If the code entered is incorrect or invalid, enter the old activation code . ▶ Have your Endress+Hauser sales organization check the new activation code remembering to specify the serial number or ask for the code again. <i>Example for a software option</i> Order code for "Application package", option EA "Extended HistoROM"  The software options currently enabled are displayed in the Software option overview parameter (→  41). <i>Web browser</i>  Once a software option has been activated, the page must be loaded again in the Web browser.

Software option overview

Navigation	  Expert → System → Administration → SW option overv. (0015)
Description	Displays all the software options that are enabled in the device.
User interface	■ Extended HistoROM ■ Electrode cleaning circuit ■ Heartbeat Verification ■ Heartbeat Monitoring ■ Custody transfer

Additional information

Description

Displays all the options that are available if ordered by the customer.

"Extended HistoROM" option

Order code for "Application package", option **EA** "Extended HistoROM"

"Electrode cleaning circuit" option

Order code for "Application package", option **EC** "ECC electrode cleaning"

"Heartbeat Verification" option and "Heartbeat Monitoring" option

Order code for "Application package", option **EB** "Heartbeat Verification + Monitoring"

"Custody transfer" option

The measuring device has an approval for custody transfer measurement.



Detailed information on the national and international approvals for custody transfer that are currently available can be supplied by your Endress+Hauser sales organization.

3.2 "Sensor" submenu

Navigation

Expert → Sensor

► Sensor

► Measured values

→ 42

► System units

→ 52

► Process parameters

→ 65

► External compensation

→ 77

► Sensor adjustment

→ 82

► Calibration

→ 87

3.2.1 "Measured values" submenu

Navigation

Expert → Sensor → Measured val.

► Measured values

► Process variables

→ 43

► Totalizer

→ 45

► Input values	→ 47
► Output values	→ 48

"Process variables" submenu

Navigation   Expert → Sensor → Measured val. → Process variab.

► Process variables	
Volume flow (1838)	→ 43
Mass flow (1847)	→ 43
Conductivity (1850)	→ 44
Corrected volume flow (1851)	→ 44
Temperature (1852)	→ 44
Corrected conductivity (1853)	→ 45

Volume flow

Navigation	  Expert → Sensor → Measured val. → Process variab. → Volume flow (1838)
Description	Displays the volume flow that is currently measured.
User interface	Signed floating-point number
Additional information	

Mass flow

Navigation	  Expert → Sensor → Measured val. → Process variab. → Mass flow (1847)
Description	Displays the mass flow currently calculated.
User interface	Signed floating-point number
Additional information	Dependency  The unit is taken from the Mass flow unit parameter (→ 57)

Conductivity

Navigation	 Expert → Sensor → Measured val. → Process variab. → Conductivity (1850)
Description	Displays the conductivity currently measured.
User interface	Signed floating-point number
Additional information	<i>Dependency</i>  The unit is taken from the Conductivity unit parameter (→  56)

Corrected volume flow

Navigation	 Expert → Sensor → Measured val. → Process variab. → Correct.vol.flow (1851)
Description	Displays the corrected volume flow currently measured.
User interface	Signed floating-point number
Additional information	<i>Dependency</i>  The unit is taken from the Corrected volume flow unit parameter (→  59)

Temperature

Navigation	 Expert → Sensor → Measured val. → Process variab. → Temperature (1852)
Prerequisite	One of the following conditions is met: ■ Order code for "Sensor option", option CI "Medium temperature measurement" - or ■ The temperature is read into the flowmeter from an external device.
Description	Displays the temperature currently calculated.
User interface	Positive floating-point number
Additional information	<i>Dependency</i>  The unit is taken from the Temperature unit parameter (→  56)

Corrected conductivity

Navigation	 Expert → Sensor → Measured val. → Process variab. → CorrConductivity (1853)
Prerequisite	One of the following conditions is met: ▪ Order code for "Sensor option", option CI "Medium temperature measurement" or ▪ The temperature is read into the flowmeter from an external device.
Description	Displays the conductivity currently corrected.
User interface	Positive floating-point number
Additional information	<i>Dependency</i>  The unit is taken from the Conductivity unit parameter (→  56)

"Totalizer" submenu

Navigation  Expert → Sensor → Measured val. → Totalizer

 Totalizer		
Totalizer value 1 to n (0911-1 to n)	→	 45
Totalizer overflow 1 to n (0910-1 to n)	→	 46

Totalizer value 1 to n

Navigation	 Expert → Sensor → Measured val. → Totalizer → Totalizer val. 1 to n (0911-1 to n)
Prerequisite	A process variable is selected in the Assign process variable parameter (→  155) of the Totalizer 1 to n submenu.
Description	Displays the current totalizer reading.
User interface	Signed floating-point number

Additional information*Description*

As it is only possible to display a maximum of 7 digits in the operating tool, the current counter value is the sum of the totalizer value and the overflow value from the **Totalizer overflow 1 to n** parameter if the display range is exceeded.

 In the event of an error, the totalizer adopts the mode defined in the **Failure mode** parameter (→  158).

User interface

The value of the process variable totalized since measuring began can be positive or negative. This depends on the settings in the **Totalizer operation mode** parameter (→  157).

 The unit of the selected process variable is specified for the totalizer in the **Unit totalizer** parameter (→  156).

Example

Calculation of the current totalizer reading when the value exceeds the 7-digit display range of the operating tool:

- Value in the **Totalizer value 1** parameter: 1 968 457 m³
- Value in the **Totalizer overflow 1** parameter: $1 \cdot 10^7$ (1 overflow) = 10 000 000 [m³]
- Current totalizer reading: 11 968 457 m³

Totalizer overflow 1 to n**Navigation**

  Expert → Sensor → Measured val. → Totalizer → Tot. overflow 1 to n (0910–1 to n)

Prerequisite

A process variable is selected in the **Assign process variable** parameter (→  155) of the **Totalizer 1 to n** submenu.

Description

Displays the current totalizer overflow.

User interface

Integer with sign

Additional information*Description*

If the current totalizer reading exceeds 7 digits, which is the maximum value range that can be displayed by the operating tool, the value above this range is output as an overflow. The current totalizer value is therefore the sum of the overflow value and the totalizer value from the **Totalizer value 1 to n** parameter.

User interface

 The unit of the selected process variable is specified for the totalizer in the **Unit totalizer** parameter (→  156).

Example

Calculation of the current totalizer reading when the value exceeds the 7-digit display range of the operating tool:

- Value in the **Totalizer value 1** parameter: 1 968 457 m³
- Value in the **Totalizer overflow 1** parameter: $2 \cdot 10^7$ (2 overflows) = 20 000 000 [m³]
- Current totalizer reading: 21 968 457 m³

"Input values" submenu

Navigation  Expert → Sensor → Measured val. → Input values

► Input values

► Current input 1 to n

→  47

► Value status input 1 to n

→  48

"Current input 1 to n" submenu

Navigation  Expert → Sensor → Measured val. → Input values → Current input 1 to n

► Current input 1 to n

Measured values 1 to n (1603-1 to n)

→  47

Measured current 1 to n (1604-1 to n)

→  47

Measured values 1 to n

Navigation	 Expert → Sensor → Measured val. → Input values → Current input 1 to n → Measured val. 1 to n (1603-1 to n)
Description	Displays the current input value.
User interface	Signed floating-point number

Measured current 1 to n

Navigation	 Expert → Sensor → Measured val. → Input values → Current input 1 to n → Measur. curr. 1 to n (1604-1 to n)
Description	Displays the current value of the current input.
User interface	0 to 22.5 mA

"Value status input 1 to n" submenu

Navigation  Expert → Sensor → Measured val. → Input values → Val.stat.inp. 1 to n

► Value status input 1 to n

Value status input (1353-1 to n) →  48

Value status input

Navigation	 Expert → Sensor → Measured val. → Input values → Val.stat.inp. 1 to n → Val.stat.inp. (1353-1 to n)
Description	Displays the current input signal level.
User interface	■ High ■ Low

"Output values" submenu

Navigation  Expert → Sensor → Measured val. → Output values

► Output values

► Value current output 1 to n →  48

► Pulse/frequency/switch output 1 to n →  49

► Relay output 1 to n →  51

► Double pulse output →  52

"Value current output 1 to n" submenu

Navigation  Expert → Sensor → Measured val. → Output values → Value curr.out 1 to n

► Value current output 1 to n

Output current 1 to n (0361-1 to n) →  49

Measured current 1 to n (0366-1 to n) →  49

Output current 1 to n	
Navigation	  Expert → Sensor → Measured val. → Output values → Value curr.out 1 to n → Output curr. 1 to n (0361-1 to n)
Description	Displays the current value currently calculated for the current output.
User interface	0 to 22.5 mA

Measured current 1 to n	
Navigation	  Expert → Sensor → Measured val. → Output values → Value curr.out 1 to n → Measur. curr. 1 to n (0366-1 to n)
Description	Use this function to display the actual measured value of the output current.
User interface	0 to 30 mA

"Pulse/frequency/switch output 1 to n" submenu

Navigation   Expert → Sensor → Measured val. → Output values → PFS output 1 to n

► Pulse/frequency/switch output 1 to n	
Output frequency 1 to n (0471-1 to n)	→  49
Value per pulse (0455-1 to n)	→  50
Switch status 1 to n (0461-1 to n)	→  50

Output frequency 1 to n	
Navigation	  Expert → Sensor → Measured val. → Output values → PFS output 1 to n → Output freq. 1 to n (0471-1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Frequency option is selected.
Description	Displays the actual value of the output frequency which is currently measured.
User interface	0.0 to 12 500.0 Hz

Value per pulse 	
Navigation	 Expert → Sensor → Measured val. → Output values → PFS output 1 to n → Value per pulse (0455-1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Pulse option is selected and one of the following options is selected in the Assign pulse output parameter (→  114): ■ Mass flow ■ Volume flow ■ Corrected volume flow
Description	Use this function to enter the value for the measured value that a pulse is equivalent to.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter →  199
Additional information	<i>User entry</i> Weighting of the pulse output with a quantity. The lower the pulse value, the ■ better the resolution. ■ the higher the frequency of the pulse response.

Switch status 1 to n	
Navigation	 Expert → Sensor → Measured val. → Output values → PFS output 1 to n → Switch status 1 to n (0461-1 to n)
Prerequisite	The Switch option is selected in the Operating mode parameter (→  112).
Description	Displays the current switch status of the status output.
User interface	■ Open ■ Closed
Additional information	<i>User interface</i> ■ Open The switch output is not conductive. ■ Closed The switch output is conductive.

"Relay output 1 to n" submenu

Navigation



Expert → Sensor → Measured val. → Output values → Relay output 1 to n

► Relay output 1 to n

Switch status (0801-1 to n)

→  51

Switch cycles (0815-1 to n)

→  51

Max. switch cycles number (0817-1 to n)

→  51

Switch status

Navigation	 Expert → Sensor → Measured val. → Output values → Relay output 1 to n → Switch status (0801-1 to n)
Description	Displays the current status of the relay output.
User interface	■ Open ■ Closed
Additional information	User interface ■ Open The relay output is not conductive. ■ Closed The relay output is conductive.

Switch cycles

Navigation	 Expert → Sensor → Measured val. → Output values → Relay output 1 to n → Switch cycles (0815-1 to n)
Description	Displays all the switch cycles performed.
User interface	Positive integer

Max. switch cycles number

Navigation	 Expert → Sensor → Measured val. → Output values → Relay output 1 to n → Max. cycles no. (0817-1 to n)
Description	Displays the maximum number of guaranteed switch cycles.

User interface Positive integer

"Double pulse output" submenu

Navigation  Expert → Sensor → Measured val. → Output values → Double pulse out

► Double pulse output

Pulse output (0987) →  52

Pulse output

Navigation  Expert → Sensor → Measured val. → Output values → Double pulse out → Pulse output (0987)

Description Displays the pulse frequency of the double pulse output which is currently output.

User interface Positive floating-point number

Additional information  For a detailed description and example: **Pulse output** parameter (→  117)

3.2.2 "System units" submenu

Navigation  Expert → Sensor → System units

► System units

Volume flow unit (0553) →  53

Volume unit (0563) →  55

Conductivity unit (0582) →  56

Temperature unit (0557) →  56

Mass flow unit (0554) →  57

Mass unit (0574) →  58

Density unit (0555) →  58

Corrected volume flow unit (0558)	→ 59
Corrected volume unit (0575)	→ 60
Date/time format (2812)	→ 60
► User-specific units	→ 61

Volume flow unit

Navigation  Expert → Sensor → System units → Volume flow unit (0553)

Description Use this function to select the unit for the volume flow.

Selection*SI units*

- cm³/s
- cm³/min
- cm³/h
- cm³/d
- dm³/s
- dm³/min
- dm³/h
- dm³/d
- m³/s
- m³/min
- m³/h
- m³/d
- ml/s
- ml/min
- ml/h
- ml/d
- l/s
- l/min
- l/h
- l/d
- hl/s
- hl/min
- hl/h
- hl/d
- Ml/s
- Ml/min
- Ml/h
- Ml/d

US units

- af/s
- af/min
- af/h
- af/d
- ft³/s
- ft³/min
- ft³/h
- ft³/d
- fl oz/s (us)
- fl oz/min (us)
- fl oz/h (us)
- fl oz/d (us)
- gal/s (us)
- gal/min (us)
- gal/h (us)
- gal/d (us)
- kgal/s (us)
- kgal/min (us)
- kgal/h (us)
- kgal/d (us)
- Mgal/s (us)
- Mgal/min (us)
- Mgal/h (us)
- Mgal/d (us)
- bbl/s (us;liq.)
- bbl/min (us;liq.)
- bbl/h (us;liq.)
- bbl/d (us;liq.)
- bbl/s (us;beer)
- bbl/min (us;beer)
- bbl/h (us;beer)
- bbl/d (us;beer)
- bbl/s (us;oil)
- bbl/min (us;oil)
- bbl/h (us;oil)
- bbl/d (us;oil)
- bbl/s (us;tank)
- bbl/min (us;tank)
- bbl/h (us;tank)
- bbl/d (us;tank)

Imperial units

- gal/s (imp)
- gal/min (imp)
- gal/h (imp)
- gal/d (imp)
- Mgal/s (imp)
- Mgal/min (imp)
- Mgal/h (imp)
- Mgal/d (imp)
- bbl/s (imp;beer)
- bbl/min (imp;beer)
- bbl/h (imp;beer)
- bbl/d (imp;beer)
- bbl/s (imp;oil)
- bbl/min (imp;oil)
- bbl/h (imp;oil)
- bbl/d (imp;oil)

Custom-specific units

- User vol./s
- User vol./min
- User vol./h
- User vol./d

Factory setting

Country-specific:

- l/h
- gal/min (us)

Additional information

Result

The selected unit applies for:
Volume flow parameter (→ ⓘ 43)

Selection

ⓘ For an explanation of the abbreviated units: → ⓘ 203

Customer-specific units

ⓘ The unit for the customer-specific volume is specified in the **User volume text** parameter (→ ⓘ 61).

Volume unit ⓘ

Navigation

ⓘ ⓘ Expert → Sensor → System units → Volume unit (0563)

Description

Use this function to select the unit for the volume.

Selection

SI units

- cm³
- dm³
- m³
- ml
- l
- hl
- Ml Mega

US units

- af
- ft³
- fl oz (us)
- gal (us)
- kgal (us)
- Mgal (us)
- bbl (us;oil)
- bbl (us;liq.)
- bbl (us;beer)
- bbl (us;tank)

Imperial units

- gal (imp)
- Mgal (imp)
- bbl (imp;beer)
- bbl (imp;oil)

Custom-specific units

User vol.

Factory setting

Country-specific:

- m³
- gal (us)

Additional information

Selection

ⓘ For an explanation of the abbreviated units: → ⓘ 203

Customer-specific units

ⓘ The unit for the customer-specific volume is specified in the **User volume text** parameter (→ ⓘ 61).

Conductivity unit

Navigation	Expert → Sensor → System units → Conductiv. unit (0582)
Prerequisite	The On option is selected in the Conductivity measurement parameter (→ 68) parameter.
Description	Use this function to select the unit for the conductivity.
Selection	<i>SI units</i> ■ nS/cm ■ µS/cm ■ µS/m ■ µS/mm ■ mS/m ■ mS/cm ■ S/cm ■ S/m ■ kS/m ■ MS/m
Factory setting	µS/cm
Additional information	<i>Effect</i> The selected unit applies for: ■ Conductivity parameter (→ 44) ■ Corrected conductivity parameter (→ 45) <i>Selection</i> For an explanation of the abbreviated units: → 203

Temperature unit

Navigation	Expert → Sensor → System units → Temperature unit (0557)
Description	Use this function to select the unit for the temperature.
Selection	<i>SI units</i> ■ °C ■ K <i>US units</i> ■ °F ■ °R
Factory setting	Country-specific: ■ °C ■ °F
Additional information	<i>Result</i> The selected unit applies for: ■ Temperature parameter (→ 44) ■ Maximum value parameter (→ 185) ■ Minimum value parameter (→ 185)

- **External temperature** parameter (→  78)
- **Maximum value** parameter (→  186)
- **Minimum value** parameter (→  186)

Selection

 For an explanation of the abbreviated units: →  203

Mass flow unit**Navigation**

  Expert → Sensor → System units → Mass flow unit (0554)

Description

Use this function to select the unit for the mass flow.

Selection*SI units*

- g/s
- g/min
- g/h
- g/d
- kg/s
- kg/min
- kg/h
- kg/d
- t/s
- t/min
- t/h
- t/d

US units

- oz/s
- oz/min
- oz/h
- oz/d
- lb/s
- lb/min
- lb/h
- lb/d
- STon/s
- STon/min
- STon/h
- STon/d

Custom-specific units

- User mass/s
- User mass/min
- User mass/h
- User mass/d

Factory setting

Country-specific:

- kg/h
- lb/min

Additional information*Result*

The selected unit applies for:
Mass flow parameter (→  43)

Selection

 For an explanation of the abbreviated units: →  203

Customer-specific units

 The unit for the customer-specific mass is specified in the **User mass text** parameter (→  62).

Mass unit

Navigation	Expert → Sensor → System units → Mass unit (0574)		
Description	Use this function to select the unit for the mass.		
Selection	<i>SI units</i> ■ g ■ kg ■ t <i>US units</i> ■ oz ■ lb ■ STon <i>Custom-specific units</i> User mass		
Factory setting	Country-specific: ■ kg ■ lb		
Additional information	<i>Selection</i> For an explanation of the abbreviated units: → 203 <i>Customer-specific units</i> The unit for the customer-specific mass is specified in the User mass text parameter (→ 62).		

Density unit

Navigation	Expert → Sensor → System units → Density unit (0555)		
Description	Use this function to select the unit for the density.		
Selection	<i>SI units</i> ■ g/cm³ ■ g/m³ ■ kg/dm³ ■ kg/l ■ kg/m³ ■ SD4°C ■ SD15°C ■ SD20°C ■ SG4°C ■ SG15°C ■ SG20°C <i>US units</i> ■ lb/ft³ ■ lb/gal (us) ■ lb/bbl (us;liq.) ■ lb/bbl (us;beer) ■ lb/bbl (us;oil) ■ lb/bbl (us;tank) <i>Imperial units</i> ■ lb/gal (imp) ■ lb/bbl (imp;beer) ■ lb/bbl (imp;oil)		
Factory setting	Country-specific: ■ kg/l ■ lb/ft³		

Additional information*Result*

The selected unit applies for:

- **External density** parameter (→  79)
- **Fixed density** parameter (→  79)

Selection

- SD = specific density

The specific density is the ratio of the fluid density to the water density at a water temperature of +4 °C (+39 °F), +15 °C (+59 °F), +20 °C (+68 °F).

- SG = specific gravity

The specific gravity is the ratio of the fluid density to the water density at a water temperature of +4 °C (+39 °F), +15 °C (+59 °F), +20 °C (+68 °F).



For an explanation of the abbreviated units: →  203

Corrected volume flow unit**Navigation**

  Expert → Sensor → System units → Cor.volflow unit (0558)

Description

Use this function to select the unit for the corrected volume flow.

Selection*SI units*

- l/s
- l/min
- l/h
- l/d
- m³/s
- m³/min
- m³/h
- m³/d
- m³/s
- m³/min
- m³/h
- m³/d

US units

- Sft³/s
- Sft³/min
- Sft³/h
- Sft³/d
- Sgal/s (us)
- Sgal/min (us)
- Sgal/h (us)
- Sgal/d (us)
- Sbbbl/s (us;liq.)
- Sbbbl/min (us;liq.)
- Sbbbl/h (us;liq.)
- Sbbbl/d (us;liq.)

Imperial units

- Sgal/s (imp)
- Sgal/min (imp)
- Sgal/h (imp)
- Sgal/d (imp)

Custom-specific units

- UserCrVol./s
- UserCrVol./min
- UserCrVol./h
- UserCrVol./d

Factory setting

Country-specific:

- l/h
- Sft³/h

Additional information	<i>Result</i> The selected unit applies for: Corrected volume flow parameter (→ ⓘ 44) <i>Selection</i> ⓘ For an explanation of the abbreviated units: → ⓘ 203 <i>Customer-specific units</i> ⓘ The unit for the customer-specific corrected volume is defined in the User corrected volume text parameter (→ ⓘ 64).
------------------------	---

Corrected volume unit	ⓘ
-----------------------	---

Navigation	📖📖 Expert → Sensor → System units → Corr. vol. unit (0575)		
Description	Use this function to select the unit for the corrected volume.		
Selection	<i>SI units</i> ■ Nl ■ Nm³ ■ Sm³ <i>Custom-specific units</i> UserCrVol.	<i>US units</i> ■ Sft³ ■ Sgal (us) ■ Sbbl (us;liq.)	<i>Imperial units</i> Sgal (imp)
Factory setting	Country-specific: ■ Nm³ ■ Sft³		
Additional information	<i>Selection</i> ⓘ For an explanation of the abbreviated units: → ⓘ 203 <i>Customer-specific units</i> ⓘ The unit for the customer-specific corrected volume is defined in the User corrected volume text parameter (→ ⓘ 64).		

Date/time format	ⓘ
------------------	---

Navigation	📖📖 Expert → Sensor → System units → Date/time format (2812)
Description	Use this function to select the desired time format for calibration history.
Selection	■ dd.mm.yy hh:mm ■ dd.mm.yy hh:mm am/pm ■ mm/dd/yy hh:mm ■ mm/dd/yy hh:mm am/pm
Factory setting	dd.mm.yy hh:mm

Additional information

Selection

 For an explanation of the abbreviated units: →  203

"User-specific units" submenu

Navigation

  Expert → Sensor → System units → User-spec. units

► User-specific units		
User volume text (0567)	→	 61
User volume factor (0568)	→	 62
User volume offset (0569)	→	 62
User mass text (0560)	→	 62
User mass factor (0561)	→	 63
User mass offset (0562)	→	 63
User corrected volume text (0592)	→	 64
User corrected volume factor (0590)	→	 64
User corrected volume offset (0602)	→	 64

User volume text



Navigation

  Expert → Sensor → System units → User-spec. units → Volume text (0567)

Description

Use this function to enter a text for the user-specific unit of volume and volume flow. The corresponding time units (s, min, h, d) for volume flow are generated automatically.

User entry

Max. 10 characters such as letters, numbers or special characters (@, %, /)

Factory setting

User vol.

Additional information*Result*

The defined unit is shown as an option in the choose list of the following parameters:

- **Volume flow unit** parameter (→ 53)
- **Volume unit** parameter (→ 55)

Example

If the text GLAS is entered, the choose list of the **Volume flow unit** parameter (→ 53) shows the following options:

- GLAS/s
- GLAS/min
- GLAS/h
- GLAS/d

User volume factor**Navigation**

Expert → Sensor → System units → User-spec. units → Volume factor (0568)

Description

Use this function to enter a quantity factor (without time) for the user-specific volume and volume flow unit.

User entry

Signed floating-point number

Factory setting

1.0

User volume offset**Navigation**

Expert → Sensor → System units → User-spec. units → Volume offset (0569)

Description

Use this function to enter the offset for adapting the user-specific volume unit and volume flow unit (without time).

User entry

Signed floating-point number

Factory setting

0

Additional information*Description*

Value in user-specific unit = (factor × value in base unit) + offset

User mass text**Navigation**

Expert → Sensor → System units → User-spec. units → Mass text (0560)

Description

Use this function to enter a text for the user-specific unit of mass and mass flow. The corresponding time units (s, min, h, d) for mass flow are generated automatically.

User entry	Max. 10 characters such as letters, numbers or special characters (@, %, /)
Factory setting	User mass
Additional information	<i>Result</i>  The defined unit is shown as an option in the choose list of the following parameters: ■ Mass flow unit parameter (→  57) ■ Mass unit parameter (→  58) <i>Example</i> If the text GLAS is entered, the following options are displayed in the picklist for the Mass flow unit parameter (→  57): ■ GLAS/s ■ GLAS/min ■ GLAS/h ■ GLAS/d

User mass factor

Navigation	 Expert → Sensor → System units → User-spec. units → Mass factor (0561)
Description	Use this function to enter a quantity factor (without time) for the user-specific mass and mass flow unit.
User entry	Signed floating-point number
Factory setting	1.0

User mass offset

Navigation	 Expert → Sensor → System units → User-spec. units → Mass offset (0562)
Description	Use this function to enter the offset for adapting the user-specific mass unit and mass flow unit (without time).
User entry	Signed floating-point number
Factory setting	0
Additional information	<i>Description</i>  Value in user-specific unit = (factor × value in base unit) + offset

User corrected volume text		
Navigation	 Expert → Sensor → System units → User-spec. units → Corr. vol. text (0592)	
Description	Use this function to enter a text for the user-specific unit of the corrected volume and corrected volume flow. The corresponding time units (s, min, h, d) for mass flow are generated automatically.	
User entry	Max. 10 characters such as letters, numbers or special characters (@, %, /)	
Factory setting	UserCrVol.	
Additional information	<i>Result</i>  The defined unit is shown as an option in the choose list of the following parameters: ▪ Corrected volume flow unit parameter (→  59) ▪ Corrected volume unit parameter (→  60) <i>Example</i> If the text GLAS is entered, the choose list of the Corrected volume flow unit parameter (→  59) shows the following options: ▪ GLAS/s ▪ GLAS/min ▪ GLAS/h ▪ GLAS/d	
User corrected volume factor		
Navigation	 Expert → Sensor → System units → User-spec. units → Cor.vol. factor (0590)	
Description	Use this function to enter a quantity factor (without time) for the user-specific corrected volume unit and corrected volume flow unit.	
User entry	Signed floating-point number	
Factory setting	1.0	
User corrected volume offset		
Navigation	 Expert → Sensor → System units → User-spec. units → Corr vol. offset (0602)	
Description	Use this function to enter the offset for adapting the user-specific corrected volume unit and corrected volume flow unit (without time).  Value in user-specific unit = (factor × value in base unit) + offset	
User entry	Signed floating-point number	

Factory setting 0

3.2.3 "Process parameters" submenu

Navigation   Expert → Sensor → Process param.

► Process parameters		
Filter options (6710)	→ 	65
Flow damping (6661)	→ 	67
Flow override (1839)	→ 	67
Conductivity damping (1803)	→ 	68
Temperature damping (1886)	→ 	68
Conductivity measurement (6514)	→ 	68
Conductivity temperature coefficient (1891)	→ 	69
Reference density (1885)	→ 	69
► Low flow cut off	→ 	70
► Empty pipe detection	→ 	73
► Electrode cleaning circuit	→ 	75

Filter options

Navigation   Expert → Sensor → Process param. → Filter options (6710)

Description Use this function to select a filter option.

- Selection
- Standard
 - Standard CIP on
 - Dynamic
 - Dynamic CIP on
 - Binomial filter

Factory setting Binomial filter

Additional information*Description*

The user can choose from a range of filter combinations which can optimize the measurement result depending on the application. Each change in the filter setting affects the output signal of the measuring device. The response time of the output signal increases as the filter depth increases.

Selection■ **Standard**

- Strong flow damping with a short output signal response time.
- Some time is needed before a stable output signal can be generated.
- Not suitable for pulsating flow as the average flow can be different here.

■ **Dynamic**

- Average flow damping with a delayed output signal response time.
- The average flow is displayed correctly over a measuring interval determined over a long period.

■ **Binominal**

- Weak flow damping with a short output signal response time.
- The average flow is displayed correctly over a measuring interval determined over a long period.

■ **CIP**

- This filter is also available for the **Standard** and **Dynamic** filter options.
- If the CIP filter has detected a change in the medium (abrupt increase in the noise level, e.g. quickly changing medium conductivity values during CIP cleaning), flow damping is greatly increased and the raw value (before flow damping) is limited by the mean value (delimiter). This eliminates extremely high measured errors (up to several 100 m/s).
- If the CIP filter is enabled, the response time of the entire measuring system increases and the output signal is delayed accordingly.

*Examples**Possible applications for the filters*

Application	Standard	Standard CIP	Dynamic	Dynamic CIP	Binomial
Pulsating flow (flow is negative intermittently)	---	---	++	--	++
Flow changes frequently (flow is dynamic)	-	--	++	-	++
Clear signal, quick control loop (< 1 s)	--	--	+ ¹⁾		++
Poor signal, slow control loop (response time of a few seconds)	++	-	--	---	---
Permanently bad signal	++	--	-	---	-
Short and severe signal distortion after a while		++		++	
Promag 50/53 replacement: Promag 100 system damping = 0.5 * Promag 50/53					+++
Promag 10 replacement: Promag 100 system damping = Promag 10 + 2			+++		
For a stable flow signal (no other requirements)	+++				

1) Flow damping value < 6

Flow damping



Navigation	Expert → Sensor → Process param. → Flow damping (6661)
Description	Use this function to enter flow damping. Reduction of the variability of the flow measured value (in relation to interference). For this purpose, the depth of the flow filter is adjusted: when the filter setting increases, the reaction time of the device also increases.
User entry	0 to 15
Factory setting	4
Additional information	<i>User entry</i> - Value = 0: no damping - Value > 0: damping is increased 0 is a weak damping and 15 a strong one. - A damping of 0 is not recommended, as the measuring signal is then so noisy that it is almost impossible to carry out a measurement. - The damping depends on the measuring period and the filter type selected. - An increase or decrease in the damping depends on the application. <i>Effect</i> The damping affects the following variables of the device: - Outputs → 96 - Low flow cut off → 70 - Totalizers → 155

Flow override



Navigation	Expert → Sensor → Process param. → Flow override (1839)
Description	Use this function to select whether to interrupt the evaluation of measured values. This is useful for the cleaning processes of a pipeline, for example.
Selection	- Off - On
Factory setting	Off
Additional information	<i>Description</i> Flow override is active - The diagnostic message diagnostic message C453 Flow override is displayed. - Output values - Temperature: proceeding output - Totalizers 1-3: Stop being totalized Positive zero return can also be enabled via the Status input: Assign status input parameter (→ 95).

Conductivity damping


Navigation	Expert → Sensor → Process param. → Conduct. damping (1803)
Prerequisite	In the Conductivity measurement parameter (→ 68), the On option is selected.
Description	Use this function to enter a time constant for conductivity damping (PT1 element).
User entry	0 to 999.9 s
Factory setting	0 s
Additional information	<i>Description</i> The damping is performed by a PT1 element ²⁾. <i>User entry</i> ■ Value = 0: no damping ■ Value > 0: damping is increased Damping is switched off if 0 is entered (factory setting).

Temperature damping


Navigation	Expert → Sensor → Process param. → Temp. damping (1886)
Prerequisite	One of the following conditions is met: ■ Order code for "Sensor option", option CI "Medium temperature measurement" or ■ The temperature is read into the flowmeter from an external device.
Description	Use this function to enter the time constant for temperature damping.
User entry	0 to 999.9 s
Factory setting	0 s

Conductivity measurement


Navigation	Expert → Sensor → Process param. → Conduct. measur. (6514)
Prerequisite	The On option is selected in the Conductivity measurement parameter (→ 68) parameter.
Description	Use this function to enable and disable conductivity measurement.

2) Proportional behavior with first-order lag

Selection	■ Off ■ On
Factory setting	Off
Additional information	<i>Description</i>  For conductivity measurement to work, the medium must have a minimum conductivity of 5 µS/cm.

Conductivity temperature coefficient

Navigation	  Expert → Sensor → Process param. → Cond. temp.coeff (1891)
Prerequisite	One of the following conditions is met: ■ Order code for "Sensor option", option CI "Medium temperature measurement" or ■ The temperature is read into the flowmeter from an external device.
Description	Use this function to enter the temperature coefficient for the conductivity.
User entry	Signed floating-point number
Factory setting	2.1 %/K

Reference density

Navigation	  Expert → Sensor → Process param. → Ref.density (1885)
Description	Use this function to enter a fixed value for the reference density.
User entry	Positive floating-point number
Factory setting	Country-specific: ■ 1 kg/l ■ 1 lb/ft³
Additional information	<i>Dependency</i>  The unit is taken from the Density unit parameter (→  58)

"Low flow cut off" submenu

Navigation  Expert → Sensor → Process param. → Low flow cut off

► Low flow cut off

Assign process variable (1837)

→  70

On value low flow cutoff (1805)

→  70

Off value low flow cutoff (1804)

→  71

Pressure shock suppression (1806)

→  71

Assign process variable

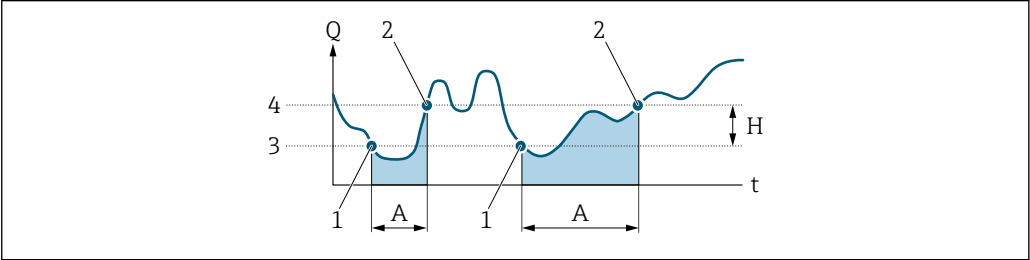
Navigation	 Expert → Sensor → Process param. → Low flow cut off → Assign variable (1837)
Description	Use this function to select the process variable for low flow cutoff detection.
Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow
Factory setting	Volume flow

On value low flow cutoff

Navigation	 Expert → Sensor → Process param. → Low flow cut off → On value (1805)
Prerequisite	One of the following options is selected in the Assign process variable parameter (→  70): ■ Volume flow ■ Mass flow
Description	Use this function to enter a switch-on value for low flow cut off. Low flow cut off is activated if the value entered is not equal to 0 →  71.
User entry	Positive floating-point number
Factory setting	Depends on country and nominal diameter →  199
Additional information	Dependency  The unit depends on the process variable selected in the Assign process variable parameter (→  70).

Off value low flow cutoff

Navigation	Expert → Sensor → Process param. → Low flow cut off → Off value (1804)
Prerequisite	One of the following options is selected in the Assign process variable parameter (→ 70): ■ Volume flow ■ Mass flow ■ Corrected volume flow
Description	Use this function to enter a switch-off value for low flow cut off. The off value is entered as a positive hysteresis from the on value→ 70.
User entry	0 to 100.0 %
Factory setting	50 %
Additional information	Example



A0012887

- Q Flow
- t Time
- H Hysteresis
- A Low flow cut off active
- 1 Low flow cut off is activated
- 2 Low flow cut off is deactivated
- 3 On value entered
- 4 Off value entered

Pressure shock suppression

Navigation	Expert → Sensor → Process param. → Low flow cut off → Pres. shock sup. (1806)
Prerequisite	One of the following options is selected in the Assign process variable parameter (→ 70): ■ Volume flow ■ Mass flow ■ Corrected volume flow
Description	Use this function to enter the time interval for signal suppression (= active pressure shock suppression).
User entry	0 to 100 s
Factory setting	0 s

Additional information*Description***Pressure shock suppression is enabled**

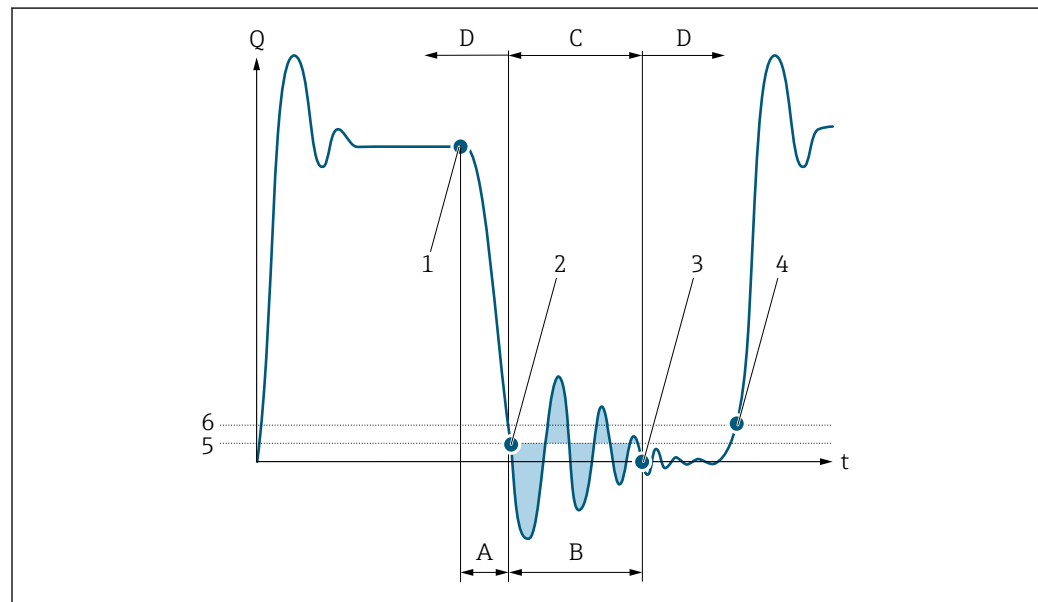
- Prerequisite:
Flow rate < on-value of low flow cut off
- Output values
 - Flow displayed: 0
 - Totalizer: the totalizers are pegged at the last correct value

Pressure shock suppression is disabled

- Prerequisite: the time interval set in this function has elapsed.
- If the flow also exceeds the switch-off value for low flow cut off, the device starts processing the current flow value again and displays it.

Example

When closing a valve, momentarily strong fluid movements may occur in the pipeline, which are registered by the measuring system. These totalized flow values lead to a false totalizer status, particularly during batching processes.



A0012888

- Q Flow
- t Time
- A Drip
- B Pressure shock
- C Pressure shock suppression active as specified by the time entered
- D Pressure shock suppression inactive
- 1 Valve closes
- 2 Flow falls below the on-value of the low flow cut off: pressure shock suppression is activated
- 3 The time entered has elapsed: pressure shock suppression is deactivated
- 4 The actual flow value is now displayed and output
- 5 On value for low flow cut off
- 6 Off value for low flow cut off

"Empty pipe detection" submenu

Navigation   Expert → Sensor → Process param. → Empty pipe det.

► Empty pipe detection		
Empty pipe detection (1860)	→ 	73
Switch point empty pipe detection (6562)	→ 	73
Response time empty pipe detection (1859)	→ 	74
New adjustment (6560)	→ 	74
Progress (6571)	→ 	74
Empty pipe adjust value (6527)	→ 	75
Full pipe adjust value (6548)	→ 	75
Measured value EPD (6559)	→ 	75

Empty pipe detection

Navigation	  Expert → Sensor → Process param. → Empty pipe det. → Empty pipe det. (1860)
Description	Use this function to switch empty pipe detection on and off.
Selection	■ Off ■ On
Factory setting	Off

Switch point empty pipe detection

Navigation	  Expert → Sensor → Process param. → Empty pipe det. → Switch point EPD (6562)
Prerequisite	The On option is selected in the Empty pipe detection parameter (→  73).
Description	Use this function to enter the percentage threshold value of the resistance in relation to the adjustment values.
User entry	0 to 100 %
Factory setting	10 %

Response time empty pipe detection


Navigation	Expert → Sensor → Process param. → Empty pipe det. → Response time (1859)
Prerequisite	In the Empty pipe detection parameter (→ 73), the On option is selected.
Description	Use this function to enter the minimum length of time (debouncing time) the signal must be present for the diagnostic message △S862 Empty pipe to be triggered if the measuring pipe is empty or partially full.
User entry	0 to 100 s
Factory setting	1 s

New adjustment


Navigation	Expert → Sensor → Process param. → Empty pipe det. → New adjustment (6560)
Prerequisite	The On option is selected in the Empty pipe detection parameter (→ 73).
Description	For selecting whether to perform an empty pipe or full pipe adjustment.
Selection	■ Cancel ■ Empty pipe adjust ■ Full pipe adjust
Factory setting	Cancel

Progress

Navigation	Expert → Sensor → Process param. → Empty pipe det. → Progress (6571)
Prerequisite	The On option is selected in the Empty pipe detection parameter (→ 73).
Description	Use this function to view the progress.
User interface	■ Ok ■ Busy ■ Not ok

Empty pipe adjust value

Navigation	  Expert → Sensor → Process param. → Empty pipe det. → Empty pipe value (6527)
Prerequisite	■ In the Empty pipe detection parameter (→  73), the On option is selected. ■ Adjustment value > full pipe value.
Description	Displays the adjustment value when the measuring pipe is empty.
User interface	Positive floating-point number

Full pipe adjust value

Navigation	  Expert → Sensor → Process param. → Empty pipe det. → Full pipe value (6548)
Prerequisite	■ In the Empty pipe detection parameter (→  73), the On option is selected. ■ Adjustment value < empty pipe value.
Description	Displays the adjustment value when the measuring pipe is full.
User interface	Positive floating-point number

Measured value EPD

Navigation	  Expert → Sensor → Process param. → Empty pipe det. → Meas. value EPD (6559)
Prerequisite	In the Empty pipe detection parameter (→  73), the On option is selected.
Description	Displays the current measured value.
User interface	Positive floating-point number

"Electrode cleaning circuit" submenu

Navigation   Expert → Sensor → Process param. → ECC

► Electrode cleaning circuit	
Electrode cleaning circuit (6528)	→  76
ECC duration (6555)	→  76
ECC recovery time (6556)	→  76

ECC cleaning cycle (6557)	→ 77
ECC Polarity (6631)	→ 77

Electrode cleaning circuit

Navigation	  Expert → Sensor → Process param. → ECC → ECC (6528)
Prerequisite	For the following order code: "Application package", option EC "ECC electrode cleaning"
Description	Use this function to enable and disable cyclic electrode cleaning.
Selection	■ Off ■ On
Factory setting	Off

ECC duration

Navigation	  Expert → Sensor → Process param. → ECC → ECC duration (6555)
Prerequisite	For the following order code: "Application package", option EC "ECC electrode cleaning"
Description	Use this function to enter the duration of electrode cleaning in seconds.
User entry	0.01 to 30 s
Factory setting	2 s

ECC recovery time

Navigation	  Expert → Sensor → Process param. → ECC → ECC recov. time (6556)
Prerequisite	For the following order code: "Application package", option EC "ECC electrode cleaning"
Description	Use this function to enter the recovery time after electrode cleaning to prevent signal output interference. The current output values are frozen in the meanwhile.
User entry	1 to 600 s
Factory setting	60 s

ECC cleaning cycle		
Navigation	  Expert → Sensor → Process param. → ECC → ECC clean. cycle (6557)	
Prerequisite	For the following order code: "Application package", option EC "ECC electrode cleaning"	
Description	Use this function to enter the pause duration until the next electrode cleaning.	
User entry	0.5 to 168 h	
Factory setting	0.5 h	

ECC Polarity	
Navigation	  Expert → Sensor → Process param. → ECC → ECC Polarity (6631)
Prerequisite	For the following order code: "Application package", option EC "ECC electrode cleaning"
Description	Displays the polarity of the electrode cleaning circuit.
User interface	■ Positive ■ Negative
Factory setting	Depends on the electrode material: ■ Platinum: Negative option ■ Tantalum, Alloy C22, stainless steel: Positive option

3.2.4 "External compensation" submenu

Navigation   Expert → Sensor → External comp.

► External compensation	
Temperature source (6712)	→  78
External temperature (6673)	→  78
Density source (6615)	→  79
External density (6630)	→  79
Fixed density (6623)	→  79

Reference temperature (1816)	→ 80
Reference density (expansion.) (1892)	→ 80
Linear expansion coefficient (1817)	→ 81
Square expansion coefficient (1818)	→ 82

Temperature source

Navigation   Expert → Sensor → External comp. → Temp. source (6712)

Description Use this function to select the temperature source.

- Selection**
- Internal temperature sensor
 - Current input 1 *
 - Current input 2 *
 - Current input 3 *
 - Off

Factory setting Off

External temperature

Navigation   Expert → Sensor → External comp. → External temp. (6673)

Prerequisite The **External value** option is selected in the **Temperature source** parameter (→  78).

Description Use this function to enter the temperature read in by the external device.

User interface Floating point number with sign

Factory setting -273.15 °C

Additional information *Dependency*
 The unit is taken from the **Temperature unit** parameter (→  56)

* Visibility depends on order options or device settings

Density source

Navigation	  Expert → Sensor → External comp. → Density source (6615)
Description	Use this function to select the density source.
Selection	■ Fixed density ■ Current input 1 * ■ Current input 2 * ■ Current input 3 * ■ Calculated value
Factory setting	Fixed density

External density

Navigation	  Expert → Sensor → External comp. → External density (6630)
Prerequisite	In the Density source parameter (→  79), the External density option is selected.
Description	Use this function to enter the density read in from the external device.
User interface	Positive floating-point number
Factory setting	0 kg/l
Additional information	<i>Dependency</i>  The unit is taken from the Density unit parameter (→  58)

Fixed density

Navigation	  Expert → Sensor → External comp. → Fixed density (6623)
Description	Use this function to enter a fixed value for the density.
User entry	Positive floating-point number
Factory setting	Country-specific: ■ 1 000 kg/l ■ 1 000 lb/ft³
Additional information	<i>Dependency</i>  The unit is taken from the Density unit parameter (→  58)

* Visibility depends on order options or device settings

Reference temperature



Navigation	Expert → Sensor → External comp. → Ref. temperature (1816)
Prerequisite	The Calculated value option is selected in the Density source parameter (→ 79)
Description	Use this function to enter a reference temperature for calculating the reference density.
User interface	-273.15 to 99 999 °C
Factory setting	Country-specific: ■ +20 °C ■ +68 °F
Additional information	<i>Dependency</i> The unit is taken from the Temperature unit parameter (→ 56)

Reference density calculation

$$\rho_n = \rho \cdot (1 + \alpha \cdot \Delta t + \beta \cdot \Delta t^2)$$

A0023403

- ρ_n : reference density
- ρ : fluid density currently measured
- t : fluid temperature currently measured
- t_n : reference temperature at which the reference density is calculated (e.g. 20 °C)
- Δt : $t - t_n$
- α : linear expansion coefficient of the fluid, unit = [1/K]; K = Kelvin
- β : square expansion coefficient of the fluid, unit = [1/K²]

Reference density (expansion.)



Navigation	Expert → Sensor → External comp. → Reference density (expansion.) (1892)
Prerequisite	The Calculated value option is selected in the Density source parameter (→ 79)
Description	Displays the reference density.
User interface	Positive floating-point number
Additional information	<i>Description</i> The reference density is required for density calculation. <i>Deviation of the process temperature from the reference temperature:</i> $\Delta T = \quad T - T_{ref}$ $\Delta T:$ <i>Deviation</i>

T : Process temperature
 T_{ref} : Reference temperature ($\rightarrow$  80)

Temperature-compensated density:

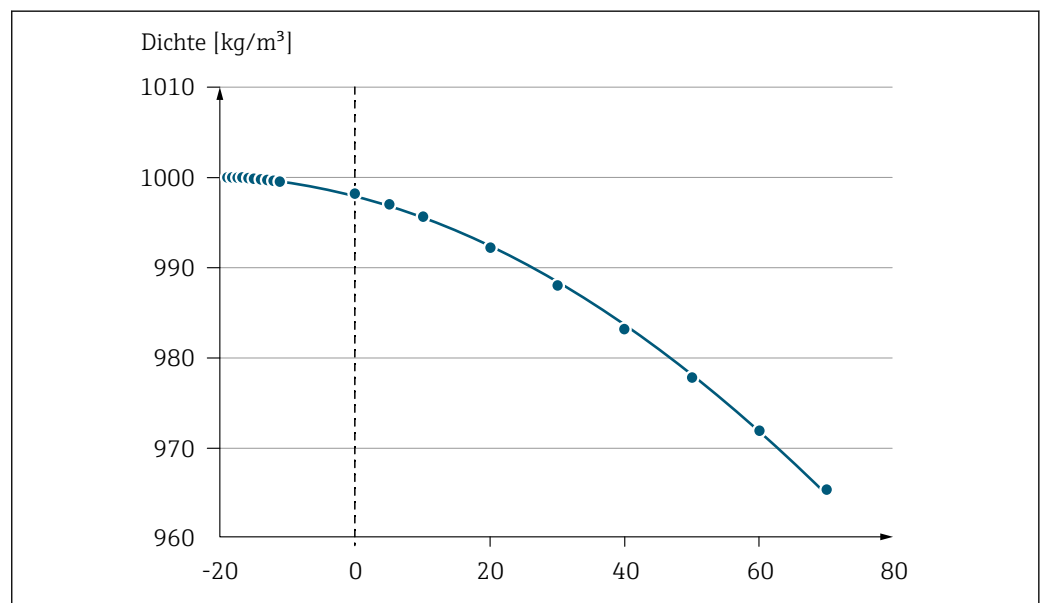
$\rho_{comp} = \rho_{ref}(1 + \alpha\Delta T + \beta\Delta T^2)$
 ρ_{comp} : Calculated density
 ρ_{ref} : Reference density
 ΔT : Deviation of the process temperature from the reference temperature
 α : Linear expansion coefficient ($\rightarrow$  81)
 β : Square expansion coefficient ($\rightarrow$  82)

Example for water (factory setting)

For a reference temperature of $T_{ref} = 20\text{ °C}$

A quadratic fit of a number of density values results in the following coefficients:

- $\alpha = -2.0295 \cdot 10^{-4} \text{ 1/K}$
- $\beta = -3.8436 \cdot 10^{-6} \text{ 1/K}^2$
- $\rho_{ref} = 997.82 \text{ kg/m}^3$



 2 Quadratic fit

Dependency

 The unit is taken from the **Density unit** parameter ($\rightarrow$  58)

Linear expansion coefficient



Navigation

  Expert $\rightarrow$ Sensor $\rightarrow$ External comp. $\rightarrow$ Linear exp coeff (1817)

Prerequisite

The **Calculated value** option is selected in the **Density source** parameter ($\rightarrow$  79) parameter.

Description	Use this function to enter a linear, fluid-specific expansion coefficient for calculating the reference density.
User interface	Signed floating-point number
Factory setting	$-2.0295 \cdot 10^{-04}$

Square expansion coefficient

Navigation	  Expert → Sensor → External comp. → Square exp coeff (1818)
Prerequisite	The Calculated value option is selected in the Density source parameter (→  79) parameter.
Description	For fluid with a non-linear expansion pattern: use this function to enter a quadratic, fluid-specific expansion coefficient for calculating the reference density.
User interface	Signed floating-point number
Factory setting	$-3.8436 \cdot 10^{-06}$

3.2.5 "Sensor adjustment" submenu

Navigation   Expert → Sensor → Sensor adjustm.

► Sensor adjustment

Installation direction (1809)

Integration time (6533)

Measuring period (6536)

► Process variable adjustment

→  82

→  83

→  83

→  83

Installation direction

Navigation	  Expert → Sensor → Sensor adjustm. → Install. direct. (1809)
Description	Use this function to change the sign of the medium flow direction.
Selection	■ Flow in arrow direction ■ Flow against arrow direction

Factory setting Flow in arrow direction

Additional information *Description*



Before changing the sign: ascertain the actual direction of fluid flow with reference to the direction indicated by the arrow on the sensor nameplate.

Integration time



Navigation Expert → Sensor → Sensor adjustm. → Integration time (6533)

Description Display the duration of an integration cycle.

User interface 1 to 65 ms

Measuring period



Navigation Expert → Sensor → Sensor adjustm. → Measuring period (6536)

Description Display the time of a full measuring period.

User interface 0 to 1 000 ms

"Process variable adjustment" submenu

Navigation Expert → Sensor → Sensor adjustm. → Variable adjust

► Process variable adjustment		
Volume flow offset (1831)	→	84
Volume flow factor (1832)	→	84
Mass flow offset (1841)	→	84
Mass flow factor (1846)	→	85
Conductivity offset (1848)	→	85
Conductivity factor (1849)	→	85
Corrected volume flow offset (1866)	→	86
Corrected volume flow factor (1867)	→	86

Temperature offset (1868)	→ 86
Temperature factor (1869)	→ 87

Volume flow offset

Navigation	  Expert → Sensor → Sensor adjustm. → Variable adjust → Vol. flow offset (1831)
Description	Use this function to enter the zero point shift for the volume flow trim. The volume flow unit on which the shift is based is m ³ /s.
User entry	Signed floating-point number
Factory setting	0 m ³ /s
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

Volume flow factor

Navigation	  Expert → Sensor → Sensor adjustm. → Variable adjust → Vol. flow factor (1832)
User entry	Positive floating-point number
Factory setting	1
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

Mass flow offset

Navigation	  Expert → Sensor → Sensor adjustm. → Variable adjust → Mass flow offset (1841)
Description	Use this function to enter the zero point shift for the mass flow trim. The mass flow unit on which the shift is based is kg/s.
User entry	Signed floating-point number
Factory setting	0 kg/s
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

Mass flow factor



Navigation	 Expert → Sensor → Sensor adjustm. → Variable adjust → Mass flow factor (1846)
Description	Use this function to enter a quantity factor (without time) for the mass flow. This multiplication factor is applied over the mass flow range.
User entry	Positive floating-point number
Factory setting	1
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

Conductivity offset



Navigation	 Expert → Sensor → Sensor adjustm. → Variable adjust → Conduct. offset (1848)
Prerequisite	The On option is selected in the Conductivity measurement parameter (→  68) parameter.
Description	Use this function to enter the zero point shift for the conductivity trim. The conductivity unit on which the shift is based is S/m.
User entry	Signed floating-point number
Factory setting	0 S/m
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

Conductivity factor



Navigation	 Expert → Sensor → Sensor adjustm. → Variable adjust → Conduct. factor (1849)
Prerequisite	The On option is selected in the Conductivity measurement parameter (→  68) parameter.
Description	Use this function to enter a quantity factor for the conductivity. This multiplication factor is applied over the conductivity range.
User entry	Positive floating-point number
Factory setting	1

Additional information *Description*

 Corrected value = (factor × value) + offset

Corrected volume flow offset

Navigation  Expert → Sensor → Sensor adjustm. → Variable adjust → Corr. vol offset (1866)

Description Use this function to enter the zero point shift for the corrected volume flow trim. The corrected volume flow unit on which the shift is based is 1 Nm³/s.

User entry Signed floating-point number

Factory setting 0 Nm³/s

Additional information *Description*

 Corrected value = (factor × value) + offset

Corrected volume flow factor

Navigation  Expert → Sensor → Sensor adjustm. → Variable adjust → Corr. vol factor (1867)

Description Use this function to enter a quantity factor (without time) for the corrected volume flow. This multiplication factor is applied over the corrected volume flow range.

User entry Positive floating-point number

Factory setting 1

Additional information *Description*

 Corrected value = (factor × value) + offset

Temperature offset

Navigation  Expert → Sensor → Sensor adjustm. → Variable adjust → Temp. offset (1868)

Prerequisite One of the following conditions is met:

- Order code for "Sensor option", option **CI** "Medium temperature measurement" or
- The temperature is read into the flowmeter from an external device.

Description Use this function to enter the zero point shift for the temperature trim. The temperature unit on which the shift is based is 1 K.

User entry	Signed floating-point number
Factory setting	0 K
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

Temperature factor

Navigation	 Expert → Sensor → Sensor adjustm. → Variable adjust → Temp. factor (1869)
Prerequisite	One of the following conditions is met: ■ Order code for "Sensor option", option CI "Medium temperature measurement" or ■ The temperature is read into the flowmeter from an external device.
Description	Use this function to enter a quantity factor (without time) for the temperature. This multiplication factor is applied over the temperature range.
User entry	Positive floating-point number
Factory setting	1
Additional information	<i>Description</i>  Corrected value = (factor × value) + offset

3.2.6 "Calibration" submenu

Navigation  Expert → Sensor → Calibration

► Calibration

Nominal diameter (2807)

Calibration factor (6522)

Zero point (6546)

Conductivity calibration factor (6718)

→  88

→  88

→  88

→  88

Nominal diameter

Navigation	 Expert → Sensor → Calibration → Nominal diameter (2807)
Description	Displays the nominal diameter of the sensor.
User interface	DNxx / x"
Factory setting	Depends on the size of the sensor
Additional information	<i>Description</i>  The value is also specified on the sensor nameplate.

Calibration factor

Navigation	 Expert → Sensor → Calibration → Cal. factor (6522)
Description	Displays the current calibration factor for the sensor.
User interface	Positive floating-point number
Factory setting	Depends on nominal diameter and calibration.

Zero point



Navigation	 Expert → Sensor → Calibration → Zero point (6546)
Description	This function shows the zero point correction value for the sensor.
User interface	Signed floating-point number
Factory setting	Depends on nominal diameter and calibration

Conductivity calibration factor



Navigation	 Expert → Sensor → Calibration → Cond. cal. fact. (6718)
Prerequisite	The On option is selected in the Conductivity measurement parameter (→  68) parameter.
Description	Displays the calibration factor for the conductivity measurement.

User interface 0 to 10000

3.3 "I/O configuration" submenu

Navigation   Expert → I/O config.

I/O configuration

I/O module 1 to n terminal numbers
(3902-1 to n)

I/O module 1 to n information
(3906-1 to n)

I/O module 1 to n type (3901-1 to n)

Apply I/O configuration (3907)

Conversion code (2762)

→  89

→  89

→  90

→  90

→  91

I/O module 1 to n terminal numbers

Navigation   Expert → I/O config. → I/O 1 to n terminals (3902-1 to n)

Description Displays the terminal numbers used by the I/O module.

User interface

- Not used
- 26-27 (I/O 1)
- 24-25 (I/O 2)
- 22-23 (I/O 3)

I/O module 1 to n information

Navigation   Expert → I/O config. → I/O 1 to n info (3906-1 to n)

Description Displays information about the plugged in I/O module.

User interface

- Not plugged
- Invalid
- Not configurable
- Configurable
- Fieldbus

Additional information	<i>"Not plugged" option</i>
	The I/O module is not plugged in.
	<i>"Invalid" option</i>
	The I/O module is not plugged correctly.
	<i>"Not configurable" option</i>
	The I/O module is not configurable.
	<i>"Configurable" option</i>
	The I/O module is configurable.
	<i>"Fieldbus" option</i>
	The I/O module is configured for the fieldbus.

I/O module 1 to n type


Navigation	Expert → I/O config. → I/O 1 to n type (3901–1 to n)
Prerequisite	For the following order code: ■ "Output; input 2", option D "Configurable I/O initial setting off" ■ "Output; input 3", option D "Configurable I/O initial setting off"
Description	Use this function to select the I/O module type for the configuration of the I/O module.
Selection	■ Off ■ Current output * ■ Current input * ■ Status input * ■ Pulse/frequency/switch output *
Factory setting	Off

Apply I/O configuration


Navigation	Expert → I/O config. → Apply I/O config (3907)
Description	Use this function to activate the newly configured I/O module type.
Selection	■ No ■ Yes
Factory setting	No

* Visibility depends on order options or device settings

Conversion code		
Navigation	Expert → I/O config. → Conversion code (2762)	
Description	Use this function to enter the ordered activation code to activate the I/O configuration change.	
User entry	Positive integer	
Factory setting	0	
Additional information	<i>Description</i> The I/O configuration is changed in the I/O module type parameter (→ 90).	

3.4 "Input" submenu

Navigation Expert → Input

► Input

► Current input 1 to n

→ 91

► Status input 1 to n

→ 94

3.4.1 "Current input 1 to n" submenu

Navigation Expert → Input → Current input 1 to n

► Current input 1 to n

Terminal number (1611-1 to n)

→ 92

Signal mode (1610-1 to n)

→ 92

Current span (1605-1 to n)

→ 92

0/4 mA value (1606-1 to n)

→ 93

20 mA value (1607-1 to n)

→ 93

Failure mode (1601-1 to n)

→ 93

Failure value (1602-1 to n)

→ 94

Terminal number	
Navigation	 Expert → Input → Current input 1 to n → Terminal no. (1611–1 to n)
Description	Displays the terminal numbers used by the current input module.
User interface	■ Not used ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)
Additional information	<i>"Not used" option</i> The current input module does not use any terminal numbers.
Signal mode 	
Navigation	 Expert → Input → Current input 1 to n → Signal mode (1610–1 to n)
Prerequisite	The measuring device is not approved for use in the hazardous area with type of protection Ex-i.
Description	Use this function to select the signal mode for the current input.
Selection	■ Passive ■ Active
Factory setting	Passive
Current span 	
Navigation	 Expert → Input → Current input 1 to n → Current span (1605–1 to n)
Description	Use this function to select the current range for the process value output and the upper and lower level for signal on alarm.
Selection	■ 4...20 mA ■ 4...20 mA NAMUR ■ 4...20 mA US ■ 0...20 mA
Factory setting	Country-specific: ■ 4...20 mA NAMUR ■ 4...20 mA US
Additional information	<i>Examples</i>  Sample values for the current range: Current span parameter (→  98)

0/4 mA value 	
Navigation	  Expert → Input → Current input 1 to n → 0/4 mA value (1606-1 to n)
Description	Use this function to enter a value for the 4 mA current.
User entry	Signed floating-point number
Factory setting	0
Additional information	<i>Current input behavior</i> The current input behaves differently depending on the settings configured in the following parameters: ■ Current span (→  92) ■ Failure mode (→  93) <i>Configuration examples</i>  Pay attention to the configuration examples for 4 mA value parameter (→  100).
20 mA value 	
Navigation	  Expert → Input → Current input 1 to n → 20 mA value (1607-1 to n)
Description	Use this function to enter a value for the 20 mA current.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter
Additional information	<i>Configuration examples</i>  Pay attention to the configuration examples for 4 mA value parameter (→  100).
Failure mode 	
Navigation	  Expert → Input → Current input 1 to n → Failure mode (1601-1 to n)
Description	Use this function to select the input behavior when measuring a current outside the configured Current span parameter (→  92).
Selection	■ Alarm ■ Last valid value ■ Defined value
Factory setting	Alarm

Additional information	Options
	■ Alarm An error message is set. ■ Last valid value The last valid measured value is used. ■ Defined value A user-defined measured value is used (Failure value parameter (→  94)).
Failure value 	
Navigation	  Expert → Input → Current input 1 to n → Failure value (1602–1 to n)
Prerequisite	In the Failure mode parameter (→  93), the Defined value option is selected.
Description	Use this function to enter the value that the device uses if it does not receive an input signal from the external device, or if the input signal is invalid.
User entry	Signed floating-point number
Factory setting	0

3.4.2 "Status input 1 to n" submenu

Navigation   Expert → Input → Status input 1 to n

► Status input 1 to n

Terminal number (1358–1 to n)

→  94

Assign status input (1352–1 to n)

→  95

Value status input (1353–1 to n)

→  95

Active level (1351–1 to n)

→  96

Response time status input (1354–1 to n)

→  96

Terminal number	
Navigation	  Expert → Input → Status input 1 to n → Terminal no. (1358–1 to n)
Description	Displays the terminal numbers used by the status input module.

User interface	■ Not used ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)
Additional information	<i>"Not used" option</i> The status input module does not use any terminal numbers.

Assign status input



Navigation	Expert → Input → Status input 1 to n → Assign stat.inp. (1352-1 to n)
Description	Use this function to select the function for the status input.
Selection	■ Off ■ Reset totalizer 1 ■ Reset totalizer 2 ■ Reset totalizer 3 ■ Reset all totalizers ■ Flow override
Factory setting	Off
Additional information	<i>Selection</i> ■ Off The status input is switched off. ■ Reset totalizer 1...3 The individual totalizers are reset. ■ Reset all totalizers All totalizers are reset. ■ Flow override The Flow override (→ 67) is activated. Note on the Flow override (→ 67): ■ The Flow override (→ 67) is enabled as long as the level is at the status input (continuous signal). ■ All other assignments react to a change in level (pulse) at the status input.

Value status input

Navigation	Expert → Input → Status input 1 to n → Val.stat.inp. (1353-1 to n)
Description	Displays the current input signal level.
User interface	■ High ■ Low

Active level

Navigation	 Expert → Input → Status input 1 to n → Active level (1351–1 to n)
Description	Use this function to determine the input signal level at which the assigned function is activated.
Selection	■ High ■ Low
Factory setting	High

Response time status input

Navigation	 Expert → Input → Status input 1 to n → Response time (1354–1 to n)
Description	Use this function to enter the minimum time period for which the input signal level must be present before the selected function is activated.
User entry	5 to 200 ms
Factory setting	50 ms

3.5 "Output" submenu

Navigation  Expert → Output

► Output

► Current output 1 to n

→  97

► Pulse/frequency/switch output 1 to n

→  110

► Relay output 1 to n

→  130

► Double pulse output

→  137

3.5.1 "Current output 1 to n" submenu

Navigation  Expert → Output → Curr.output 1 to n

► Current output 1 to n		
Terminal number (0379-1 to n)	→ 	97
Signal mode (0377-1 to n)	→ 	98
Assign current output 1 to n (0359-1 to n)	→ 	98
Current span (0353-1 to n)	→ 	98
Fixed current (0365-1 to n)	→ 	99
0/4 mA value (0367-1 to n)	→ 	100
20 mA value (0372-1 to n)	→ 	101
Measuring mode (0351-1 to n)	→ 	102
Damping output 1 to n (0363-1 to n)	→ 	106
Response time (0378-1 to n)	→ 	107
Failure mode (0364-1 to n)	→ 	108
Failure current (0352-1 to n)	→ 	109
Output current 1 to n (0361-1 to n)	→ 	109
Measured current 1 to n (0366-1 to n)	→ 	110

Terminal number

Navigation	 Expert → Output → Curr.output 1 to n → Terminal no. (0379-1 to n)
Description	Displays the terminal numbers used by the current output module.
User interface	■ Not used ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)
Additional information	"Not used" option The current output module does not use any terminal numbers.

Signal mode

Navigation  Expert → Output → Curr.output 1 to n → Signal mode (0377–1 to n)

Description Use this function to select the signal mode for the current output.

Selection

- Passive
- Active

Factory setting Passive

Assign current output 1 to n

Navigation  Expert → Output → Curr.output 1 to n → Assign curr. 1 to n (0359–1 to n)

Description Use this function to select a process variable for the current output.

Selection

- Off
- Volume flow
- Mass flow
- Corrected volume flow
- Flow velocity
- Conductivity^{*}
- Corrected conductivity^{*}
- Temperature^{*}
- Electronic temperature

Factory setting Volume flow

Current span

Navigation  Expert → Output → Curr.output 1 to n → Current span (0353–1 to n)

Description Use this function to select the current range for the process value output and the upper and lower level for signal on alarm.

Selection

- 4...20 mA NAMUR
- 4...20 mA US
- 4...20 mA
- 0...20 mA
- Fixed current

Factory setting Country-specific:

- 4...20 mA NAMUR
- 4...20 mA US

* Visibility depends on order options or device settings

Additional information

Description

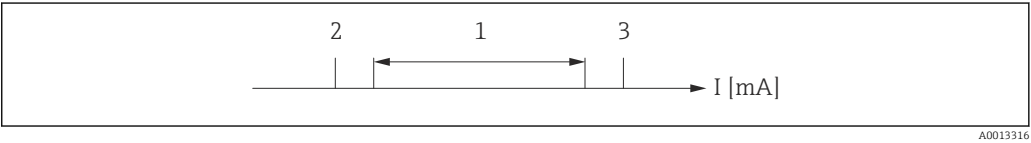
- In the event of a device alarm, the current output adopts the value specified in the **Failure mode** parameter (→ ⓘ 108).
- If the measured value is outside the measuring range, the diagnostic message **△S441 Current output 1 to n** is displayed.
- The measuring range is specified via the **0/4 mA value** parameter (→ ⓘ 100) and **20 mA value** parameter (→ ⓘ 101).

"Fixed current" option

The current value is set via the **Fixed current** parameter (→ ⓘ 99).

Example

Shows the relationship between the current span for the output of the process variable and the lower and upper alarm levels:



- I Current
- 1 Current span for process value
- 2 Lower level for signal on alarm
- 3 Upper level for signal on alarm

Selection

Selection	1	2	3
4...20 mA NAMUR	3.8 to 20.5 mA	< 3.6 mA	> 21.95 mA
4...20 mA US	3.9 to 20.8 mA US	< 3.6 mA	> 21.95 mA
4...20 mA	4 to 20.5 mA	< 3.6 mA	> 21.95 mA
0...20 mA	0 to 20.5 mA	< 0 mA	> 21.95 mA

- If the flow exceeds or falls below the upper or lower signal on alarm level, the diagnostic message **△S441 Current output 1 to n** is displayed.

Fixed current



Navigation ⓘ ⓘ Expert → Output → Curr.output 1 to n → Fixed current (0365-1 to n)

Prerequisite In the **Current span** parameter (→ ⓘ 98), the **Fixed current** option is selected.

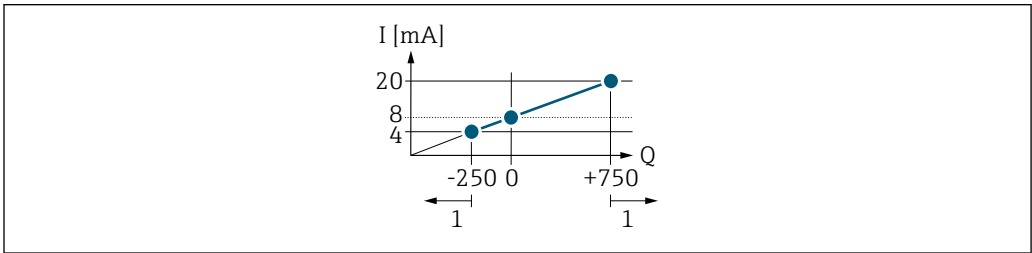
Description Use this function to enter a constant current value for the current output.

User entry 0 to 22.5 mA

Factory setting 22.5 mA

0/4 mA value

Navigation	Expert → Output → Curr.output 1 to n → 0/4 mA value (0367-1 to n)
Prerequisite	One of the following options is selected in the Current span parameter (→ 98): ■ 4...20 mA NAMUR ■ 4...20 mA US ■ 4...20 mA ■ 0...20 mA
Description	Use this function to enter a value for the 0/4 mA current.
User entry	Signed floating-point number
Factory setting	Country-specific: ■ 0 l/h ■ 0 gal/min (us)
Additional information	<i>Description</i> Positive and negative values are permitted depending on the process variable assigned in the Assign current output parameter (→ 98). In addition, the value can be greater than or smaller than the value assigned for the 20 mA current in the 20 mA value parameter (→ 101). <i>Dependency</i> The unit depends on the process variable selected in the Assign current output parameter (→ 98). <i>Current output behavior</i> The current output behaves differently depending on the settings configured in the following parameters: ■ Current span (→ 98) ■ Failure mode (→ 108) <i>Configuration examples</i> Some examples of parameter settings and their effect on the current output are given in the following section. Configuration example A Measuring mode with Forward flow option ■ 0/4 mA value parameter (→ 100) = not equal to zero flow (e.g. -250 m³/h) ■ 20 mA value parameter (→ 101) = not equal to zero flow (e.g. +750 m³/h) ■ Calculated current value = 8 mA at zero flow

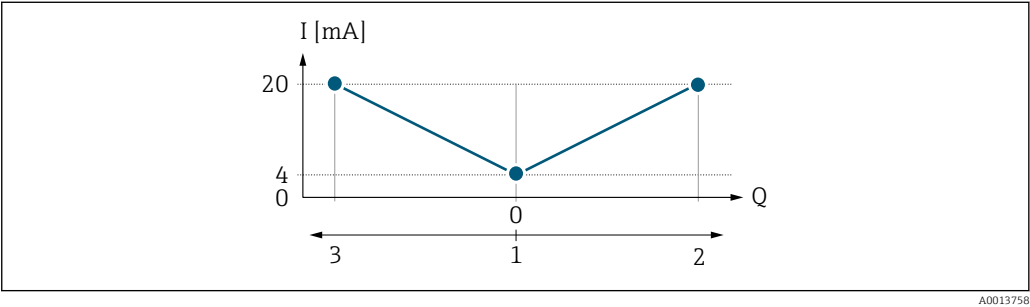


A0013757

Q Flow
I Current
1 Measuring range is exceeded or undershot

The operational range of the measuring device is defined by the values entered for the **0/4 mA value** parameter (→  100) and **20 mA value** parameter (→  101). If the effective flow exceeds or falls below this operational range, the diagnostic message **△S441 Current output 1 to n** is displayed.

Configuration example B
Measuring mode with **Forward/Reverse flow** option



- I* Current
- Q* Flow
- 1* Value assigned to the 0/4 mA current
- 2* Forward flow
- 3* Reverse flow

The current output signal is independent of the direction of flow (absolute amount of the measured variable). The values for the **0/4 mA value** parameter (→  100) and **20 mA value** parameter (→  101) must have the same sign. The value for the **20 mA value** parameter (→  101) (e.g. reverse flow) corresponds to the mirrored value for the **20 mA value** parameter (→  101) (e.g. forward flow).

Configuration example C
Measuring mode with **Reverse flow compensation** option
If flow is characterized by severe fluctuations (e.g. when using reciprocating pumps), flow components outside the measuring range are buffered, balanced and output after a maximum delay of 60 s → 102.

20 mA value 	
Navigation	 Expert → Output → Curr.output 1 to n → 20 mA value (0372-1 to n)
Prerequisite	One of the following options is selected in the Current span parameter (→  98): ■ 4...20 mA NAMUR ■ 4...20 mA US ■ 4...20 mA ■ 0...20 mA
Description	Use this function to enter a value for the 20 mA current.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter →  198
Additional information	<i>Description</i> Positive and negative values are permitted depending on the process variable assigned in the Assign current output parameter (→  98). In addition, the value can be greater

than or smaller than the value assigned for the 0/4 mA current in the **0/4 mA value** parameter (→  100).

Dependency

 The unit depends on the process variable selected in the **Assign current output** parameter (→  98).

Example

- Value assigned to 0/4 mA = -250 m³/h
- Value assigned to 20 mA = +750 m³/h
- Calculated current value = 8 mA (at zero flow)

If the **Forward/Reverse flow** option is selected in the **Measuring mode** parameter (→  102), different signs cannot be entered for the values of the **0/4 mA value** parameter (→  100) and **20 mA value** parameter (→  101). The diagnostic message **△S441 Current output 1 to n** is displayed.

Configuration examples

 Observe the configuration examples for the **0/4 mA value** parameter (→  100).

Measuring mode

Navigation

  Expert → Output → Curr.output 1 to n → Measuring mode (0351-1 to n)

Prerequisite

One of the following options is selected in the **Assign current output** parameter (→  98):

- Volume flow
- Mass flow
- Corrected volume flow
- Flow velocity
- Conductivity^{*}
- Corrected conductivity^{*}
- Temperature^{*}
- Electronic temperature

One of the following options is selected in the **Current span** parameter (→  98):

- 4...20 mA NAMUR
- 4...20 mA US
- 4...20 mA
- 0...20 mA

Description

Use this function to select the measuring mode for the current output.

Selection

- Forward flow
- Forward/Reverse flow
- Reverse flow compensation

Factory setting

Forward flow

* Visibility depends on order options or device settings

Additional information*Description*

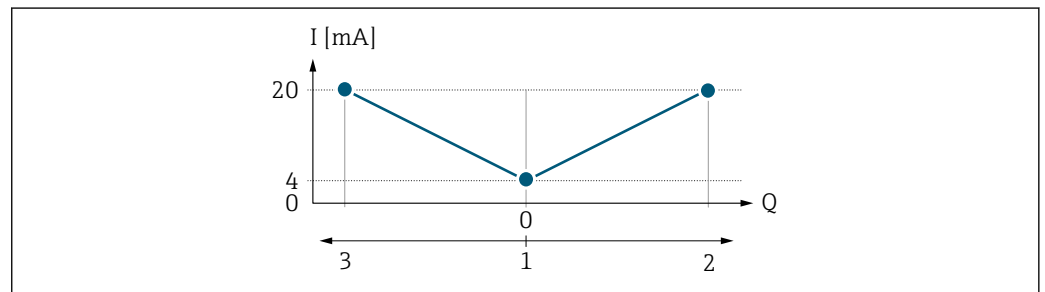
The process variable that is assigned to the current output via the **Assign current output** parameter (→ 98) is displayed below the parameter.

"Forward flow" option

The current output signal is proportional to the process variable assigned. The measuring range is defined by the values that are assigned to the 0/4 mA and 20 mA current value.

The flow components outside the scaled measuring range are taken into account for signal output as follows:

- Both values are defined such that they are not equal to zero flow e.g.:
 - 0/4 mA current value = $-5 \text{ m}^3/\text{h}$
 - 20 mA current value = $10 \text{ m}^3/\text{h}$
- If the effective flow exceeds or falls below this measuring range, the diagnostic message **△S441 Current output 1 to n** is displayed.

"Forward/Reverse flow" option

A0013758

<i>I</i>	Current
<i>Q</i>	Flow
<i>1</i>	Value assigned to the 0/4 mA current
<i>2</i>	Forward flow
<i>3</i>	Reverse flow

- The current output signal is independent of the direction of flow (absolute amount of the measured variable). The values for the **0/4 mA value** parameter (→ 100) and **20 mA value** parameter (→ 101) must have the same sign.
- The value for the **20 mA value** parameter (→ 101) (e.g. reverse flow) corresponds to the mirrored value for the **20 mA value** parameter (→ 101) (e.g. forward flow).

"Reverse flow compensation" option

The **Reverse flow compensation** option is primarily used to compensate for abrupt reverse flow which can occur in connection with positive displacement pumps as a result of wear or high viscosity. The reverse flows are recorded in a buffer and balanced against forward flow the next time flow is in the forward direction.

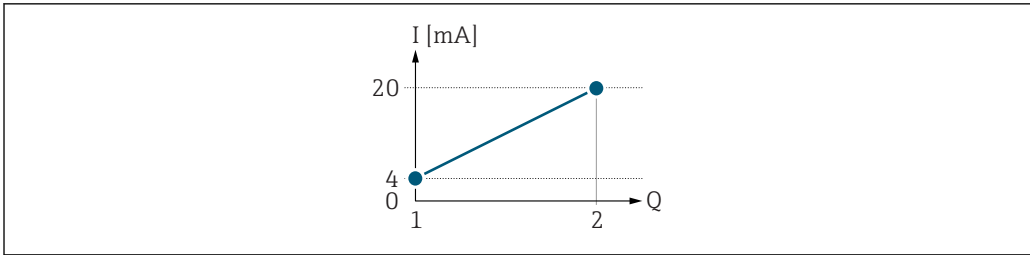
If buffering cannot be processed within approx. 60 s, the diagnostic message **△S441 Current output 1 to n** is displayed.

Flow values can aggregate in the buffer in the event of prolonged and unwanted fluid reverse flow. However, these flows are not taken into consideration by the current output configuration, i.e. the reverse flow is not compensated.

If this option is set, the measuring device does not attenuate the flow signal. The flow signal is not attenuated.

*Examples of how the current output behaves***Example 1**

Defined measuring range: lower range value and upper range value with the **same** sign

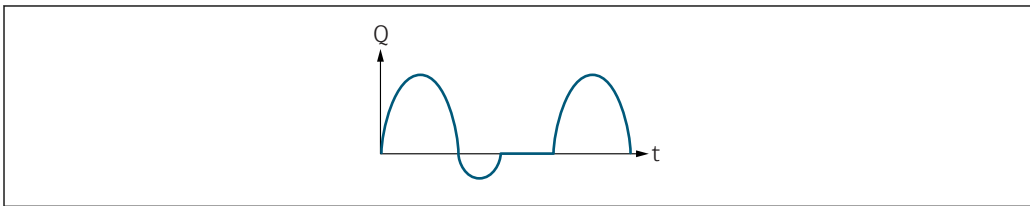


A0028084

3 Measuring range

- I Current
- Q Flow
- 1 Lower range value (value assigned to 0/4 mA current)
- 2 Upper range value (value assigned to 20 mA current)

With the following flow response:



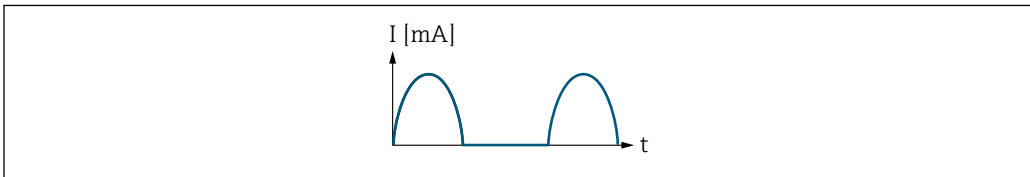
A0028091

4 Flow response

- Q Flow
- t Time

With the **Forward flow** option

The current output signal is proportional to the process variable assigned. The flow components outside the scaled measuring range are not taken into account for signal output:.

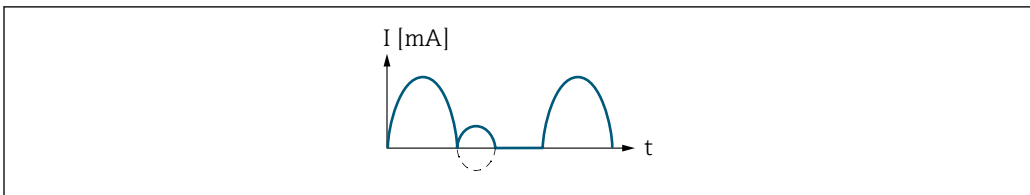


A0028092

- I Current
- t Time

With the **Forward/Reverse flow** option

The current output signal is independent of the direction of flow.

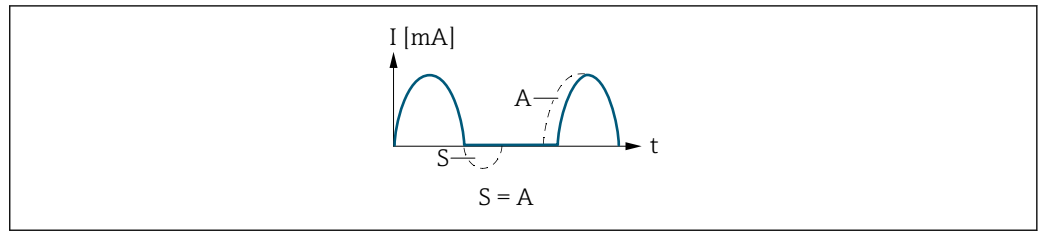


A0028093

- I Current
- t Time

With the **Reverse flow compensation** option

Flow components outside the span are buffered, balanced and output after a maximum delay of 60 s.

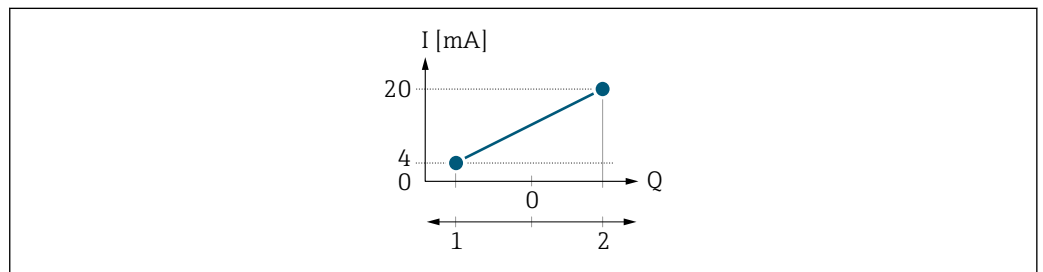


A0028094

I Current
 t Time
 S Flow components saved
 A Balancing of saved flow components

Example 2

Defined measuring range: lower range value and upper range value with **different** signs

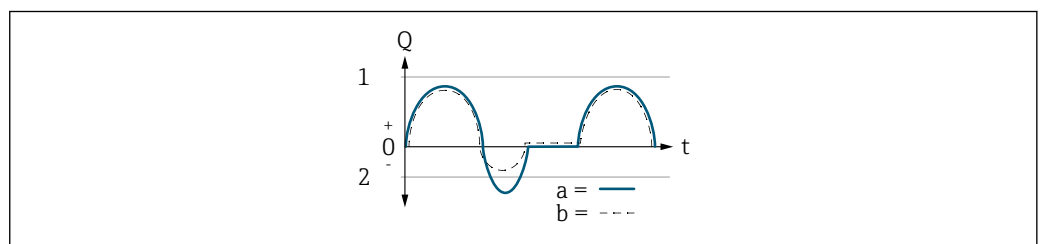


A0028095

5 Measuring range

I Current
 Q Flow
 1 Lower range value (value assigned to 0/4 mA current)
 2 Upper range value (value assigned to 20 mA current)

With flow a (–) outside, b (– –) inside the measuring range

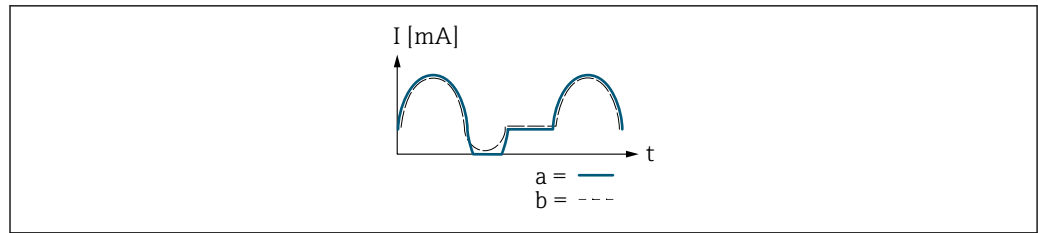


A0028098

Q Flow
 t Time
 1 Lower range value (value assigned to 0/4 mA current)
 2 Upper range value (value assigned to 20 mA current)

With the **Forward flow** option

- a (–): The flow components outside the scaled measuring range cannot be taken into account for signal output.
 The diagnostic message $\triangle S441$ Current output 1 to n is displayed.
- b (– –): The current output signal is proportional to the process variable assigned.



A0028100

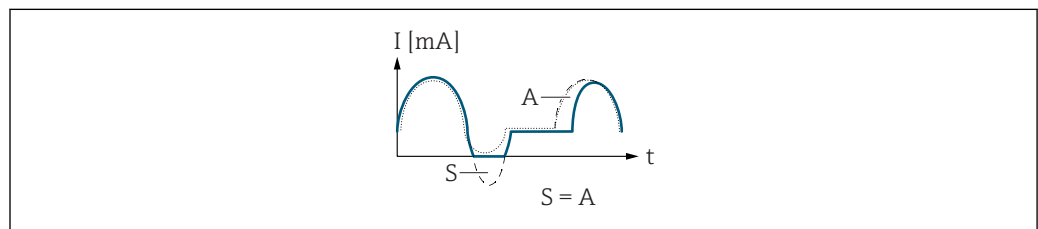
I Current
 t Time

With the **Forward/Reverse flow** option

This option is not possible in this case as the values for the **0/4 mA value** parameter (→ 100) and **20 mA value** parameter (→ 101) have different signs.

With the **Reverse flow compensation** option

Flow components outside the span are buffered, balanced and output after a maximum delay of 60 s.



A0028101

I Current
 t Time
 S Flow components saved
 A Balancing of saved flow components

Damping output 1 to n



Navigation

Expert → Output → Curr.output 1 to n → Damping out. 1 to n (0363-1 to n)

Prerequisite

One of the following options is selected in the **Assign current output** parameter (→ 98):

- Volume flow
- Mass flow
- Corrected volume flow
- Flow velocity
- Conductivity*
- Corrected conductivity*
- Temperature*
- Electronic temperature

One of the following options is selected in the **Current span** parameter (→ 98):

- 4...20 mA NAMUR
- 4...20 mA US
- 4...20 mA
- 0...20 mA

* Visibility depends on order options or device settings

Description	Use this function to enter a time constant for the reaction time of the current output signal to fluctuations in the measured value caused by process conditions.
User entry	0.0 to 999.9 s
Factory setting	1.0 s
Additional information	<i>User entry</i> Use this function to enter a time constant (PT1 element ³⁾) for current output damping: ■ If a low time constant is entered, the current output reacts particularly quickly to fluctuating measured variables. ■ On the other hand, the current output reacts more slowly if a high time constant is entered.  Damping is switched off if 0 is entered (factory setting).

Response time

Navigation	  Expert → Output → Curr.output 1 to n → Response time (0378-1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity * ■ Corrected conductivity * ■ Temperature * ■ Electronic temperature One of the following options is selected in the Current span parameter (→  98): ■ 4...20 mA NAMUR ■ 4...20 mA US ■ 4...20 mA ■ 0...20 mA
Description	Displays the response time. This specifies how quickly the current output reaches the measured value change of 63 % of 100 % of the measured value change.
User interface	Positive floating-point number
Additional information	<i>Description</i>  The response time is made up of the time specified for the following dampings: ■ Current output damping →  106 and ■ Depending on the measured variable assigned to the output. Flow damping

³⁾ proportional transmission behavior with first order delay

* Visibility depends on order options or device settings

Failure mode



Navigation

  Expert → Output → Curr.output 1 to n → Failure mode (0364-1 to n)

Prerequisite

One of the following options is selected in the **Assign current output** parameter (→  98):

- Volume flow
- Mass flow
- Corrected volume flow
- Flow velocity
- Conductivity *
- Corrected conductivity *
- Temperature *
- Electronic temperature

One of the following options is selected in the **Current span** parameter (→  98):

- 4...20 mA NAMUR
- 4...20 mA US
- 4...20 mA
- 0...20 mA

Description

Use this function to select the value of the current output in the event of a device alarm.

Selection

- Min.
- Max.
- Last valid value
- Actual value
- Defined value

Factory setting

Max.

* Visibility depends on order options or device settings

Additional information*Description*

This setting does not affect the failsafe mode of other outputs and totalizers. This is specified in separate parameters.

"Min." option

The current output adopts the value of the lower level for signal on alarm.



The signal on alarm level is defined via the **Current span** parameter (→ 98).

"Max." option

The current output adopts the value of the upper level for signal on alarm.



The signal on alarm level is defined via the **Current span** parameter (→ 98).

"Last valid value" option

The current output adopts the last measured value that was valid before the device alarm occurred.

"Actual value" option

The current output adopts the measured value on the basis of the current flow measurement; the device alarm is ignored.

"Defined value" option

The current output adopts a defined measured value.



The measured value is defined via the **Failure current** parameter (→ 109).

Failure current**Navigation**

Expert → Output → Curr.output 1 to n → Failure current (0352–1 to n)

Prerequisite

In the **Failure mode** parameter (→ 108), the **Defined value** option is selected.

Description

Use this function to enter a fixed value that the current output adopts in the event of a device alarm.

User entry

0 to 22.5 mA

Factory setting

22.5 mA

Output current 1 to n**Navigation**

Expert → Output → Curr.output 1 to n → Output curr. 1 to n (0361–1 to n)

Description

Displays the current value currently calculated for the current output.

User interface

3.59 to 22.5 mA

Measured current 1 to n

Navigation	Expert → Output → Curr.output 1 to n → Measur. curr. 1 to n (0366-1 to n)
Description	Use this function to display the actual measured value of the output current.
User interface	0 to 30 mA

3.5.2 "Pulse/frequency/switch output 1 to n" submenu

Navigation Expert → Output → PFS output 1 to n

► Pulse/frequency/switch output 1 to n		
Terminal number (0492-1 to n)	→	111
Signal mode (0490-1 to n)	→	112
Operating mode (0469-1 to n)	→	112
Assign pulse output 1 to n (0460-1 to n)	→	114
Value per pulse (0455-1 to n)	→	114
Pulse width (0452-1 to n)	→	115
Measuring mode (0457-1 to n)	→	115
Failure mode (0480-1 to n)	→	116
Pulse output 1 to n (0456-1 to n)	→	117
Assign frequency output (0478-1 to n)	→	118
Minimum frequency value (0453-1 to n)	→	118
Maximum frequency value (0454-1 to n)	→	119
Measuring value at minimum frequency (0476-1 to n)	→	119
Measuring value at maximum frequency (0475-1 to n)	→	120

Measuring mode (0479-1 to n)	→  120
Damping output 1 to n (0477-1 to n)	→  121
Response time (0491-1 to n)	→  122
Failure mode (0451-1 to n)	→  122
Failure frequency (0474-1 to n)	→  123
Output frequency 1 to n (0471-1 to n)	→  123
Switch output function (0481-1 to n)	→  124
Assign diagnostic behavior (0482-1 to n)	→  124
Assign limit (0483-1 to n)	→  125
Switch-on value (0466-1 to n)	→  127
Switch-off value (0464-1 to n)	→  127
Assign flow direction check (0484-1 to n)	→  128
Assign status (0485-1 to n)	→  128
Switch-on delay (0467-1 to n)	→  128
Switch-off delay (0465-1 to n)	→  129
Failure mode (0486-1 to n)	→  129
Switch status 1 to n (0461-1 to n)	→  129
Invert output signal (0470-1 to n)	→  130

Terminal number

Navigation

  Expert → Output → PFS output 1 to n → Terminal no. (0492-1 to n)

Description

Displays the terminal numbers used by the pulse/frequency/switch output module.

User interface

- Not used
- 24-25 (I/O 2)
- 22-23 (I/O 3)

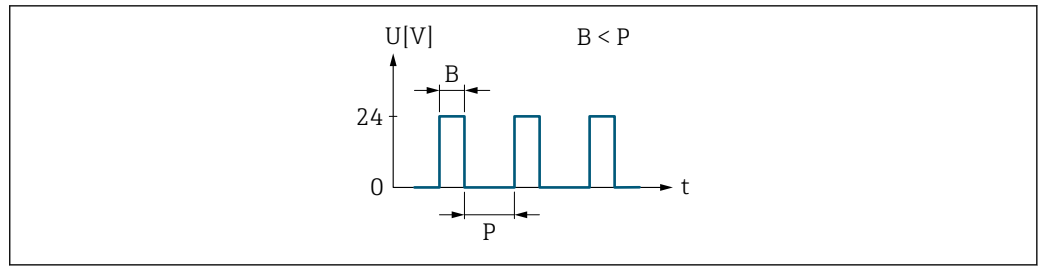
Additional information	<i>"Not used" option</i> The pulse/frequency/switch output module does not use any terminal numbers.
-------------------------------	--

Signal mode	
--------------------	---

Navigation	 Expert → Output → PFS output 1 to n → Signal mode (0490–1 to n)
Description	Use this function to select the signal mode for the pulse/frequency/switch output.
Selection	■ Passive ■ Active
Factory setting	Passive

Operating mode	
-----------------------	---

Navigation	 Expert → Output → PFS output 1 to n → Operating mode (0469–1 to n)
Description	Use this function to select the operating mode of the output as a pulse, frequency or switch output.
Selection	■ Pulse ■ Frequency ■ Switch
Factory setting	Pulse
Additional information	<i>"Pulse" option</i> Quantity-dependent pulse with configurable pulse width ■ Whenever a specific mass, volume or corrected volume is reached (pulse value), a pulse is output, the duration of which was set previously (pulse width). ■ The pulses are never shorter than the set duration. Example ■ Flow rate approx. 100 g/s ■ Pulse value 0.1 g ■ Pulse width 0.05 ms ■ Pulse rate 1 000 Impuls/s



A0026883

6 Quantity-proportional pulse (pulse value) with pulse width to be configured

B Pulse width entered

P Pauses between the individual pulses

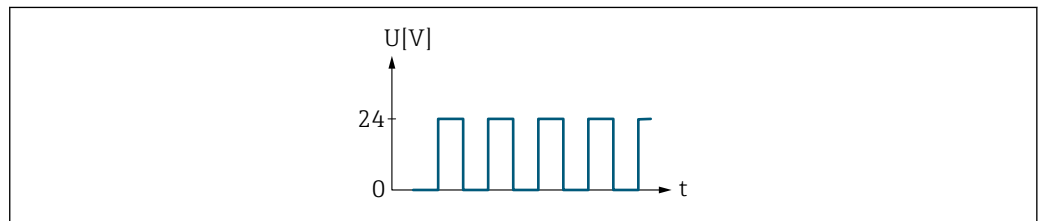
"Frequency" option

Flow-proportional frequency output with 1:1 on/off ratio

An output frequency is output that is proportional to the value of a process variable, such as volume flow, mass flow, corrected volume flow, flow velocity, conductivity, corrected conductivity, temperature or electronic temperature.

Example

- Flow rate approx. 100 g/s
- Max. frequency 10 kHz
- Flow rate at max. frequency 1 000 g/s
- Output frequency approx. 1 000 Hz



A0026886

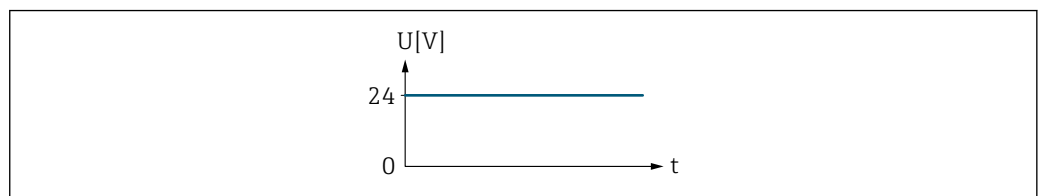
7 Flow-proportional frequency output

"Switch" option

Contact for displaying a condition (e.g. alarm or warning if a limit value is reached)

Example

Alarm response without alarm

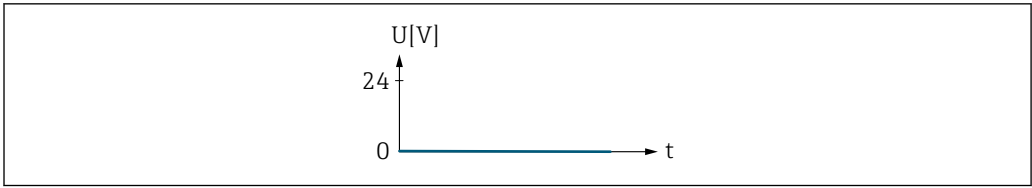


A0026884

8 No alarm, high level

Example

Alarm response in case of alarm



9 Alarm, low level

A002685

Assign pulse output 1 to n

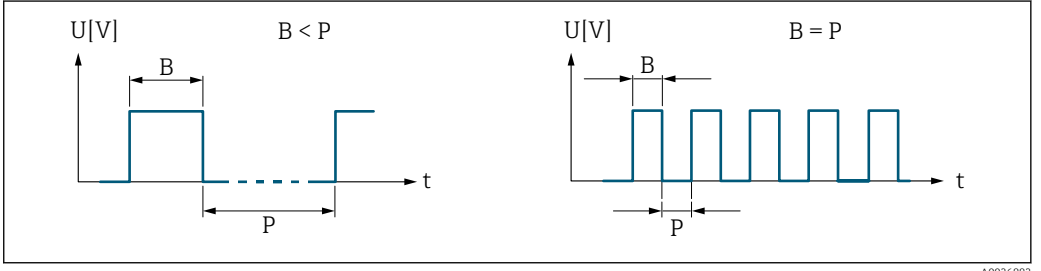


Navigation	 Expert → Output → PFS output 1 to n → Assign pulse 1 to n (0460–1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Pulse option is selected.
Description	Use this function to select the process variable for the pulse output.
Selection	■ Off ■ Mass flow ■ Volume flow ■ Corrected volume flow
Factory setting	Off

Value per pulse



Navigation	 Expert → Output → PFS output 1 to n → Value per pulse (0455–1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Pulse option is selected and one of the following options is selected in the Assign pulse output parameter (→  114): ■ Mass flow ■ Volume flow ■ Corrected volume flow
Description	Use this function to enter the value for the measured value that a pulse is equivalent to.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter →  199
Additional information	<i>User entry</i> Weighting of the pulse output with a quantity. The lower the pulse value, the ■ better the resolution. ■ the higher the frequency of the pulse response.

Pulse width	
Navigation	  Expert → Output → PFS output 1 to n → Pulse width (0452-1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Pulse option is selected and one of the following options is selected in the Assign pulse output parameter (→  114): ■ Mass flow ■ Volume flow ■ Corrected volume flow
Description	Use this function to enter the duration of the output pulse.
User entry	0.05 to 2 000 ms
Factory setting	100 ms
Additional information	<i>Description</i> ■ Define how long a pulse is (duration). ■ The maximum pulse rate is defined by $f_{\max} = 1 / (2 \times \text{pulse width})$. ■ The interval between two pulses lasts at least as long as the set pulse width. ■ The maximum flow is defined by $Q_{\max} = f_{\max} \times \text{pulse value}$. ■ If the flow exceeds these limit values, the measuring device displays the diagnostic message △S443 Pulse output 1 to n.  <i>B Pulse width entered</i> <i>P Pauses between the individual pulses</i>
Example	■ Pulse value: 0.1 g ■ Pulse width: 0.1 ms ■ $f_{\max}: 1 / (2 \times 0.1 \text{ ms}) = 5 \text{ kHz}$ ■ $Q_{\max}: 5 \text{ kHz} \times 0.1 \text{ g} = 0.5 \text{ kg/s}$

Measuring mode	
Navigation	  Expert → Output → PFS output 1 to n → Measuring mode (0457-1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Pulse option is selected and one of the following options is selected in the Assign pulse output parameter (→  114): ■ Mass flow ■ Volume flow ■ Corrected volume flow

Description	Use this function to select the measuring mode for the pulse output.
Selection	■ Forward flow ■ Forward/Reverse flow ■ Reverse flow ■ Reverse flow compensation
Factory setting	Forward flow
Additional information	<i>Selection</i> ■ Forward flow Positive flow is output, negative flow is not output. ■ Forward/Reverse flow Positive and negative flow are output (absolute value), but a distinction is not made between positive and negative flow. ■ Reverse flow Negative flow is output, positive flow is not output. ■ Reverse flow compensation The flow components outside the span are buffered, balanced and output after a maximum delay of 60 s.  For a detailed description of the options available, see the Measuring mode parameter (→  102) <i>Examples</i>  For a detailed description of the configuration examples, see the Measuring mode parameter (→  102)

Failure mode

Navigation	  Expert → Output → PFS output 1 to n → Failure mode (0480–1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Pulse option is selected and one of the following options is selected in the Assign pulse output parameter (→  114): ■ Mass flow ■ Volume flow ■ Corrected volume flow
Description	Use this function to select the failure mode of the pulse output in the event of a device alarm.
Selection	■ Actual value ■ No pulses
Factory setting	No pulses

Additional information*Description*

The dictates of safety render it advisable to ensure that the pulse output shows a predefined behavior in the event of a device alarm.

Selection

- Actual value

In the event of a device alarm, the pulse output continues on the basis of the current flow measurement. The fault is ignored.

- No pulses

In the event of a device alarm, the pulse output is "switched off".

NOTICE! A device alarm is a measuring device error that must be taken seriously. It can affect the measurement quality such that the quality can no longer be guaranteed. The **Actual value** option is only recommended if it can be guaranteed that all possible alarm conditions will not affect the measurement quality.

Pulse output 1 to n**Navigation**

📁📁 Expert → Output → PFS output 1 to n → Pulse output 1 to n (0456–1 to n)

Prerequisite

In the **Operating mode** parameter (→ 📄 112), the **Pulse** option is selected.

Description

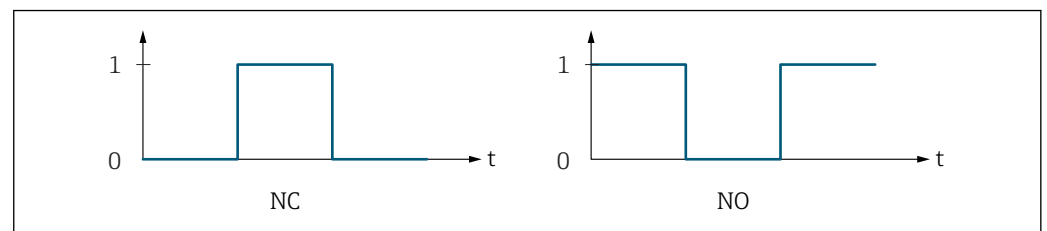
Displays the pulse frequency currently output.

User interface

Positive floating-point number

Additional information*Description*

- The pulse output is an open collector output.
- This is configured at the factory in such a way that the transistor is conductive for the duration of the pulse (NO contact) and is safety-oriented.



A0028726

0 Non-conductive

1 Conductive

NC NC contact (normally closed)

NO NO contact (normally open)

The output behavior can be reversed via the **Invert output signal** parameter (→ 📄 130) i.e. the transistor does not conduct for the duration of the pulse.

In addition, the behavior of the output in the event of a device alarm (**Failure mode** parameter (→ 📄 116)) can be configured.

Assign frequency output


Navigation	Expert → Output → PFS output 1 to n → Assign freq. (0478–1 to n)
Prerequisite	In the Operating mode parameter (→ 112), the Frequency option is selected.
Description	Use this function to select the process variable for the frequency output.
Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Factory setting	Off

Minimum frequency value


Navigation	Expert → Output → PFS output 1 to n → Min. freq. value (0453–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→ 98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Description	Use this function to enter the start value frequency.
User entry	0.0 to 10 000.0 Hz
Factory setting	0.0 Hz

* Visibility depends on order options or device settings

Maximum frequency value 	
Navigation	  Expert → Output → PFS output 1 to n → Max. freq. value (0454–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity * ■ Corrected conductivity * ■ Temperature * ■ Electronic temperature
Description	Use this function to enter the end value frequency.
User entry	0.0 to 10 000.0 Hz
Factory setting	10 000.0 Hz

Measuring value at minimum frequency 	
Navigation	  Expert → Output → PFS output 1 to n → Val. at min.freq (0476–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity * ■ Corrected conductivity * ■ Temperature * ■ Electronic temperature
Description	Use this function to enter the measured value for the start value frequency.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter
Additional information	<i>Dependency</i>  The entry depends on the process variable selected in the Assign frequency output parameter (→  118).

* Visibility depends on order options or device settings

Measuring value at maximum frequency


Navigation	Expert → Output → PFS output 1 to n → Val. at max.freq (0475–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→ 98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Description	Use this function to enter the measured value for the end value frequency.
User entry	Signed floating-point number
Factory setting	Depends on country and nominal diameter
Additional information	<i>Description</i> Use this function to enter the maximum measured value at the maximum frequency. The selected process variable is output as a proportional frequency. <i>Dependency</i> The entry depends on the process variable selected in the Assign frequency output parameter (→ 118).

Measuring mode


Navigation	Expert → Output → PFS output 1 to n → Measuring mode (0479–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→ 98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Description	Use this function to select the measuring mode for the frequency output.
Selection	■ Forward flow ■ Forward/Reverse flow ■ Reverse flow compensation

* Visibility depends on order options or device settings

Factory setting	Forward flow
Additional information	<i>Selection</i>  For a detailed description of the options available, see the Measuring mode parameter (→  102) <i>Examples</i>  For a detailed description of the configuration examples, see the Measuring mode parameter (→  102)

Damping output 1 to n

Navigation	  Expert → Output → PFS output 1 to n → Damping out. 1 to n (0477–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Description	Use this function to enter a time constant for the reaction time of the output signal to fluctuations in the measured value.
User entry	0 to 999.9 s
Factory setting	0.0 s
Additional information	<i>User entry</i> Use this function to enter a time constant (PT1 element⁴⁾) for frequency output damping: ■ If a low time constant is entered, the current output reacts particularly quickly to fluctuating measured variables. ■ On the other hand, the current output reacts more slowly if a high time constant is entered.  Damping is switched off if 0 is entered (factory setting). The frequency output is subject to separate damping that is independent of all preceding time constants.

* Visibility depends on order options or device settings

4) proportional transmission behavior with first order delay

Response time

Navigation	 Expert → Output → PFS output 1 to n → Response time (0491–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Description	Displays the response time. This specifies how quickly the pulse/frequency/switch output reaches the measured value change of 63 % of 100 % of the measured value change.
User interface	Positive floating-point number
Additional information	<i>Description</i>  The response time is made up of the time specified for the following dampings: ■ Damping of pulse/frequency/switch output →  106 and ■ Depending on the measured variable assigned to the output. Flow damping

Failure mode



Navigation	 Expert → Output → PFS output 1 to n → Failure mode (0451–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*] ■ Electronic temperature
Description	Use this function to select the failure mode of the frequency output in the event of a device alarm.
Selection	■ Actual value ■ Defined value ■ 0 Hz
Factory setting	0 Hz

* Visibility depends on order options or device settings

Additional information	<i>Selection</i> ■ Actual value In the event of a device alarm, the frequency output continues on the basis of the current flow measurement. The device alarm is ignored. ■ Defined value In the event of a device alarm, the frequency output continues on the basis of a predefined value. The Failure frequency (→  123) replaces the current measured value, making it possible to bypass the device alarm. The actual measurement is switched off for the duration of the device alarm. ■ 0 Hz In the event of a device alarm, the frequency output is "switched off". NOTICE! A device alarm is a measuring device error that must be taken seriously. It can affect the measurement quality such that the quality can no longer be guaranteed. The Actual value option is only recommended if it can be guaranteed that all possible alarm conditions will not affect the measurement quality.
-------------------------------	---

Failure frequency

Navigation	  Expert → Output → PFS output 1 to n → Failure freq. (0474–1 to n)
Prerequisite	One of the following options is selected in the Assign current output parameter (→  98): ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity * ■ Corrected conductivity * ■ Temperature * ■ Electronic temperature
Description	Use this function to enter the value for the frequency output in the event of a device alarm in order to bypass the alarm.
User entry	0.0 to 12 500.0 Hz
Factory setting	0.0 Hz

Output frequency 1 to n

Navigation	  Expert → Output → PFS output 1 to n → Output freq. 1 to n (0471–1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Frequency option is selected.
Description	Displays the actual value of the output frequency which is currently measured.
User interface	0.0 to 12 500.0 Hz

* Visibility depends on order options or device settings

Switch output function 	
Navigation	 Expert → Output → PFS output 1 to n → Switch out funct (0481–1 to n)
Prerequisite	In the Operating mode parameter (→  112) the Switch option is selected.
Description	Use this function to select a function for the switch output.
Selection	■ Off ■ On ■ Diagnostic behavior ■ Limit ■ Flow direction check ■ Status
Factory setting	Off
Additional information	<i>Selection</i> ■ Off The switch output is permanently switched off (open, non-conductive). ■ On The switch output is permanently switched on (closed, conductive). ■ Diagnostic behavior Indicates if the diagnostic event is present or not. Is used to output diagnostic information and to react to it appropriately at the system level. ■ Limit Indicates if a specified limit value has been reached for the process variable. Is used to output diagnostic information relating to the process and to react to it appropriately at the system level. ■ Flow direction check Indicates the flow direction (forward or reverse flow). ■ Status Indicates the device status depending on whether empty pipe detection or low flow cut off is selected.
Assign diagnostic behavior 	
Navigation	 Expert → Output → PFS output 1 to n → Assign diag. beh (0482–1 to n)
Prerequisite	■ In the Operating mode parameter (→  112), the Switch option is selected. ■ In the Switch output function parameter (→  124), the Diagnostic behavior option is selected.
Description	Use this function to select the diagnostic event category that is displayed for the switch output.
Selection	■ Alarm ■ Alarm or warning ■ Warning
Factory setting	Alarm

Additional information*Description*

If no diagnostic event is pending, the switch output is closed and conductive.

Selection

- **Alarm**
The switch output signals only diagnostic events in the alarm category.
- **Alarm or warning**
The switch output signals diagnostic events in the alarm and warning category.
- **Warning**
The switch output signals only diagnostic events in the warning category.

Assign limit**Navigation**

Expert → Output → PFS output 1 to n → Assign limit (0483–1 to n)

Prerequisite

- In the **Operating mode** parameter (→ 112), the **Switch** option is selected.
- In the **Switch output function** parameter (→ 124), the **Limit** option is selected.

Description

Use this function to select a process variable for the limit function.

Selection

- Off
- Volume flow
- Mass flow
- Corrected volume flow
- Flow velocity
- Conductivity *
- Corrected conductivity *
- Totalizer 1
- Totalizer 2
- Totalizer 3
- Temperature *
- Electronic temperature

Factory setting

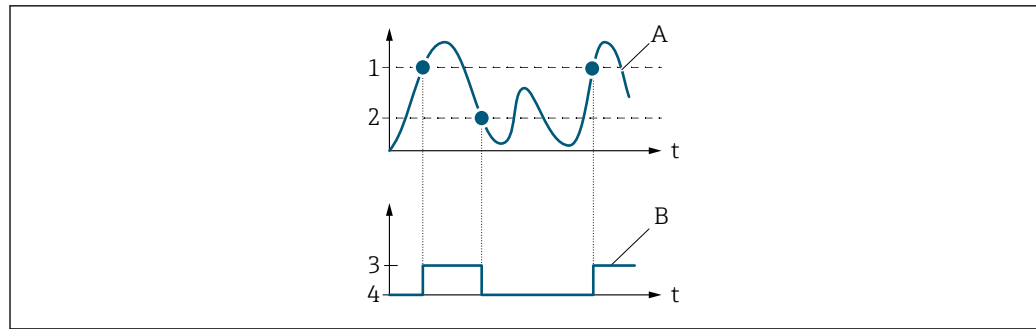
Volume flow

Additional information*Description*

Behavior of status output when Switch-on value > Switch-off value:

- Process variable > Switch-on value: transistor is conductive
- Process variable < Switch-off value: transistor is non-conductive

* Visibility depends on order options or device settings

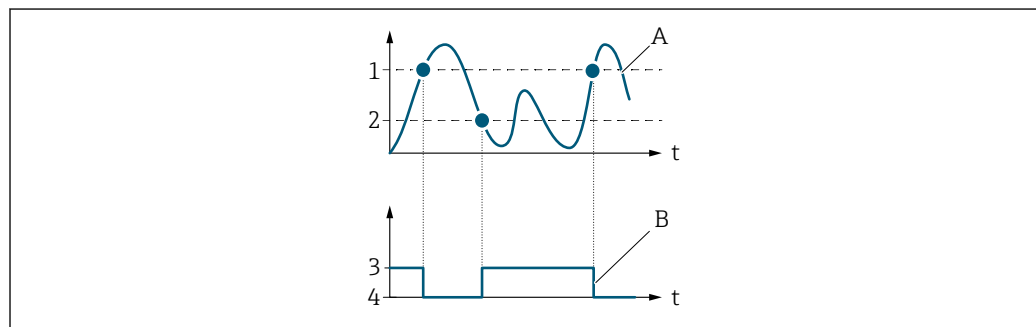


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- 1 Switch-on value
- 2 Switch-off value
- 3 Conductive
- 4 Non-conductive
- A Process variable
- B Status output

Behavior of status output when Switch-on value < Switch-off value:

- Process variable < Switch-on value: transistor is conductive
- Process variable > Switch-off value: transistor is non-conductive

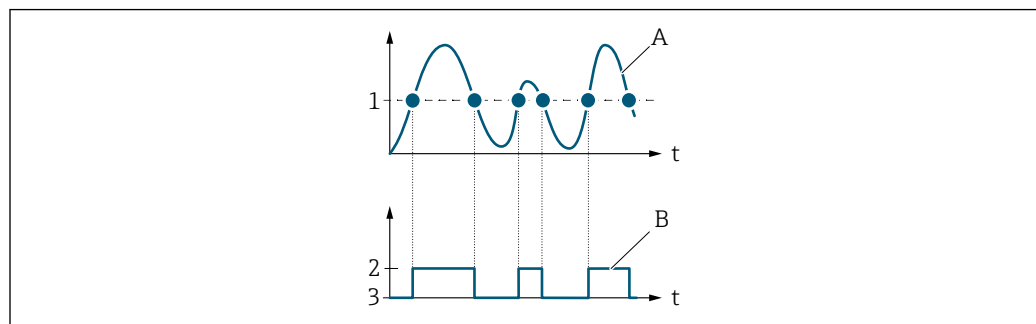


A0026892

- 1 Switch-off value
- 2 Switch-on value
- 3 Conductive
- 4 Non-conductive
- A Process variable
- B Status output

Behavior of status output when Switch-on value = Switch-off value:

- Process variable > Switch-on value: transistor is conductive
- Process variable < Switch-off value: transistor is non-conductive



A0026893

- 1 Switch-on value = Switch-off value
- 2 Conductive
- 3 Non-conductive
- A Process variable
- B Status output

Switch-on value 	
Navigation	  Expert → Output → PFS output 1 to n → Switch-on value (0466–1 to n)
Prerequisite	■ In the Operating mode parameter (→  112), the Switch option is selected. ■ In the Switch output function parameter (→  124), the Limit option is selected.
Description	Use this function to enter the measured value for the switch-on point.
User entry	Signed floating-point number
Factory setting	Country-specific: ■ 0 l/h ■ 0 gal/min (us)
Additional information	<i>Description</i> Use this function to enter the limit value for the switch-on value (process variable > switch-on value = closed, conductive).  When using a hysteresis: Switch-on value > Switch-off value. <i>Dependency</i>  The unit depends on the process variable selected in the Assign limit parameter (→  125).
Switch-off value 	
Navigation	  Expert → Output → PFS output 1 to n → Switch-off value (0464–1 to n)
Prerequisite	■ In the Operating mode parameter (→  112), the Switch option is selected. ■ In the Switch output function parameter (→  124), the Limit option is selected.
Description	Use this function to enter the measured value for the switch-off point.
User entry	Signed floating-point number
Factory setting	Country-specific: ■ 0 l/h ■ 0 gal/min (us)
Additional information	<i>Description</i> Use this function to enter the limit value for the switch-off value (process variable < switch-off value = open, non-conductive).  When using a hysteresis: Switch-on value > Switch-off value. <i>Dependency</i>  The unit depends on the process variable selected in the Assign limit parameter (→  125).

Assign flow direction check 	
Navigation	 Expert → Output → PFS output 1 to n → Assign dir.check (0484-1 to n)
Prerequisite	■ The Switch option is selected in the Operating mode parameter (→  112). ■ The Flow direction check option is selected in the Switch output function parameter (→  124).
Description	Use this function to select a process variable for monitoring the flow direction.
Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow
Factory setting	Volume flow
Assign status 	
Navigation	 Expert → Output → PFS output 1 to n → Assign status (0485-1 to n)
Prerequisite	■ The Switch option is selected in the Operating mode parameter (→  112). ■ The Status option is selected in the Switch output function parameter (→  124).
Description	Use this function to select a device status for the switch output.
Selection	■ Empty pipe detection ■ Low flow cut off
Factory setting	Empty pipe detection
Additional information	<i>Options</i> If empty pipe detection or low flow cut off are enabled, the output is conductive. Otherwise, the switch output is non-conductive.
Switch-on delay 	
Navigation	 Expert → Output → PFS output 1 to n → Switch-on delay (0467-1 to n)
Prerequisite	■ The Switch option is selected in the Operating mode parameter (→  112). ■ The Limit option is selected in the Switch output function parameter (→  124).
Description	Use this function to enter a delay time for switching on the switch output.
User entry	0.0 to 100.0 s
Factory setting	0.0 s

Switch-off delay



Navigation	Expert → Output → PFS output 1 to n → Switch-off delay (0465–1 to n)
Prerequisite	▪ The Switch option is selected in the Operating mode parameter (→ 112). ▪ The Limit option is selected in the Switch output function parameter (→ 124).
Description	Use this function to enter a delay time for switching off the switch output.
User entry	0.0 to 100.0 s
Factory setting	0.0 s

Failure mode



Navigation	Expert → Output → PFS output 1 to n → Failure mode (0486–1 to n)
Description	Use this function to select a failsafe mode for the switch output in the event of a device alarm.
Selection	▪ Actual status ▪ Open ▪ Closed
Factory setting	Open
Additional information	<i>Options</i> ▪ Actual status In the event of a device alarm, faults are ignored and the current behavior of the input value is output by the switch output. The Actual status option behaves in the same way as the current input value. ▪ Open In the event of a device alarm, the switch output's transistor is set to non-conductive. ▪ Closed In the event of a device alarm, the switch output's transistor is set to conductive.

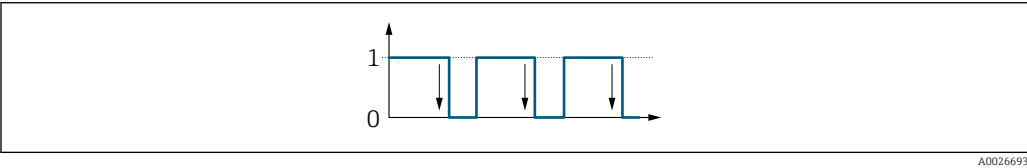
Switch status 1 to n

Navigation	Expert → Output → PFS output 1 to n → Switch status 1 to n (0461–1 to n)
Prerequisite	The Switch option is selected in the Operating mode parameter (→ 112).
Description	Displays the current switch status of the status output.
User interface	▪ Open ▪ Closed

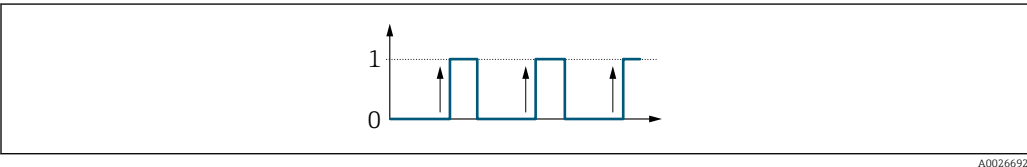
Additional information	User interface ■ Open The switch output is not conductive. ■ Closed The switch output is conductive.
------------------------	--

Invert output signal

Navigation	  Expert → Output → PFS output 1 to n → Invert outp.sig. (0470-1 to n)
Description	Use this function to select whether to invert the output signal.
Selection	■ No ■ Yes
Factory setting	No
Additional information	Selection No option (passive - negative)



Yes option (passive - positive)



3.5.3 "Relay output 1 to n" submenu

Navigation   Expert → Output → Relay output 1 to n

► Relay output 1 to n

Terminal number

→  131

Relay output function

→  131

Assign flow direction check

→  132

Assign limit	→  132
Assign diagnostic behavior	→  133
Assign status	→  133
Switch-off value	→  134
Switch-off delay	→  134
Switch-on value	→  134
Switch-on delay	→  135
Failure mode	→  135
Switch status	→  136
Actual relay state	→  136

Terminal number

Navigation	 Expert → Output → Relay output 1 to n → Terminal no. (0812–1 to n)
Description	Displays the terminal numbers used by the relay output module.
User interface	■ Not used ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)
Additional information	<i>"Not used" option</i> The relay output module does not use any terminal numbers.

Relay output function

Navigation	 Expert → Output → Relay output 1 to n → Relay outp.func. (0804–1 to n)
Description	Use this function to select an output function for the relay output.
Selection	■ Closed ■ Open ■ Diagnostic behavior ■ Limit ■ Flow direction check ■ Digital Output

Factory setting	Closed
Additional information	<i>Selection</i> ■ Closed The relay output is permanently switched on (closed, conductive). ■ Open The relay output is permanently switched off (open, non-conductive). ■ Diagnostic behavior Indicates if the diagnostic event is present or not. Is used to output diagnostic information and to react to it appropriately at the system level. ■ Limit Indicates if a specified limit value has been reached for the process variable. Is used to output diagnostic information relating to the process and to react to it appropriately at the system level. ■ Flow direction check Indicates the flow direction (forward or reverse flow). ■ Digital Output Indicates the device status depending on whether empty pipe detection or low flow cut off is selected.

Assign flow direction check



Navigation	Expert → Output → Relay output 1 to n → Assign dir.check (0808–1 to n)
Prerequisite	In the Relay output function parameter (→ 131), the Flow direction check option is selected.
Description	Use this function to select a process variable for monitoring the flow direction.
Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow
Factory setting	Volume flow

Assign limit



Navigation	Expert → Output → Relay output 1 to n → Assign limit (0807–1 to n)
Prerequisite	In the Relay output function parameter (→ 131), the Limit option is selected.
Description	Use this function to select a process variable for the limit value function.
Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity

- Conductivity *
- Corrected conductivity *
- Totalizer 1
- Totalizer 2
- Totalizer 3
- Temperature *
- Electronic temperature

Factory setting Volume flow

Assign diagnostic behavior

Navigation   Expert → Output → Relay output 1 to n → Assign diag. beh (0806–1 to n)

Prerequisite In the **Relay output function** parameter (→  131), the **Diagnostic behavior** option is selected.

Description Use this function to select the category of the diagnostic events that are displayed for the relay output.

Selection

- Alarm
- Alarm or warning
- Warning

Factory setting Alarm

Additional information *Description*
 If no diagnostic event is pending, the relay output is closed and conductive.

Selection

- Alarm
The relay output signals only diagnostic events in the alarm category.
- Alarm or warning
The relay output signals diagnostic events in the alarm and warning category.
- Warning
The relay output signals only diagnostic events in the warning category.

Assign status

Navigation   Expert → Output → Relay output 1 to n → Assign status (0805–1 to n)

Prerequisite In the **Relay output function** parameter (→  131), the **Digital Output** option is selected.

Description Use this function to select the device status for the relay output.

* Visibility depends on order options or device settings

Selection

- Partially filled pipe detection
- Low flow cut off

Factory setting Partially filled pipe detection

Switch-off value

Navigation   Expert → Output → Relay output 1 to n → Switch-off value (0809–1 to n)

Prerequisite In the **Relay output function** parameter (→  131), the **Limit** option is selected.

Description Use this function to enter the measured value for the switch-off point.

User entry Signed floating-point number

Factory setting Country-specific:

- 0 l/h
- 0 gal(us)/min

Additional information *Description*

Use this function to enter the limit value for the switch-off value (process variable < switch-off value = open, non-conductive).

 When using a hysteresis: Switch-on value > Switch-off value.

Dependency

 The unit is dependent on the process variable selected in the **Assign limit** parameter (→  132).

Switch-off delay

Navigation   Expert → Output → Relay output 1 to n → Switch-off delay (0813–1 to n)

Prerequisite In the **Relay output function** parameter (→  131), the **Limit** option is selected.

Description Use this function to enter a delay time for switching off the switch output.

User entry 0.0 to 100.0 s

Factory setting 0.0 s

Switch-on value

Navigation   Expert → Output → Relay output 1 to n → Switch-on value (0810–1 to n)

Prerequisite In the **Relay output function** parameter (→  131), the **Limit** option is selected.

Description	Use this function to enter the measured value for the switch-on point.
User entry	Signed floating-point number
Factory setting	Country-specific: ■ 0 l/h ■ 0 gal(us)/min
Additional information	<i>Description</i> Use this function to enter the limit value for the switch-on value (process variable > switch-on value = closed, conductive).  When using a hysteresis: Switch-on value > Switch-off value. <i>Dependency</i>  The unit is dependent on the process variable selected in the Assign limit parameter (→  132).

Switch-on delay

Navigation	  Expert → Output → Relay output 1 to n → Switch-on delay (0814-1 to n)
Prerequisite	In the Relay output function parameter (→  131), the Limit option is selected.
Description	Use this function to enter a delay time for switching on the switch output.
User entry	0.0 to 100.0 s
Factory setting	0.0 s

Failure mode

Navigation	  Expert → Output → Relay output 1 to n → Failure mode (0811-1 to n)
Description	Use this function to select the failure mode of the relay output in the event of a device alarm.
Selection	■ Actual status ■ Open ■ Closed
Factory setting	Open

Additional information	<i>Selection</i> ■ Actual status In the event of a device alarm, faults are ignored and the current behavior of the input value is output by the relay output. The Actual status option behaves in the same way as the current input value. ■ Open In the event of a device alarm, the relay output's transistor is set to non-conductive. ■ Closed In the event of a device alarm, the relay output's transistor is set to conductive.
-------------------------------	--

Switch status

Navigation  Expert → Output → Relay output 1 to n → Switch status (0801–1 to n)

Description Displays the current status of the relay output.

User interface

- Open
- Closed

Additional information *User interface*

- Open
The relay output is not conductive.
- Closed
The relay output is conductive.

Actual relay state



Navigation  Expert → Output → Relay output 1 to n → Act. relay state (0816–1 to n)

Description Use this function to select the quiescent state for the relay output.

Selection

- Open
- Closed

Factory setting Open

Additional information *Selection*

- Open
The relay output is not conductive.
- Closed
The relay output is conductive.

3.5.4 "Double pulse output" submenu

Navigation   Expert → Output → Double pulse out

► Double pulse output		
Master terminal number (0981)	→ 	137
Slave terminal number (0990)	→ 	137
Signal mode (0991)	→ 	138
Assign pulse output 1 (0982-1)	→ 	138
Value per pulse (0983)	→ 	138
Pulse width (0986)	→ 	139
Phase shift (0992)	→ 	139
Measuring mode (0984)	→ 	139
Failure mode (0985)	→ 	140
Pulse output (0987)	→ 	141
Invert output signal (0993)	→ 	141

Master terminal number

Navigation	  Expert → Output → Double pulse out → Master term. no. (0981)
Description	Displays the master terminal number for the double pulse output.
User interface	■ Not used ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)
Additional information	"Not used" option The double pulse output does not use any terminal numbers.

Slave terminal number

Navigation	  Expert → Output → Double pulse out → Slave term. no. (0990)
Description	Displays the slave terminal number for the double pulse output.

- User interface**
- Not used
 - 24-25 (I/O 2)
 - 22-23 (I/O 3)

Additional information *"Not used" option*
 The double pulse output does not use any terminal numbers.

Signal mode



Navigation Expert → Output → Double pulse out → Signal mode (0991)

Description Use this function to select the signal mode for the double pulse output.

- Selection**
- Passive
 - Active
 - Passive NAMUR

Factory setting Passive

Assign pulse output 1



Navigation Expert → Output → Double pulse out → Assign pulse 1 (0982-1)

Description Use this function to select a process variable for the double pulse output.

- Selection**
- Off
 - Mass flow
 - Volume flow
 - Corrected volume flow

Factory setting Off

Value per pulse



Navigation Expert → Output → Double pulse out → Value per pulse (0983)

Description Use this function to enter the value for the measured value that a pulse is equivalent to.

User entry Signed floating-point number

Factory setting Depends on country and nominal diameter → 199

Additional information *User entry*
 Weighting of the pulse output with a quantity.

- The lower the pulse value, the
- better the resolution.
 - the higher the frequency of the pulse response.

Pulse width



Navigation	Expert → Output → Double pulse out → Pulse width (0986)
Description	Use this function to enter the duration of the output pulse.
User entry	0.5 to 2 000 ms
Factory setting	0.5 ms
Additional information	For a detailed description and example: Pulse width parameter (→ 115)

Phase shift



Navigation	Expert → Output → Double pulse out → Phase shift (0992)
Description	Use this function to select the degree of phase shift.
Selection	■ 90° ■ 180°
Factory setting	90°
Additional information	<i>Selection</i> ■ 90° Phase shift by a quarter period. ■ 180° Phase shift by a half period, which is equivalent to a phase reversal.

Measuring mode



Navigation	Expert → Output → Double pulse out → Measuring mode (0984)
Description	Use this function to select the measuring mode for the double pulse output.
Selection	■ Forward flow ■ Forward/Reverse flow ■ Reverse flow ■ Reverse flow compensation
Factory setting	Forward flow

Additional information*Selection*

- Forward flow
Positive flow is output, negative flow is not output.
- Forward/Reverse flow
Positive and negative flow are output (absolute value), but a distinction is not made between positive and negative flow.
- Reverse flow
Negative flow is output, positive flow is not output.
- Reverse flow compensation
The flow components outside the span are buffered, balanced and output after a maximum delay of 60 s.

 For a detailed description of the options available, see the **Measuring mode** parameter (→  102)

Examples

 For a detailed description of the configuration examples, see the **Measuring mode** parameter (→  102)

Failure mode**Navigation**

  Expert → Output → Double pulse out → Failure mode (0985)

Description

Use this function to select the failure mode of the double pulse output in the event of a device alarm.

Selection

- Actual value
- No pulses

Factory setting

No pulses

Additional information*Description*

The dictates of safety render it advisable to ensure that the double pulse output shows a predefined behavior in the event of a device alarm.

Selection

- Actual value
In the event of a device alarm, the double pulse output continues on the basis of the current flow measurement. The fault is ignored.
- No pulses
In the event of a device alarm, the double pulse output is "switched off".

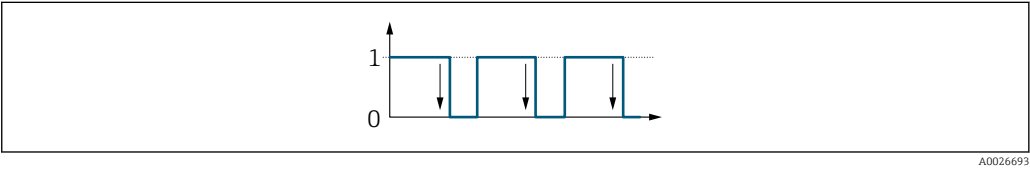
NOTICE! A device alarm is a measuring device error that must be taken seriously. It can affect the measurement quality such that the quality can no longer be guaranteed. The **Actual value** option is only recommended if it can be guaranteed that all possible alarm conditions will not affect the measurement quality.

Pulse output

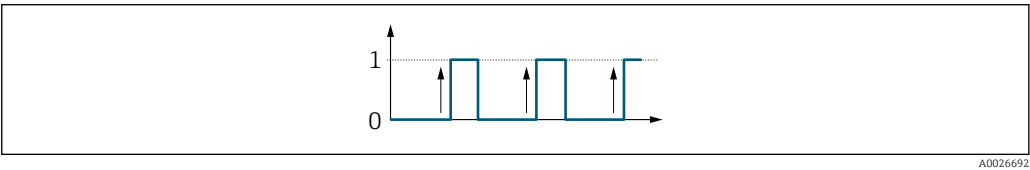
Navigation	 Expert → Output → Double pulse out → Pulse output (0987)
Description	Displays the pulse frequency of the double pulse output which is currently output.
User interface	Positive floating-point number
Additional information	 For a detailed description and example: Pulse output parameter (→  117)

Invert output signal

Navigation	 Expert → Output → Double pulse out → Invert outp.sig. (0993)
Description	Use this function to select whether to invert the output signal.
Selection	■ No ■ Yes
Factory setting	No
Additional information	<i>Selection</i> No option (passive - negative)



Yes option (passive - positive)



3.6 "Communication" submenu

Navigation  Expert → Communication

► Communication

► Modbus configuration

→  142

► Modbus information	→ 146
► Modbus data map	→ 147
► Web server	→ 147
► WLAN settings	→ 150

3.6.1 "Modbus configuration" submenu

Navigation  Expert → Communication → Modbus config.

► Modbus configuration	
Bus address (7112)	→ 142
Baudrate (7111)	→ 143
Data transfer mode (7115)	→ 143
Parity (7122)	→ 143
Byte order (7113)	→ 144
Telegram delay (7146)	→ 144
Failure mode (7116)	→ 144
Bus termination (7155)	→ 145
Fieldbus writing access (7156)	→ 145

Bus address	
-------------	---

Navigation	  Expert → Communication → Modbus config. → Bus address (7112)
Description	For entering the device address.
User entry	1 to 247
Factory setting	247

Baudrate



Navigation Expert → Communication → Modbus config. → Baudrate (7111)

Description Use this function to select a transmission rate.

Selection

- 1200 BAUD
- 2400 BAUD
- 4800 BAUD
- 9600 BAUD
- 19200 BAUD
- 38400 BAUD
- 57600 BAUD
- 115200 BAUD

Factory setting 19200 BAUD

Data transfer mode



Navigation Expert → Communication → Modbus config. → Data trans. mode (7115)

Description Use this function to select the data transmission mode.

Selection

- ASCII
- RTU

Factory setting RTU

Additional information *Options*

- ASCII
Transmission of data in the form of readable ASCII characters. Error protection via LRC.
- RTU
Transmission of data in binary form. Error protection via CRC16.

Parity



Navigation Expert → Communication → Modbus config. → Parity (7122)

Description Use this function to select the parity bit.

Selection

- Odd
- Even
- None / 1 stop bit
- None / 2 stop bits

Factory setting Even

Additional information	<i>Options</i> Picklist ASCII option: ■ 0 = Even option ■ 1 = Odd option Picklist RTU option: ■ 0 = Even option ■ 1 = Odd option ■ 2 = None / 1 stop bit option ■ 3 = None / 2 stop bits option
-------------------------------	--

Byte order	
-------------------	---

Navigation	  Expert → Communication → Modbus config. → Byte order (7113)
Description	Use this function to select the sequence in which the bytes are transmitted. The transmission sequence must be coordinated with the Modbus master.
Selection	■ 0-1-2-3 ■ 3-2-1-0 ■ 1-0-3-2 ■ 2-3-0-1
Factory setting	1-0-3-2

Telegram delay	
-----------------------	---

Navigation	  Expert → Communication → Modbus config. → Telegram delay (7146)
Description	Use this function to enter a delay time after which the measuring device replies to the request telegram of the Modbus master. This allows communication to be adapted to slow Modbus RS485 masters.
User entry	0 to 100 ms
Factory setting	6 ms

Failure mode	
---------------------	---

Navigation	  Expert → Communication → Modbus config. → Failure mode (7116)
Description	Use this function to select the measured value output in the event of a diagnostic message via Modbus communication.
Selection	■ NaN value ■ Last valid value

Factory setting	NaN value
Additional information	<i>Options</i> ■ NaN value The device outputs the NaN value ⁵⁾. ■ Last valid value The device outputs the last valid measured value before the fault occurred.  This effect of this parameter depends on the option selected in the Assign diagnostic behavior parameter.

Bus termination

Navigation	 Expert → Communication → Modbus config. → Bus termination (7155)
Description	Displays whether the terminating resistor is enabled or disabled.
User interface	■ Off ■ On
Factory setting	Off
Additional information	<i>Selection</i> ■ Off The terminating resistor is disabled. ■ On The terminating resistor is enabled.  For detailed information about enabling the terminating resistor, see the Operating Instructions for the device, "Enabling the terminating resistor" section

Fieldbus writing access

Navigation	 Expert → Communication → Modbus config. → Fieldb.writ.acc. (7156)
Description	Use this function to restrict access to the measuring device via fieldbus (Modbus protocol).
Selection	■ Read + write ■ Read only
Factory setting	Read + write

5) Not a Number

Additional information	<i>Description</i> If read and/or write protection is enabled, the parameter can only be controlled and reset via local operation. Access is no longer possible via operating tools.  This does not affect cyclic measured value transmission to the higher-order system, which is always guaranteed. <i>Selection</i> ■ Read + write The parameters are read and write parameters. ■ Read only The parameters are read only parameters.
------------------------	---

3.6.2 "Modbus information" submenu

Navigation  Expert → Communication → Modbus info

► Modbus information

Device ID (7153)

→  146

Device revision (7154)

→  146

Device ID

Navigation	 Expert → Communication → Modbus info → Device ID (7153)
Description	Displays the device ID for identifying the measuring device.
User interface	4-digit hexadecimal number

Device revision

Navigation	 Expert → Communication → Modbus info → Device revision (7154)
Description	Displays the device revision.
User interface	4-digit hexadecimal number

3.6.3 "Modbus data map" submenu

Navigation  Expert → Communication → Modbus data map

► Modbus data map

Scan list register 0 to 15 (7114)

→  147

Scan list register 0 to 15



Navigation	 Expert → Communication → Modbus data map → Scan list reg.0 to 15 (7114)
Description	Use this function to enter the scan list register. By entering the register address (1-based), up to 16 device parameters can be grouped by assigning them to the scan list registers 0 to 15. The data of the device parameters assigned here are read out via the register addresses 5051 to 5081.
User entry	1 to 65 535
Factory setting	1

3.6.4 "Web server" submenu

Navigation  Expert → Communication → Web server

► Web server

Web server language (7221)

→  148

MAC address (7214)

→  148

IP address (7209)

→  149

Subnet mask (7211)

→  149

Default gateway (7210)

→  149

Web server functionality (7222)

→  149

Login page (7273)

→  150

Web server language

Navigation  Expert → Communication → Web server → Webserv.language (7221)

Description Use this function to select the web server language setting.

Selection

- English *
- Deutsch *
- Français *
- Español *
- Italiano *
- Nederlands *
- Portuguesa *
- Polski *
- русский язык (Russian) *
- Svenska *
- Türkçe *
- 中文 (Chinese) *
- 日本語 (Japanese) *
- 한국어 (Korean) *
- العربية (Arabic) *
- Bahasa Indonesia *
- ภาษาไทย (Thai) *
- tiếng Việt (Vietnamese) *
- čeština (Czech) *

Factory setting English

MAC address

Navigation  Expert → Communication → Web server → MAC Address (7214)

Description Displays the MAC ⁶⁾ address of the measuring device.

User interface Unique 12-digit character string comprising letters and numbers

Factory setting Each measuring device is given an individual address.

Additional information *Example*
For the display format
00:07:05:10:01:5F

* Visibility depends on order options or device settings

6) Media Access Control

IP address



Navigation  Expert → Communication → Web server → IP address (7209)

Description Displays the IP address of the device's web server.

User interface 4 octet: 0 to 255 (in the particular octet)

Factory setting 192.168.1.212

Subnet mask



Navigation  Expert → Communication → Web server → Subnet mask (7211)

Description Displays the subnet mask.

User interface 4 octet: 0 to 255 (in the particular octet)

Factory setting 255.255.255.0

Default gateway



Navigation  Expert → Communication → Web server → Default gateway (7210)

Description Displays the default gateway.

User interface 4 octet: 0 to 255 (in the particular octet)

Factory setting 0.0.0.0

Web server functionality



Navigation  Expert → Communication → Web server → Webserver funct. (7222)

Description Use this function to switch the Web server on and off.

Selection

- Off
- On

Factory setting On

Additional information

Description

 Once disabled, the Web server functionality can only be re-enabled via or the operating tool FieldCare.

Selection

Option	Description
Off	▪ The web server is completely disabled. ▪ Port 80 is locked.
On	▪ The complete functionality of the web server is available. ▪ JavaScript is used. ▪ The password is transferred in an encrypted state. ▪ Any change to the password is also transferred in an encrypted state.

Login page



Navigation

 Expert → Communication → Web server → Login page (7273)

Description

Use this function to select the format of the login page.

Selection

- Without header
- With header

Factory setting

With header

3.6.5 "WLAN settings" submenu

Navigation  Expert → Communication → WLAN settings

▶ WLAN settings		
WLAN (2702)		→  151
WLAN IP address (2711)		→  151
WLAN subnet mask (2709)		→  151
WLAN MAC address (2703)		→  152
Security type (2705)		→  152
WLAN passphrase (2706)		→  152
Assign SSID name (2708)		→  153
SSID name (2707)		→  153

Select antenna (2713)	→ 153
2.4 GHz WLAN channel (2704)	→ 154
Apply changes (2712)	→ 154

WLAN

Navigation	 Expert → Communication → WLAN settings → WLAN (2702)
Description	Use this function to enable and disable the WLAN connection.
Selection	■ Disable ■ Enable
Factory setting	Enable

WLAN IP address

Navigation	 Expert → Communication → WLAN settings → WLAN IP address (2711)
Description	Use this function to enter the IP address of the measuring device's WLAN connection.
User entry	4 octet: 0 to 255 (in the particular octet)
Factory setting	192.168.1.212

WLAN subnet mask

Navigation	 Expert → Communication → WLAN settings → WLAN subnet mask (2709)
Description	Use this function to enter the subnet mask.
User entry	4 octet: 0 to 255 (in the particular octet)
Factory setting	255.255.255.0

WLAN MAC address

Navigation	 Expert → Communication → WLAN settings → WLAN MAC address (2703)
Description	Displays the MAC ⁷⁾ address of the measuring device.
User interface	Unique 12-digit character string comprising letters and numbers
Factory setting	Each measuring device is given an individual address.
Additional information	<i>Example</i> For the display format 00:07:05:10:01:5F

Security type



Navigation	 Expert → Communication → WLAN settings → Security type (2705)
Description	Use this function to select the type of security for the WLAN interface.
Selection	■ Unsecured ■ WPA2-PSK
Factory setting	WPA2-PSK
Additional information	<i>Selection</i> ■ Unsecured Access the WLAN connection without identification. ■ WPA2-PSK Access the WLAN connection with a network key.

WLAN passphrase



Navigation	 Expert → Communication → WLAN settings → WLAN passphrase (2706)
Prerequisite	In the Security type parameter (→  152), the WPA2-PSK option is selected.
Description	Use this function to enter the network key.
User entry	8 to 32-digit character string comprising numbers, letters and special characters
Factory setting	Serial number of the measuring device (e.g. L100A802000)

7) Media Access Control

Assign SSID name

Navigation   Expert → Communication → WLAN settings → Assign SSID name (2708)

Description Use this function to select which name is used for the SSID ⁸⁾.

Selection

- Device tag
- User-defined

Factory setting User-defined

Additional information *Selection*

- Device tag
The device tag name is used as the SSID.
- User-defined
A user-defined name is used as the SSID.

SSID name

Navigation   Expert → Communication → WLAN settings → SSID name (2707)

Prerequisite In the **Assign SSID name** parameter (→  153), the **User-defined** option is selected.

Description Use this function to enter a user-defined SSID name.

User entry Max. 32-digit character string comprising numbers, letters and special characters

Factory setting EH_device designation_last 7 digits of the serial number (e.g. EH_Promag_300_A802000)

Select antenna

Navigation   Expert → Communication → WLAN settings → Select antenna (2713)

Description Use this function to select whether the external or internal antenna is used for reception.

Selection

- External antenna
- Internal antenna

Factory setting Internal antenna

8) Service Set Identifier

2.4 GHz WLAN channel



Navigation	 Expert → Communication → WLAN settings → WLAN channel (2704)
Description	Use this function to enter the 2.4 GHz WLAN channel.
User entry	1 to 11
Factory setting	6
Additional information	<i>Description</i>  - It is only necessary to enter a 2.4 GHz WLAN channel if multiple WLAN devices are in use. - If just one measuring device is in use, it is recommended to keep the factory setting.

Apply changes



Navigation	 Expert → Communication → WLAN settings → Apply changes (2712)
Description	Use this function to adopt modified WLAN settings.
Selection	- Cancel - Ok
Factory setting	Cancel
Additional information	<i>Selection</i> - Cancel No action is executed and the user exits the parameter. - Ok The measuring device adopts the modified WLAN settings.

3.7 "Application" submenu

Navigation  Expert → Application

► Application	
Reset all totalizers (2806)	→  155
► Totalizer 1 to n	→  155
► Custody transfer	→  159

Reset all totalizers

Navigation	 Expert → Application → Reset all tot. (2806)						
Description	Use this function to reset all totalizers to the value 0 and restart the totaling process. This deletes all the flow values previously totalized.						
Selection	■ Cancel ■ Reset + totalize						
Factory setting	Cancel						
Additional information	<i>Selection</i> <table><tr><th>Options</th><th>Description</th></tr><tr><td>Cancel</td><td>No action is executed and the user exits the parameter.</td></tr><tr><td>Reset + totalize</td><td>Resets all totalizers to 0 and restarts the totaling process. This deletes all the flow values previously totalized.</td></tr></table>	Options	Description	Cancel	No action is executed and the user exits the parameter.	Reset + totalize	Resets all totalizers to 0 and restarts the totaling process. This deletes all the flow values previously totalized.
Options	Description						
Cancel	No action is executed and the user exits the parameter.						
Reset + totalize	Resets all totalizers to 0 and restarts the totaling process. This deletes all the flow values previously totalized.						

3.7.1 "Totalizer 1 to n" submenu

Navigation  Expert → Application → Totalizer 1 to n

► Totalizer 1 to n		
Assign process variable (0914-1 to n)	→	 155
Unit totalizer 1 to n (0915-1 to n)	→	 156
Totalizer operation mode (0908-1 to n)	→	 157
Control Totalizer 1 to n (0912-1 to n)	→	 157
Preset value 1 to n (0913-1 to n)	→	 158
Failure mode (0901-1 to n)	→	 158

Assign process variable



Navigation	 Expert → Application → Totalizer 1 to n → Assign variable (0914-1 to n)
Description	Use this function to select a process variable for the Totalizer 1 to n.

Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow
Factory setting	Volume flow
Additional information	Description i If the option selected is changed, the device resets the totalizer to 0. Selection If the Off option is selected, only Assign process variable parameter (→  155) is still displayed in the Totalizer 1 to n submenu. All other parameters in the submenu are hidden.

Unit totalizer 1 to n

Navigation	  Expert → Application → Totalizer 1 to n → Unit totalizer 1 to n (0915–1 to n)		
Description	Use this function to select the process variable unit for the Totalizer 1 to n (→  155).		
Selection	SI units ■ g ■ kg ■ t Custom-specific units User mass or SI units ■ cm³ ■ dm³ ■ m³ ■ ml ■ l ■ hl ■ Ml Mega Custom-specific units User vol. or US units ■ oz ■ lb ■ STon US units ■ af ■ ft³ ■ fl oz (us) ■ gal (us) ■ kgal (us) ■ Mgal (us) ■ bbl (us;liq.) ■ bbl (us;beer) ■ bbl (us;oil) ■ bbl (us;tank) Imperial units ■ gal (imp) ■ Mgal (imp) ■ bbl (imp;beer) ■ bbl (imp;oil)		

	<i>SI units</i> ■ NI ■ Nm³ ■ Sm³ <i>Custom-specific units</i> UserCrVol.	<i>US units</i> ■ Sft³ ■ Sgal (us) ■ Sbbl (us;liq.)	<i>Imperial units</i> Sgal (imp)
Factory setting	1		
Additional information	<i>Description</i>  The unit is selected separately for each totalizer. It is independent of the selection made in the System units submenu (→  52). <i>Selection</i> The selection is dependent on the process variable selected in the Assign process variable parameter (→  155).		

Totalizer operation mode 	
Navigation	  Expert → Application → Totalizer 1 to n → Operation mode (0908–1 to n)
Description	Use this function to select how the totalizer summates the flow.
Selection	■ Net flow total ■ Forward flow total ■ Reverse flow total
Factory setting	Net flow total
Additional information	<i>Selection</i> ■ Net flow total Flow values in the forward and reverse flow direction are totalized and balanced against one another. Net flow is registered in the flow direction. ■ Forward flow total Only the flow in the forward flow direction is totalized. ■ Reverse flow total Only the flow in the reverse flow direction is totalized (= reverse flow quantity).

Control Totalizer 1 to n	
Navigation	  Expert → Application → Totalizer 1 to n → Control Tot. 1 to n (0912–1 to n)
Prerequisite	One of the following options is selected in the Assign process variable parameter (→  155) Totalizer 1 to n submenu: Volume flow
Description	Use this function to select the control of totalizer value 1-3.

- Selection
- Totalize

■ Reset + hold

■ Preset + hold

■ Reset + totalize

■ Preset + totalize

■ Hold

Factory setting

Totalize

Additional information

Selection

Options	Description
Totalize	The totalizer is started or continues running.
Reset + hold	The totaling process is stopped and the totalizer is reset to 0.
Preset + hold	The totaling process is stopped and the totalizer is set to its defined start value from the Preset value parameter.
Reset + totalize	The totalizer is reset to 0 and the totaling process is restarted.
Preset + totalize	The totalizer is set to the defined start value from the Preset value parameter and the totaling process is restarted.
Hold	Totalizing is stopped.

Preset value 1 to n

Navigation

 Expert → Application → Totalizer 1 to n → Preset value 1 to n (0913–1 to n)

Prerequisite

One of the following options is selected in the **Assign process variable** parameter (→  155)**Totalizer 1 to n** submenu:
Volume flow

Description

Use this function to enter a start value for the Totalizer 1 to n.

User entry

Signed floating-point number

Factory setting

0 l

Additional information

User entry

 The unit of the selected process variable is specified for the totalizer in the **Unit totalizer** parameter (→  156).

Example

This configuration is suitable for applications such as iterative filling processes with a fixed batch quantity.

Failure mode

Navigation

 Expert → Application → Totalizer 1 to n → Failure mode (0901–1 to n)

Description

Use this function to select how a totalizer behaves in the event of a device alarm.

Selection	■ Stop ■ Actual value ■ Last valid value
Factory setting	Stop
Additional information	<i>Description</i>  This setting does not affect the failsafe mode of other totalizers and the outputs. This is specified in separate parameters. <i>Selection</i> ■ Stop The totalizer is stopped in the event of a device alarm. ■ Actual value The totalizer continues to count based on the actual measured value; the device alarm is ignored. ■ Last valid value The totalizer continues to count based on the last valid measured value before the device alarm occurred.

3.7.2 "Custody transfer" submenu

-  Only available for Promag H.
-  For detailed information on the parameter descriptions for custody transfer measurement, see the Special Documentation for the device →  7

Navigation   Expert → Application → Custody transfer

► Custody transfer

3.8 "Diagnostics" submenu

Navigation   Expert → Diagnostics

► Diagnostics

Actual diagnostics (0691)

Previous diagnostics (0690)

Operating time from restart (0653)

Operating time (0652)

► Diagnostic list

► Event logbook

→  160

→  161

→  162

→  162

→  162

→  167

▶ Custody transfer logbook	→ 169
▶ Device information	→ 169
▶ Mainboard module	→ 173
▶ Sensor electronic module (ISEM)	→ 173
▶ I/O module 1	→ 174
▶ I/O module 2	→ 174
▶ I/O module 3	→ 175
▶ Display module	→ 176
▶ Min/max values	→ 184
▶ Data logging	→ 176
▶ Heartbeat	→ 187
▶ Simulation	→ 187

Actual diagnostics

Navigation	Expert → Diagnostics → Actual diagnos. (0691)
Prerequisite	A diagnostic event has occurred.
Description	Displays the current diagnostic message. If two or more messages occur simultaneously, the message with the highest priority is shown on the display.
User interface	Symbol for diagnostic behavior, diagnostic code and short message.
Additional information	<i>Display</i> Additional pending diagnostic messages can be viewed in the Diagnostic list submenu (→ 162). Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the key. <i>Example</i> For the display format: ⊗F271 Main electronic failure

Timestamp

Navigation	 Expert → Diagnostics → Timestamp
Description	Displays the operating time when the current diagnostic message occurred.
User interface	Days (d), hours (h), minutes (m) and seconds (s)
Additional information	<i>Display</i>  The diagnostic message can be viewed via the Actual diagnostics parameter (→  160). <i>Example</i> For the display format: 24d12h13m00s

Previous diagnostics

Navigation	  Expert → Diagnostics → Prev.diagnostics (0690)
Prerequisite	Two diagnostic events have already occurred.
Description	Displays the diagnostic message that occurred before the current message.
User interface	Symbol for diagnostic behavior, diagnostic code and short message.
Additional information	<i>Display</i>  Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the  key. <i>Example</i> For the display format: ⊗F271 Main electronic failure

Timestamp

Navigation	 Expert → Diagnostics → Timestamp
Description	Displays the operating time when the last diagnostic message before the current message occurred.
User interface	Days (d), hours (h), minutes (m) and seconds (s)

Additional information	Display  The diagnostic message can be viewed via the Previous diagnostics parameter (→  161). Example For the display format: 24d12h13m00s
------------------------	--

Operating time from restart

Navigation	  Expert → Diagnostics → Time fr. restart (0653)
Description	Use this function to display the time the device has been in operation since the last device restart.
User interface	Days (d), hours (h), minutes (m) and seconds (s)

Operating time

Navigation	  Expert → Diagnostics → Operating time (0652)
Description	Use this function to display the length of time the device has been in operation.
User interface	Days (d), hours (h), minutes (m) and seconds (s)
Additional information	User interface The maximum number of days is 9999, which is equivalent to 27 years.

3.8.1 "Diagnostic list" submenu

Navigation

  Expert → Diagnostics → Diagnostic list

► Diagnostic list	
Diagnostics 1 (0692)	→  163
Diagnostics 2 (0693)	→  163
Diagnostics 3 (0694)	→  164
Diagnostics 4 (0695)	→  165
Diagnostics 5 (0696)	→  166

Diagnostics 1

Navigation	 Expert → Diagnostics → Diagnostic list → Diagnostics 1 (0692)
Description	Displays the current diagnostics message with the highest priority.
User interface	Symbol for diagnostic behavior, diagnostic code and short message.
Additional information	<i>Display</i>  Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the  key. <i>Examples</i> For the display format: ■  F271 Main electronic failure ■  F276 I/O module failure

Timestamp

Navigation	 Expert → Diagnostics → Diagnostic list → Timestamp
Description	Displays the operating time when the diagnostic message with the highest priority occurred.
User interface	Days (d), hours (h), minutes (m) and seconds (s)
Additional information	<i>Display</i>  The diagnostic message can be viewed via the Diagnostics 1 parameter (→  163). <i>Example</i> For the display format: 24d12h13m00s

Diagnostics 2

Navigation	 Expert → Diagnostics → Diagnostic list → Diagnostics 2 (0693)
Description	Displays the current diagnostics message with the second-highest priority.
User interface	Symbol for diagnostic behavior, diagnostic code and short message.

Additional information*Display*

 Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the  key.

Examples

For the display format:

-  F271 Main electronic failure
-  F276 I/O module failure

Timestamp**Navigation**

 Expert → Diagnostics → Diagnostic list → Timestamp

Description

Displays the operating time when the diagnostic message with the second-highest priority occurred.

User interface

Days (d), hours (h), minutes (m) and seconds (s)

Additional information*Display*

 The diagnostic message can be viewed via the **Diagnostics 2** parameter (→  163).

Example

For the display format:

24d12h13m00s

Diagnostics 3**Navigation**

  Expert → Diagnostics → Diagnostic list → Diagnostics 3 (0694)

Description

Displays the current diagnostics message with the third-highest priority.

User interface

Symbol for diagnostic behavior, diagnostic code and short message.

Additional information*Display*

 Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the  key.

Examples

For the display format:

-  F271 Main electronic failure
-  F276 I/O module failure

Timestamp

Navigation	 Expert → Diagnostics → Diagnostic list → Timestamp
Description	Displays the operating time when the diagnostic message with the third-highest priority occurred.
User interface	Days (d), hours (h), minutes (m) and seconds (s)
Additional information	<i>Display</i>  The diagnostic message can be viewed via the Diagnostics 3 parameter (→  164). <i>Example</i> For the display format: 24d12h13m00s

Diagnostics 4

Navigation	  Expert → Diagnostics → Diagnostic list → Diagnostics 4 (0695)
Description	Displays the current diagnostics message with the fourth-highest priority.
User interface	Symbol for diagnostic behavior, diagnostic code and short message.
Additional information	<i>Display</i>  Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the  key. <i>Examples</i> For the display format: ■  F271 Main electronic failure ■  F276 I/O module failure

Timestamp

Navigation	 Expert → Diagnostics → Diagnostic list → Timestamp
Description	Displays the operating time when the diagnostic message with the fourth-highest priority occurred.
User interface	Days (d), hours (h), minutes (m) and seconds (s)

Additional information*Display*

The diagnostic message can be viewed via the **Diagnostics 4** parameter (→ 165).

Example

For the display format:
24d12h13m00s

Diagnostics 5**Navigation**

Expert → Diagnostics → Diagnostic list → Diagnostics 5 (0696)

Description

Displays the current diagnostics message with the fifth-highest priority.

User interface

Symbol for diagnostic behavior, diagnostic code and short message.

Additional information*Display*

Via the local display: the time stamp and corrective measures referring to the cause of the diagnostic message can be accessed via the key.

Examples

For the display format:

- F271 Main electronic failure
- F276 I/O module failure

Timestamp**Navigation**

Expert → Diagnostics → Diagnostic list → Timestamp

Description

Displays the operating time when the diagnostic message with the fifth-highest priority occurred.

User interface

Days (d), hours (h), minutes (m) and seconds (s)

Additional information*Display*

The diagnostic message can be viewed via the **Diagnostics 5** parameter (→ 166).

Example

For the display format:
24d12h13m00s

3.8.2 "Event logbook" submenu

Navigation  Expert → Diagnostics → Event logbook

► Event logbook

Filter options (0705) →  167

► Event list →  168

Filter options

Navigation	 Expert → Diagnostics → Event logbook → Filter options (0705)
Description	Use this function to select the category whose event messages are displayed in the event list of the local display.
Selection	■ All ■ Failure (F) ■ Function check (C) ■ Out of specification (S) ■ Maintenance required (M) ■ Information (I)
Factory setting	All
Additional information	Description  The status signals are categorized in accordance with VDI/VDE 2650 and NAMUR Recommendation NE 107: ■ F = Failure ■ C = Function Check ■ S = Out of Specification ■ M = Maintenance Required

Filter options

Navigation	 Expert → Diagnostics → Event logbook → Filter options
Description	Use this function to select the category whose event messages are displayed in the event list of the operating tool.
Selection	■ All ■ Failure (F) ■ Function check (C) ■ Out of specification (S) ■ Maintenance required (M) ■ Information (I)

Factory settingAll

Additional informationDescription

-  The status signals are categorized in accordance with VDI/VDE 2650 and NAMUR Recommendation NE 107:
- F = Failure
 - C = Function Check
 - S = Out of Specification
 - M = Maintenance Required

"Event list" submenu

-  The **Event list** submenu is only displayed if operating via the local display.
- If operating via the FieldCare operating tool, the event list can be read out with a separate FieldCare module.
- If operating via the Web browser, the event messages can be found directly in the **Event logbook** submenu.

Navigation Expert → Diagnostics → Event logbook → Event list



Event list

Navigation Expert → Diagnostics → Event logbook → Event list

DescriptionDisplays the history of event messages of the category selected in the **Filter options** parameter (→  167).

- User interface
- For a "Category I" event message
Information event, short message, symbol for event recording and operating time when error occurred
 - For a "Category F, C, S, M" event message (status signal)
Diagnostics code, short message, symbol for event recording and operating time when error occurred

Additional informationDescription

A maximum of 20 event messages are displayed in chronological order.

If the **Extended HistoROM** application package (order option) is enabled in the device, the event list can contain up to 100 entries .

The following symbols indicate whether an event has occurred or has ended:

- ↻: Occurrence of the event
- ⌂: End of the event

Examples

For the display format:

- I1091 Configuration modified
 ↻ 24d12h13m00s
- ⌂F271 Main electronic failure
 ↻ 01d04h12min30s

HistoROM

A HistoROM is a "non-volatile" device memory in the form of an EEPROM.

3.8.3 "Custody transfer logbook" submenu

-  Only available for Promag H.
-  For detailed information on the parameter descriptions for custody transfer measurement, see the Special Documentation for the device →  7

Navigation   Expert → Diagnostics → Cust.transf.log.

▶ Custody transfer logbook

3.8.4 "Device information" submenu

Navigation   Expert → Diagnostics → Device info

▶ Device information

Device tag

Serial number

Firmware version

Device name

Order code

Extended order code 1

Extended order code 2

Extended order code 3

ENP version

→  170

→  170

→  170

→  171

→  171

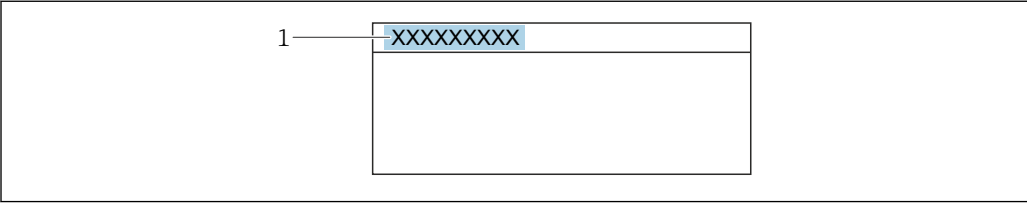
→  171

→  172

→  172

→  172

Device tag

Navigation	  Expert → Diagnostics → Device info → Device tag (0011)
Description	Displays a unique name for the measuring point so it can be identified quickly within the plant. The name is displayed in the header.
User interface	Max. 32 characters, such as letters, numbers or special characters (e.g. @, %, /).
Factory setting	Promag 300 500MB
Additional information	<i>User interface</i>  A0029422 1 Position of the header text on the display The number of characters displayed depends on the characters used.

Serial number

Navigation	  Expert → Diagnostics → Device info → Serial number (0009)
Description	Displays the serial number of the measuring device.  The number can be found on the nameplate of the sensor and transmitter.
User interface	A maximum of 11-digit character string comprising letters and numbers.
Additional information	<i>Description</i>  Uses of the serial number ▪ To identify the measuring device quickly, e.g. when contacting Endress+Hauser. ▪ To obtain specific information on the measuring device using the Device Viewer: www.endress.com/deviceviewer

Firmware version

Navigation	  Expert → Diagnostics → Device info → Firmware version (0010)
Description	Displays the device firmware version installed.
User interface	Character string in the format xx.yy.zz

Additional information*Display*

The Firmware version is also located:

- On the title page of the Operating instructions
- On the transmitter nameplate

Device name**Navigation**

Expert → Diagnostics → Device info → Device name (0020)

Description

Displays the name of the transmitter. It can also be found on the nameplate of the transmitter.

User interface

Promag300/500

Order code**Navigation**

Expert → Diagnostics → Device info → Order code (0008)

Description

Displays the device order code.

User interface

Character string composed of letters, numbers and certain punctuation marks (e.g. /).

Additional information*Description*

The order code can be found on the nameplate of the sensor and transmitter in the "Order code" field.

The order code is generated from the extended order code through a process of reversible transformation. The extended order code indicates the attributes for all the device features in the product structure. The device features are not directly readable from the order code.

**Uses of the order code**

- To order an identical spare device.
- To identify the device quickly and easily, e.g. when contacting Endress+Hauser.

Extended order code 1**Navigation**

Expert → Diagnostics → Device info → Ext. order cd. 1 (0023)

Description

Displays the first part of the extended order code.

On account of length restrictions, the extended order code is split into a maximum of 3 parameters.

User interface

Character string

Additional information	<i>Description</i> The extended order code indicates the version of all the features of the product structure for the measuring device and thus uniquely identifies the measuring device.  The extended order code can also be found on the nameplate of the sensor and transmitter in the "Ext. ord. cd." field.
-------------------------------	--

Extended order code 2



Navigation	  Expert → Diagnostics → Device info → Ext. order cd. 2 (0021)
Description	Displays the second part of the extended order code.
User interface	Character string
Additional information	For additional information, see Extended order code 1 parameter (→  171)

Extended order code 3



Navigation	  Expert → Diagnostics → Device info → Ext. order cd. 3 (0022)
Description	Displays the third part of the extended order code.
User interface	Character string
Additional information	For additional information, see Extended order code 1 parameter (→  171)

ENP version

Navigation	  Expert → Diagnostics → Device info → ENP version (0012)
Description	Displays the version of the electronic nameplate.
User interface	Character string
Factory setting	2.02.00
Additional information	<i>Description</i> This electronic nameplate stores a data record for device identification that includes more data than the nameplates attached to the outside of the device.

3.8.5 "Mainboard module" submenu

Navigation  Expert → Diagnostics → Mainboard module

► Mainboard module

Software revision (0072)

→  173

Software revision

Navigation	 Expert → Diagnostics → Mainboard module → Software rev. (0072)
Description	Use this function to display the software revision of the module.
User interface	Positive integer

3.8.6 "Sensor electronic module (ISEM)" submenu

Navigation  Expert → Diagnostics → Sens. electronic

► Sensor electronic module (ISEM)

Software revision (0072)

→  173

Software revision

Navigation	 Expert → Diagnostics → Sens. electronic → Software rev. (0072)
Description	Use this function to display the software revision of the module.
User interface	Positive integer

3.8.7 "I/O module 1" submenu

Navigation  Expert → Diagnostics → I/O module 1

► I/O module 1

I/O module 1 terminal numbers
(3902-1)

→  174

Software revision (0072)

→  174

I/O module 1 terminal numbers

Navigation	 Expert → Diagnostics → I/O module 1 → I/O 1 terminals (3902-1)
Description	Displays the terminal numbers used by the I/O module.
User interface	■ Not used ■ 26-27 (I/O 1) ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)

Software revision

Navigation	 Expert → Diagnostics → I/O module 1 → Software rev. (0072)
Description	Use this function to display the software revision of the module.
User interface	Positive integer

3.8.8 "I/O module 2" submenu

Navigation  Expert → Diagnostics → I/O module 2

► I/O module 2

I/O module 2 terminal numbers

→  175

Software revision

→  175

I/O module 2 terminal numbers

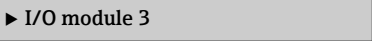
Navigation	 Expert → Diagnostics → I/O module 2 → I/O 2 terminals (3902-2)
Description	Displays the terminal numbers used by the I/O module.
User interface	■ Not used ■ 26-27 (I/O 1) ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)

Software revision

Navigation	 Expert → Diagnostics → I/O module 2 → Software rev. (0072)
Description	Use this function to display the software revision of the module.
User interface	Positive integer

3.8.9 "I/O module 3" submenu

Navigation  Expert → Diagnostics → I/O module 3

	
I/O module 3 terminal numbers	→  175
Software revision	→  176

I/O module 3 terminal numbers

Navigation	 Expert → Diagnostics → I/O module 3 → I/O 3 terminals (3902-3)
Description	Displays the terminal numbers used by the I/O module.
User interface	■ Not used ■ 26-27 (I/O 1) ■ 24-25 (I/O 2) ■ 22-23 (I/O 3)

Software revision

Navigation	  Expert → Diagnostics → I/O module 3 → Software rev. (0072)
Description	Use this function to display the software revision of the module.
User interface	Positive integer

3.8.10 "Display module" submenu

Navigation

  Expert → Diagnostics → Display module

► Display module

Software revision (0072)→  176

Software revision

Navigation	  Expert → Diagnostics → Display module → Software rev. (0072)
Description	Use this function to display the software revision of the module.
User interface	Positive integer

3.8.11 "Data logging" submenu

Navigation

  Expert → Diagnostics → Data logging

► Data logging

Assign channel 1 (0851)→  177

Assign channel 2 (0852)→  178

Assign channel 3 (0853)→  178

Assign channel 4 (0854)→  179

Logging interval (0856)→  179

Clear logging data (0855)	→  180
Data logging (0860)	→  180
Logging delay (0859)	→  180
Data logging control (0857)	→  181
Data logging status (0858)	→  181
Entire logging duration (0861)	→  182
► Display channel 1	→  182
► Display channel 2	→  183
► Display channel 3	→  183
► Display channel 4	→  184

Assign channel 1

Navigation	 Expert → Diagnostics → Data logging → Assign chan. 1 (0851)
Prerequisite	The Extended HistoROM application package is available.  The software options currently enabled are displayed in the Software option overview parameter (→  41).
Description	Use this function to select a process variable for the data logging channel.
Selection	■ Off ■ Volume flow ■ Mass flow ■ Corrected volume flow ■ Flow velocity ■ Conductivity * ■ Corrected conductivity * ■ Current output 1 ■ Current output 2 * ■ Current output 3 * ■ Temperature * ■ Electronic temperature
Factory setting	Off

* Visibility depends on order options or device settings

Additional information	<i>Description</i> A total of 1000 measured values can be logged. This means: ■ 1000 data points if 1 logging channel is used ■ 500 data points if 2 logging channels are used ■ 333 data points if 3 logging channels are used ■ 250 data points if 4 logging channels are used Once the maximum number of data points is reached, the oldest data points in the data log are cyclically overwritten in such a way that the last 1000, 500, 333 or 250 measured values are always in the log (ring memory principle).  The log contents are cleared if the option selected is changed.
-------------------------------	---

Assign channel 2

Navigation	 Expert → Diagnostics → Data logging → Assign chan. 2 (0852)
Prerequisite	The Extended HistoROM application package is available.  The software options currently enabled are displayed in the Software option overview parameter (→  41).
Description	Options for the assignment of a process variable to the data logging channel.
Selection	Picklist, see Assign channel 1 parameter (→  177)
Factory setting	Off

Assign channel 3

Navigation	 Expert → Diagnostics → Data logging → Assign chan. 3 (0853)
Prerequisite	The Extended HistoROM application package is available.  The software options currently enabled are displayed in the Software option overview parameter (→  41).
Description	Options for the assignment of a process variable to the data logging channel.
Selection	Picklist, see Assign channel 1 parameter (→  177)
Factory setting	Off

Assign channel 4

Navigation	Expert → Diagnostics → Data logging → Assign chan. 4 (0854)
Prerequisite	The Extended HistoROM application package is available. The software options currently enabled are displayed in the Software option overview parameter (→ 41).
Description	Options for the assignment of a process variable to the data logging channel.
Selection	Picklist, see Assign channel 1 parameter (→ 177)
Factory setting	Off

Logging interval

Navigation	Expert → Diagnostics → Data logging → Logging interval (0856)
Prerequisite	The Extended HistoROM application package is available. The software options currently enabled are displayed in the Software option overview parameter (→ 41).
Description	Use this function to enter the logging interval $T_{\log}$ for data logging.
User entry	0.1 to 999.0 s
Factory setting	1.0 s
Additional information	<i>Description</i> This defines the interval between the individual data points in the data log, and thus the maximum loggable process time $T_{\log}$: ■ If 1 logging channel is used: $T_{\log} = 1000 \times t_{\log}$ ■ If 2 logging channels are used: $T_{\log} = 500 \times t_{\log}$ ■ If 3 logging channels are used: $T_{\log} = 333 \times t_{\log}$ ■ If 4 logging channels are used: $T_{\log} = 250 \times t_{\log}$ Once this time elapses, the oldest data points in the data log are cyclically overwritten such that a time of $T_{\log}$ always remains in the memory (ring memory principle). The log contents are cleared if the length of the logging interval is changed. <i>Example</i> If 1 logging channel is used: ■ $T_{\log} = 1000 \times 1 \text{ s} = 1\,000 \text{ s} \approx 15 \text{ min}$ ■ $T_{\log} = 1000 \times 10 \text{ s} = 10\,000 \text{ s} \approx 3 \text{ h}$ ■ $T_{\log} = 1000 \times 80 \text{ s} = 80\,000 \text{ s} \approx 1 \text{ d}$ ■ $T_{\log} = 1000 \times 3\,600 \text{ s} = 3\,600\,000 \text{ s} \approx 41 \text{ d}$

Clear logging data



Navigation	Expert → Diagnostics → Data logging → Clear logging (0855)
Prerequisite	The Extended HistoROM application package is available. The software options currently enabled are displayed in the Software option overview parameter (→ 41).
Description	Use this function to clear the entire logging data.
Selection	■ Cancel ■ Clear data
Factory setting	Cancel
Additional information	<i>Selection</i> ■ Cancel The data is not cleared. All the data is retained. ■ Clear data The logging data is cleared. The logging process starts from the beginning.

Data logging

Navigation	Expert → Diagnostics → Data logging → Data logging (0860)
Description	Use this function to select the data logging method.
Selection	■ Overwriting ■ Not overwriting
Factory setting	Overwriting
Additional information	<i>Selection</i> ■ Overwriting The device memory applies the FIFO principle. ■ Not overwriting Data logging is canceled if the measured value memory is full (single shot).

Logging delay

Navigation	Expert → Diagnostics → Data logging → Logging delay (0859)
Prerequisite	In the Data logging parameter (→ 180), the Not overwriting option is selected.
Description	Use this function to enter the time delay for measured value logging.
User entry	0 to 999 h

Factory setting 0 h

Additional information *Description*

Once measured value logging has been started with the **Data logging control** parameter (→  181), the device does not save any data for the duration of the time delay entered.

Data logging control

Navigation   Expert → Diagnostics → Data logging → Data log.control (0857)

Prerequisite In the **Data logging** parameter (→  180), the **Not overwriting** option is selected.

Description Use this function to start and stop measured value logging.

Selection

- None
- Delete + start
- Stop

Factory setting None

Additional information *Selection*

- None
Initial measured value logging status.
- Delete + start
All the measured values recorded for all the channels are deleted and measured value logging starts again.
- Stop
Measured value logging is stopped.

Data logging status

Navigation   Expert → Diagnostics → Data logging → Data log. status (0858)

Prerequisite In the **Data logging** parameter (→  180), the **Not overwriting** option is selected.

Description Displays the measured value logging status.

User interface

- Done
- Delay active
- Active
- Stopped

Factory setting Done

Additional information	Selection ■ Done Measured value logging has been performed and completed successfully. ■ Delay active Measured value logging has been started but the logging interval has not yet elapsed. ■ Active The logging interval has elapsed and measured value logging is active. ■ Stopped Measured value logging is stopped.
------------------------	---

Entire logging duration

Navigation	Expert → Diagnostics → Data logging → Logging duration (0861)
Prerequisite	In the Data logging parameter (→ 180), the Not overwriting option is selected.
Description	Displays the total logging duration.
User interface	Positive floating-point number
Factory setting	0 s

"Display channel 1" submenu

Navigation

Expert → Diagnostics → Data logging → Displ.channel 1

► Display channel 1

Display channel 1 → 182

Display channel 1

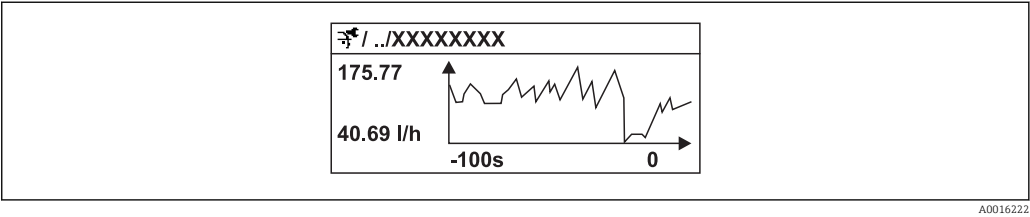
Navigation	Expert → Diagnostics → Data logging → Displ.channel 1
Prerequisite	The Extended HistoROM application package is available. The software options currently enabled are displayed in the Software option overview parameter (→ 41). One of the following options is selected in the Assign channel 1 parameter (→ 177): ■ Conductivity[*] ■ Corrected conductivity[*] ■ Temperature[*]

*

Visibility depends on order options or device settings

Description Displays the measured value trend for the logging channel in the form of a chart.

Additional information *Description*



 10 Chart of a measured value trend

- x-axis: depending on the number of channels selected displays 250 to 1000 measured values of a process variable.
- y-axis: displays the approximate measured value span and constantly adapts this to the ongoing measurement.

"Display channel 2" submenu

Navigation  Expert → Diagnostics → Data logging → Displ.channel 2

► Display channel 2

Display channel 2 →  183

Display channel 2

Navigation  Expert → Diagnostics → Data logging → Displ.channel 2

Prerequisite A process variable is defined in the **Assign channel 2** parameter.

Description See the **Display channel 1** parameter →  182

"Display channel 3" submenu

Navigation  Expert → Diagnostics → Data logging → Displ.channel 3

► Display channel 3

Display channel 3 →  184

Display channel 3

Navigation	 Expert → Diagnostics → Data logging → Displ.channel 3
Prerequisite	A process variable is defined in the Assign channel 3 parameter.
Description	See the Display channel 1 parameter →  182

"Display channel 4" submenu

Navigation

 Expert → Diagnostics → Data logging → Displ.channel 4

► Display channel 4

Display channel 4 →  184

Display channel 4

Navigation	 Expert → Diagnostics → Data logging → Displ.channel 4
Prerequisite	A process variable is defined in the Assign channel 4 parameter.
Description	See the Display channel 1 parameter →  182

3.8.12 "Min/max values" submenu

Navigation

  Expert → Diagnostics → Min/max val.

► Min/max values

Reset min/max values (6541) →  185

► Main electronic temperature →  185

► Temperature →  186

Reset min/max values 	
Navigation	  Expert → Diagnostics → Min/max val. → Reset min/max (6541)
Description	Use this function to select measured variables whose minimum, maximum and average measured values are to be reset.
Selection	Cancel
Factory setting	Cancel

"Main electronic temperature" submenu

Navigation   Expert → Diagnostics → Min/max val. → Main elect.temp.

► Main electronic temperature

Minimum value (6547)

→  185

Maximum value (6545)

→  185

Minimum value	
Navigation	  Expert → Diagnostics → Min/max val. → Main elect.temp. → Minimum value (6547)
Description	Displays the lowest previously measured temperature value of the main electronics module.
User interface	Signed floating-point number
Additional information	<i>Dependency</i>  The unit is taken from the Temperature unit parameter (→  56)

Maximum value	
Navigation	  Expert → Diagnostics → Min/max val. → Main elect.temp. → Maximum value (6545)
Description	Displays the highest previously measured temperature value of the main electronics module.
User interface	Signed floating-point number

Additional information

Dependency

 The unit is taken from the **Temperature unit** parameter (→  56)

"Temperature" submenu

Navigation  Expert → Diagnostics → Min/max val. → Temperature

► Temperature

Minimum value (6681)

→  186

Maximum value (6680)

→  186

Minimum value

Navigation

 Expert → Diagnostics → Min/max val. → Temperature → Minimum value (6681)

Prerequisite

One of the following conditions is met:

- Order code for "Sensor option", option **CI** "Medium temperature measurement" or
- The temperature is read into the flowmeter from an external device.

Description

Displays the lowest previously measured medium temperature value.

User interface

Signed floating-point number

Additional information

Dependency

 The unit is taken from the **Temperature unit** parameter (→  56)

Maximum value

Navigation

 Expert → Diagnostics → Min/max val. → Temperature → Maximum value (6680)

Prerequisite

One of the following conditions is met:

- Order code for "Sensor option", option **CI** "Medium temperature measurement" or
- The temperature is read into the flowmeter from an external device.

Description

Displays the highest previously measured medium temperature value.

User interface

Signed floating-point number

Additional information

Dependency

 The unit is taken from the **Temperature unit** parameter (→  56)

3.8.13 "Heartbeat" submenu

 For detailed information on the parameter descriptions for the **Heartbeat Verification+Monitoring** application package, refer to the Special Documentation for the device →  7

Navigation

  Expert → Diagnostics → Heartbeat

▶ Heartbeat

▶ Heartbeat base settings

▶ Performing verification

▶ Verification results

▶ Monitoring results

3.8.14 "Simulation" submenu

Navigation

  Expert → Diagnostics → Simulation

▶ Simulation

Assign simulation process variable (1810)→  188

Process variable value (1811)→  189

Status input simulation (1355)→  189

Input signal level (1356)→  190

Current input 1 to n simulation (1608-1 to n)→  190

Value current input 1 to n (1609-1 to n)→  190

Current output 1 to n simulation (0354-1 to n)→  191

Value current output 1 to n (0355-1 to n)→  191

Frequency output simulation 1 to n (0472-1 to n)	→ 191
Frequency value 1 to n (0473-1 to n)	→ 192
Pulse output simulation 1 to n (0458-1 to n)	→ 192
Pulse value 1 to n (0459-1 to n)	→ 193
Switch output simulation 1 to n (0462-1 to n)	→ 193
Switch status 1 to n (0463-1 to n)	→ 194
Relay output 1 to n simulation (0802-1 to n)	→ 194
Switch status 1 to n (0803-1 to n)	→ 195
Pulse output simulation (0988)	→ 195
Pulse value (0989)	→ 196
Device alarm simulation (0654)	→ 196
Diagnostic event category (0738)	→ 197
Diagnostic event simulation (0737)	→ 197

Assign simulation process variable



Navigation

Expert → Diagnostics → Simulation → Assign proc.var. (1810)

Description

Use this function to select a process variable for the simulation process that is activated. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.

Selection

- Off
- Volume flow
- Mass flow
- Corrected volume flow
- Conductivity*
- Corrected conductivity*
- Temperature*

Factory setting

Off

* Visibility depends on order options or device settings

Additional information	<i>Description</i>  The simulation value of the process variable selected is defined in the Process variable value parameter (→  189).
Process variable value 	
Navigation	  Expert → Diagnostics → Simulation → Proc. var. value (1811)
Description	Use this function to enter a simulation value for the selected process variable. Subsequent measured value processing and the signal output use this simulation value. In this way, users can verify whether the measuring device has been configured correctly.
User entry	Depends on the process variable selected
Factory setting	0
Additional information	<i>User entry</i>  The unit of the displayed measured value is taken from the System units submenu (→  52).
Status input simulation 	
Navigation	  Expert → Diagnostics → Simulation → Status inp. sim. (1355)
Description	Use this function to switch simulation of the status input on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.
Selection	■ Off ■ On
Factory setting	Off
Additional information	<i>Description</i>  The desired simulation value is defined in the Input signal level parameter (→  190). <i>Selection</i> ■ Off Simulation for the status input is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ On Simulation for the status input is active.

Input signal level 	
Navigation	  Expert → Diagnostics → Simulation → Signal level (1356)
Prerequisite	In the Status input simulation parameter (→  189), the On option is selected.
Description	Use this function to select the signal level for the simulation of the status input. In this way, users can verify the correct configuration of the status input and the correct function of upstream feed-in units.
Selection	■ High ■ Low
Current input 1 to n simulation 	
Navigation	  Expert → Diagnostics → Simulation → Curr.inp 1 to n sim. (1608–1 to n)
Description	Option for switching simulation of the current input on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.  The desired simulation value is defined in the Value current input 1 to n parameter.
Selection	■ Off ■ On
Factory setting	Off
Additional information	<i>Selection</i> ■ Off Current simulation is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ On Current simulation is active.
Value current input 1 to n 	
Navigation	  Expert → Diagnostics → Simulation → Value curr.inp 1 to n (1609–1 to n)
Prerequisite	In the Current input 1 to n simulation parameter, the On option is selected.
Description	Use this function to enter the current value for the simulation. In this way, users can verify the correct configuration of the current input and the correct function of upstream feed-in units.
User entry	0 to 22.5 mA

Current output 1 to n simulation 	
Navigation	  Expert → Diagnostics → Simulation → Curr.out. 1 to n sim. (0354–1 to n)
Description	Use this function to switch simulation of the current output on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.
Selection	■ Off ■ On
Factory setting	Off
Additional information	<i>Description</i>  The desired simulation value is defined in the Value current output 1 to n parameter. <i>Selection</i> ■ Off Current simulation is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ On Current simulation is active.

Value current output 1 to n 	
Navigation	  Expert → Diagnostics → Simulation → Value curr.out 1 to n (0355–1 to n)
Prerequisite	In the Current output 1 to n simulation parameter, the On option is selected.
Description	Use this function to enter a current value for the simulation. In this way, users can verify the correct adjustment of the current output and the correct function of downstream switching units.
User entry	3.59 to 22.5 mA
Additional information	<i>Dependency</i> The input range is dependent on the option selected in the Current span parameter (→  98).

Frequency output simulation 1 to n 	
Navigation	  Expert → Diagnostics → Simulation → FreqOutputSim 1 to n (0472–1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Frequency option is selected.

Description Use this function to switch simulation of the frequency output on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.

Selection

- Off
- On

Factory setting Off

Additional information *Description*

 The desired simulation value is defined in the **Frequency value 1 to n** parameter.

Selection

- Off
Frequency simulation is switched off. The device is in normal measuring mode or another process variable is being simulated.
- On
Frequency simulation is active.

Frequency value 1 to n

Navigation   Expert → Diagnostics → Simulation → Freq value 1 to n (0473–1 to n)

Prerequisite In the **Frequency output simulation 1 to n** parameter, the **On** option is selected.

Description Use this function to enter a frequency value for the simulation. In this way, users can verify the correct adjustment of the frequency output and the correct function of downstream switching units.

User entry 0.0 to 12 500.0 Hz

Pulse output simulation 1 to n

Navigation   Expert → Diagnostics → Simulation → Puls.outp.sim. 1 to n (0458–1 to n)

Prerequisite In the **Operating mode** parameter (→  112), the **Pulse** option is selected.

Description Use this function to switch simulation of the pulse output on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.

Selection

- Off
- Fixed value
- Down-counting value

Factory setting Off

Additional information	<i>Description</i>  The desired simulation value is defined in the Pulse value 1 to n parameter. <i>Selection</i> ■ Off Pulse simulation is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ Fixed value Pulses are continuously output with the pulse width specified in the Pulse width parameter (→  115). ■ Down-counting value The pulses specified in the Pulse value parameter (→  193) are output.
Pulse value 1 to n 	
Navigation	 Expert → Diagnostics → Simulation → Pulse value 1 to n (0459–1 to n)
Prerequisite	In the Pulse output simulation 1 to n parameter, the Down-counting value option is selected.
Description	Use this function to enter a pulse value for the simulation. In this way, users can verify the correct adjustment of the pulse output and the correct function of downstream switching units.
User entry	0 to 65 535
Switch output simulation 1 to n 	
Navigation	 Expert → Diagnostics → Simulation → Switch sim. 1 to n (0462–1 to n)
Prerequisite	In the Operating mode parameter (→  112), the Switch option is selected.
Description	Use this function to switch simulation of the switch output on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.
Selection	■ Off ■ On
Factory setting	Off

Additional information *Description*

The desired simulation value is defined in the **Switch status 1 to n** parameter.

Selection

- Off
Switch simulation is switched off. The device is in normal measuring mode or another process variable is being simulated.
- On
Switch simulation is active.

Switch status 1 to n**Navigation**

Expert → Diagnostics → Simulation → Switch status 1 to n (0463–1 to n)

Description

Use this function to select a switch value for the simulation. In this way, users can verify the correct adjustment of the switch output and the correct function of downstream switching units.

Selection

- Open
- Closed

Additional information*Selection*

- Open
Switch simulation is switched off. The device is in normal measuring mode or another process variable is being simulated.
- Closed
Switch simulation is active.

Relay output 1 to n simulation**Navigation**

Expert → Diagnostics → Simulation → Relay out. 1 to n sim (0802–1 to n)

Description

Use this function to switch simulation of the relay output on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.

Selection

- Off
- On

Factory setting

Off

Additional information	<i>Description</i>  The desired simulation value is defined in the Switch status 1 to n parameter. <i>Selection</i> ■ Off Relay simulation is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ On Relay simulation is active.
Switch status 1 to n 	
Navigation	 Expert → Diagnostics → Simulation → Switch status 1 to n (0803–1 to n)
Prerequisite	The On option is selected in the Switch output simulation 1 to n parameter parameter.
Description	Use this function to select a relay value for the simulation. In this way, users can verify the correct adjustment of the relay output and the correct function of downstream switching units.
Selection	■ Open ■ Closed
Additional information	<i>Selection</i> ■ Open Relay simulation is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ Closed Relay simulation is active.
Pulse output simulation 	
Navigation	 Expert → Diagnostics → Simulation → Puls.outp.sim. (0988)
Description	Use this function to switch simulation of the double pulse output on and off. The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.
Selection	■ Off ■ Fixed value ■ Down-counting value
Factory setting	Off

Additional information	Description  The desired simulation value is defined in the Pulse value parameter (→  196).
	Selection ■ Off Simulation of the double pulse output is switched off. The device is in normal measuring mode or another process variable is being simulated. ■ Fixed value Pulses are continuously output with the pulse width specified in the Pulse width parameter (→  139). ■ Down-counting value The pulses specified in the Pulse value parameter (→  196) are output.
Pulse value	
Navigation	  Expert → Diagnostics → Simulation → Pulse value (0989)
Prerequisite	In the Pulse output simulation parameter (→  195), the Down-counting value option is selected.
Description	Use this function to enter a pulse value for simulation of the double pulse output. In this way, users can verify the correct adjustment of the double pulse output and the correct function of downstream switching units.
User entry	0 to 65 535
Device alarm simulation	
Navigation	  Expert → Diagnostics → Simulation → Dev. alarm sim. (0654)
Description	Use this function to switch the device alarm on and off.
Selection	■ Off ■ On
Factory setting	Off
Additional information	Description The display alternates between the measured value and a diagnostic message of the "Function check" category (C) while simulation is in progress.

Diagnostic event category


Navigation	Expert → Diagnostics → Simulation → Event category (0738)
Description	Use this function to select the category of the diagnostic events that are displayed for the simulation in the Diagnostic event simulation parameter (→ 197).
Selection	■ Sensor ■ Electronics ■ Configuration ■ Process
Factory setting	Process

Diagnostic event simulation


Navigation	Expert → Diagnostics → Simulation → Diag. event sim. (0737)
Description	Use this function to select a diagnostic event for the simulation process that is activated.
Selection	■ Off ■ Diagnostic event picklist (depends on the category selected)
Factory setting	Off
Additional information	<i>Description</i> For the simulation, you can choose from the diagnostic events of the category selected in the Diagnostic event category parameter (→ 197).

4 Country-specific factory settings

4.1 SI units



Not valid for USA and Canada.

4.1.1 System units

Volume flow	l/h
Volume	m ³
Conductivity	µS/cm
Temperature	°C
Mass flow	kg/h
Mass	kg
Density	kg/l
Corrected volume flow	NI/h
Corrected volume	Nm ³

4.1.2 Full scale values



The factory settings apply to the following parameters:

- 20 mA value (full scale value of the current output)
- 100% bar graph value 1

Nominal diameter [mm]	(v ~ 2.5 m/s) [dm ³ /min]
2	0.5
4	2
8	8
15	25
25	75
32	125
40	200
50	300
65	500
80	750
100	1200
125	1850
150	150 m ³ /h
200	300 m ³ /h
250	500 m ³ /h
300	750 m ³ /h
350	1000 m ³ /h
400	1200 m ³ /h
450	1500 m ³ /h

Nominal diameter [mm]	($v \sim 2.5 \text{ m/s}$) [dm ³ /min]
500	2 000 m ³ /h
600	2 500 m ³ /h

4.1.3 Output current span

Current output 1 to n	4 to 20 mA NAMUR
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4.1.4 Pulse value

Nominal diameter [mm]	($\sim 2 \text{ pulse/s}$) [dm ³]
2	0.005
4	0.025
8	0.1
15	0.2
25	0.5
32	1
40	1.5
50	2.5
65	5
80	5
100	10
125	15
150	0.03 m ³
200	0.05 m ³
250	0.05 m ³
300	0.1 m ³
350	0.1 m ³
400	0.15 m ³
450	0.25 m ³
500	0.25 m ³
600	0.3 m ³

4.1.5 On value low flow cut off

 The switch-on point depends on the type of medium and the nominal diameter.

Nominal diameter [mm]	($v \sim 0.04 \text{ m/s}$) [m ³ /h]
2	0.01
4	0.05
8	0.1
15	0.5
25	1

Nominal diameter [mm]	(v ~ 0.04 m/s) [m³/h]
32	2
40	3
50	5
65	8
80	12
100	20
125	30
150	2.5
200	5
250	7.5
300	10
350	15
400	20
450	25
500	30
600	40

4.2 US units

 Only valid for USA and Canada.

4.2.1 System units

Volume flow	gal/min (us)
Volume	gal (us)
Temperature	°F
Mass flow	lb/min
Mass	lb
Density	lb/ft³
Corrected volume flow	Sft³/h
Corrected volume	Sft³

4.2.2 Full scale values

 The factory settings apply to the following parameters:

- 20 mA value (full scale value of the current output)
- 100% bar graph value 1

Nominal diameter [in]	(v ~ 2.5 m/s) [gal/min]
1/12	0.1
1/8	0.5
3/8	2
1/2	6

Nominal diameter [in]	(v ~ 2.5 m/s) [gal/min]
1	18
1½	50
2	75
3	200
4	300
5	450
6	600
8	1200
10	1500
12	2400
14	3600
15	4800
16	4800
18	6000
20	7500
24	10500

4.2.3 Output current span

Current output 1 to n	4 to 20 mA US
-----------------------	---------------

4.2.4 Pulse value

Nominal diameter [in]	(~ 2 pulse/s) [gal]
1/12	0.001
1/8	0.005
3/8	0.02
1/2	0.1
1	0.2
1½	0.5
2	0.5
3	2
4	2
5	5
6	5
8	10
10	15
12	25
14	30
15	50
16	50
18	50

Nominal diameter [in]	(~ 2 pulse/s) [gal]
20	75
24	100

4.2.5 On value low flow cut off



The switch-on point depends on the type of medium and the nominal diameter.

Nominal diameter [in]	(v ~ 0.04 m/s) [gal/min]
1/12	0.002
1/8	0.008
3/8	0.025
1/2	0.15
1	0.25
1 1/2	0.75
2	1.25
3	2.5
4	4
5	7
6	12
8	15
10	30
12	45
14	60
15	60
16	60
18	90
20	120
24	180

5 Explanation of abbreviated units

5.1 SI units

Process variable	Units	Explanation
Density	g/cm ³ , g/m ³	Gram/volume unit
	kg/dm ³ , kg/l, kg/m ³	Kilogram/volume unit
	SD4°C, SD15°C, SD20°C	Specific density: The specific density is the ratio of the density of the fluid to the density of water at a water temperature of 4 °C (39 °F), 15 °C (59 °F), 20 °C (68 °F).
	SG4°C, SG15°C, SG20°C	Specific gravity: The specific gravity is the ratio of the density of the fluid to the density of water at a water temperature of 4 °C (39 °F), 15 °C (59 °F), 20 °C (68 °F).
Conductivity	μS/mm	Microsiemens/length unit
	nS/cm, μS/cm, mS/cm, S/cm	Nano-, Micro-, Milli-, Siemens/length unit
	μS/m, mS/m, S/m, kS/m, MS/m	Micro-, Milli-, Siemens, Kilo-, Megasiemens/length unit
Mass	g, kg, t	Gram, kilogram, metric ton
Mass flow	g/s, g/min, g/h, g/d	Gram/time unit
	kg/s, kg/min, kg/h, kg/d	Kilogram/time unit
	t/s, t/min, t/h, t/d	Metric ton/time unit
Temperature	°C, K	Celsius, Kelvin
Volume	cm ³ , dm ³ , m ³	Cubic centimeter, cubic decimeter, cubic meter
	ml, l, hl, Ml Mega	Milliliter, liter, hectoliter, megaliter
Time	s, m, h, d, y	Second, minute, hour, day, year

5.2 US units

Process variable	Units	Explanation
Density	lb/ft ³ , lb/gal (us)	Pound/cubic foot, pound/gallon
	lb/bbl (us;liq.), lb/bbl (us;beer), lb/bbl (us;oil), lb/bbl (us;tank)	Pound/volume unit
Mass	oz, lb, STon	Ounce, pound, standard ton
Mass flow	oz/s, oz/min, oz/h, oz/d	Ounce/time unit
	lb/s, lb/min, lb/h, lb/d	Pound/time unit
	STon/s, STon/min, STon/h, STon/d	Standard ton/time unit
Corrected volume	Sft ³ , Sgal (us), Sbbl (us;liq.)	Standard cubic foot, standard gallon, standard barrel
Corrected volume flow	Sft ³ /s, Sft ³ /min, Sft ³ /h, Sft ³ /d	Standard cubic foot/time unit
	Sgal/s (us), Sgal/min (us), Sgal/h (us), Sgal/d (us)	Standard gallon/time unit
	Sbbl/s (us;liq.), Sbbl/min (us;liq.), Sbbl/h (us;liq.), Sbbl/d (us;liq.)	Barrel/time unit (normal liquids)
Temperature	°F, °R	Fahrenheit, Rankine
Volume	af	Acre foot
	ft ³	Cubic foot

Process variable	Units	Explanation
	fl oz (us), gal (us), kgal (us), Mgal (us)	Fluid ounce, gallon, kilogallon, million gallon
	bbl (us;liq.), bbl (us;beer), bbl (us;oil), bbl (us;tank)	Barrel (normal liquids), barrel (beer), barrel (petrochemicals), barrel (filling tanks)
Volume flow	af/s, af/min, af/h, af/d	Acre foot/time unit
	ft ³ /s, ft ³ /min, ft ³ /h, ft ³ /d	Cubic foot/time unit
	fl oz/s (us), fl oz/min (us), fl oz/h (us), fl oz/d (us)	Fluid ounce/time unit
	gal/s (us), gal/min (us), gal/h (us), gal/d (us)	Gallon/time unit
	kgal/s (us), kgal/min (us), kgal/h (us), kgal/d (us)	Kilogallon/time unit
	Mgal/s (us), Mgal/min (us), Mgal/h (us), Mgal/d (us)	Million gallon/time unit
	bbl/s (us;liq.), bbl/min (us;liq.), bbl/h (us;liq.), bbl/d (us;liq.)	Barrel/time unit (normal liquids) Normal liquids: 31.5 gal/bbl
	bbl/s (us;beer), bbl/min (us;beer), bbl/h (us;beer), bbl/d (us;beer)	Barrel /time unit (beer) Beer: 31.0 gal/bbl
	bbl/s (us;oil), bbl/min (us;oil), bbl/h (us;oil), bbl/d (us;oil)	Barrel/time unit (petrochemicals) Petrochemicals: 42.0 gal/bbl
	bbl/s (us;tank), bbl/min (us;tank), bbl/h (us;tank), bbl/d (us;tank)	Barrel/time unit (filling tank) Filling tanks: 55.0 gal/bbl
Time	s, m, h, d, y	Second, minute, hour, day, year
	am, pm	Ante meridiem (before midday), post meridiem (after midday)

5.3 Imperial units

Process variable	Units	Explanation
Density	lb/gal (imp), lb/bbl (imp;beer), lb/bbl (imp;oil)	Pound/volume unit
Corrected volume	Sgal (imp)	Standard gallon
Corrected volume flow	Sgal/s (imp), Sgal/min (imp), Sgal/h (imp), Sgal/d (imp)	Standard gallon/time unit
Volume	gal (imp), Mgal (imp)	Gallon, mega gallon
	bbl (imp;beer), bbl (imp;oil)	Barrel (beer), barrel (petrochemicals)
Volume flow	gal/s (imp), gal/min (imp), gal/h (imp), gal/d (imp)	Gallon/time unit
	Mgal/s (imp), Mgal/min (imp), Mgal/h (imp), Mgal/d (imp)	Mega gallon/time unit
	bbl/s (imp;beer), bbl/min (imp;beer), bbl/h (imp;beer), bbl/d (imp;beer)	Barrel /time unit (beer) Beer: 36.0 gal/bbl
	bbl/s (imp;oil), bbl/min (imp;oil), bbl/h (imp;oil), bbl/d (imp;oil)	Barrel/time unit (petrochemicals) Petrochemicals: 34.97 gal/bbl
Time	s, m, h, d, y	Second, minute, hour, day, year
	am, pm	Ante meridiem (before midday), post meridiem (after midday)

6 Modbus RS485 Register Information

6.1 Notes

6.1.1 Structure of the register information

The individual parts of a parameter description are described in the following section:

Navigation: navigation path to the parameter					
Parameter	Register	Data type	Access type	Selection/input	→ 
Name of parameter	Indicated in decimal numerical format	- Float length = 4 byte - Integer length = 2 byte - String length, depending on parameter	Possible type of access to parameter: - Read access via function codes 03, 04 or 23 - Write access via function codes 06, 16 or 23	Selection List of the individual options for the parameter - Option 1 Option 2 Option 3 ⁽⁺⁾  - Factory setting highlighted in bold ⁽⁺⁾ = Factory setting depends on country, order options or device settings User entry Input range for the parameter	Specified number of pages and cross-reference to standard parameter description

NOTICE

If non-volatile device parameters are modified via the MODBUS RS485 function codes 06, 16 or 23, the change is saved in the EEPROM of the measuring device.

The number of writes to the EEPROM is technically restricted to a maximum of 1 million.

- Make sure to comply with this limit since, if it is exceeded, data loss and measuring device failure will result.
- Avoid constantly writing non-volatile device parameters via the MODBUS RS485.

6.1.2 Address model

The Modbus RS485 register addresses of the measuring device are implemented in accordance with the "Modbus Applications Protocol Specification V1.1".

In addition, systems are used that work with the register address model "Modicon Modbus Protocol Reference Guide (PI-MBUS-300 Rev. J)".

Depending on the function code used, a number is added at the start of the register address with this specification:

- "3" → "Read" access
- "4" → "Write" access

Function code	Access type	Register in accordance with "Modbus Applications Protocol Specification"	Register in accordance with "Modicon Modbus Protocol Reference Guide"
03 04 23	Read	XXXX Example: mass flow = 2007	3XXXX Example: mass flow = 32007
06 16 23	Write	XXXX Example: reset totalizer = 6401	4XXXX Example: reset totalizer = 46401

6.2 Overview of the Expert operating menu

The following table provides an overview of the menu structure of the expert operating menu and its parameters. The page reference indicates where the associated description of the submenu or parameter can be found.

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6.3 Register information

Navigation: Expert					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Direct access (0106)	3878	Integer	Read / Write	0 to 65535	11
Locking status (0004)	4918	Integer	Read	256 = Hardware locked 512 = Temporarily locked 2048 = CT active - all parameters 32768 = CT active - defined parameters	12
Access status (0005)	2178	Integer	Read	0 = Operator 1 = Maintenance	13
Enter access code (0003)	2177	Integer	Read / Write	0 to 9999	13

6.3.1 "System" submenu

"Display" submenu

Navigation: Expert → System → Display					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Display language (0104)	3673	Integer	Read / Write	0 = English 1 = Deutsch * 2 = Français * 3 = Español * 4 = Italiano * 5 = Nederlands * 8 = Svenska * 10 = Bahasa Indonesia * 11 = 日本語 (Japanese) * 12 = Portuguesa * 13 = Polski * 14 = русский язык (Russian) * 15 = čeština (Czech) * 16 = 中文 (Chinese) * 17 = ภาษาไทย (Thai) * 18 = Türkçe * 19 = tiếng Việt (Vietnamese) * 20 = 한국어 (Korean) * 21 = العربية (Arabic) *	15
Format display (0098)	3625	Integer	Read / Write	0 = 1 value, max. size 1 = 1 bargraph + 1 value 2 = 2 values 3 = 1 value large + 2 values 4 = 4 values	16
Value 1 display (0107)	3963	Integer	Read / Write	1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 3 = Conductivity * 4 = Corrected conductivity * 5 = Temperature * 16 = Totalizer 1 17 = Totalizer 2 18 = Totalizer 3 37 = Flow velocity 39 = Electronic temperature 121 = Current output 1 122 = Current output 2 *	18
0% bargraph value 1 (0123)	4136	Float	Read / Write	Signed floating-point number	18
100% bargraph value 1 (0125)	4142	Float	Read / Write	Signed floating-point number	19
Decimal places 1 (0095)	3365	Integer	Read / Write	0 = x 1 = x.x 2 = x.xx 3 = x.xxx 4 = x.xxxx	19

Navigation: Expert → System → Display					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Value 2 display (0108)	3964	Integer	Read / Write	0 = Custody transfer counter * 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 4 = Conductivity * 7 = Temperature * 9 = Corrected conductivity * 16 = Totalizer 1 17 = Totalizer 2 18 = Totalizer 3 37 = Flow velocity 39 = Electronic temperature 121 = Current output 1 122 = Current output 2 * 123 = Current output 3 * 124 = Current output 4 * 251 = None	20
Decimal places 2 (0117)	4049	Integer	Read / Write	0 = x 1 = x.x 2 = x.xx 3 = x.xxx 4 = x.xxxx	21
Value 3 display (0110)	3966	Integer	Read / Write	For the picklist, see the Value 2 display parameter (→  20)	21
0% bargraph value 3 (0124)	4138	Float	Read / Write	Signed floating-point number	22
100% bargraph value 3 (0126)	4140	Float	Read / Write	Signed floating-point number	22
Decimal places 3 (0118)	4050	Integer	Read / Write	0 = x 1 = x.x 2 = x.xx 3 = x.xxx 4 = x.xxxx	23
Value 4 display (0109)	3965	Integer	Read / Write	For the picklist, see the Value 2 display parameter (→  20)	23
Decimal places 4 (0119)	4051	Integer	Read / Write	0 = x 1 = x.x 2 = x.xx 3 = x.xxx 4 = x.xxxx	24
Display interval (0096)	3604	Float	Read / Write	1 to 10 s	24
Display damping (0094)	3554	Float	Read / Write	0.0 to 999.9 s	25
Header (0097)	3624	Integer	Read / Write	0 = Device tag 1 = Free text	25
Header text (0112)	3968	String	Read / Write	Max. 12 characters such as letters, numbers or special characters (e.g. @, %, /)	26
Separator (0101)	3671	Integer	Read / Write	■ . (point) ■ , (comma)	27
Contrast display (0105)	3674	Float	Read / Write	20 to 80 %	27
Backlight (0111)	3967	Integer	Read / Write	0 = Disable 1 = Enable	27

* Visibility depends on order options or device settings

"Configuration backup" submenu

Navigation: Expert → System → Configuration backup					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Operating time (0652)	2631	Integer	Read	Days (d), hours (h), minutes (m) and seconds (s)	28
Last backup (2757)	6430	Integer	Read	Days (d), hours (h), minutes (m) and seconds (s)	28
Configuration management (2758)	5500	Integer	Read / Write	0 = Cancel 1 = Execute backup 2 = Restore 4 = Clear backup data 5 = Compare	28
Backup state (2759)	5502	Integer	Read	1 = Backup in progress 2 = Restoring in progress 4 = Delete in progress 5 = Compare in progress 6 = Restoring failed 7 = Backup failed 251 = None	29
Comparison result (2760)	5514	Integer	Read	0 = Settings identical 1 = Settings not identical 2 = No backup available 3 = Check not done 4 = Backup settings corrupt 5 = Dataset incompatible	30

"Diagnostic handling" submenu

Navigation: Expert → System → Diagnostic handling					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Alarm delay (0651)	6808	Float	Read / Write	0 to 60 s	31

"Diagnostic behavior" submenu

Navigation: Expert → System → Diagnostic handling → Diagnostic behavior					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign behavior of diagnostic no. 302 (0739)	2312	Integer	Read / Write	2 = Warning 3 = Alarm	33
Assign behavior of diagnostic no. 531 (0741)	2397	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	33
Assign behavior of diagnostic no. 441 (0657)	4742	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	33
Assign behavior of diagnostic no. 442 (0658)	4919	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	34
Assign behavior of diagnostic no. 443 (0659)	5000	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	34

Navigation: Expert → System → Diagnostic handling → Diagnostic behavior					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign behavior of diagnostic no. 444 (0740)	5120	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	34
Assign behavior of diagnostic no. 543 (0643)	2362	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	35
Assign behavior of diagnostic no. 832 (0681)	2759	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	35
Assign behavior of diagnostic no. 833 (0682)	2762	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	36
Assign behavior of diagnostic no. 834 (0700)	2761	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	36
Assign behavior of diagnostic no. 835 (0702)	2760	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	36
Assign behavior of diagnostic no. 862 (0745)	2097	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	37
Assign behavior of diagnostic no. 937 (0743)	2396	Integer	Read / Write	0 = Off 1 = Logbook entry only 2 = Warning 3 = Alarm	37

"Administration" submenu

Navigation: Expert → System → Administration					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Device reset (0000)	6817	Integer	Read / Write	0 = Cancel 0 = Restore S-DAT backup 1 = Restart device 2 = To delivery settings 5 = Delete powerfail data 21 = Delete T-DAT 22 = Reset faulty parameters 23 = Delete factory data 24 = Delete embedded HistoROM	40
Activate SW option (0029)	2795	Integer	Read / Write	Max. 10-digit string consisting of numbers.	40
Software option overview (0015)	2902	Integer	Read	1 = Extended HistoROM 32 = Electrode cleaning circuit 128 = Custody transfer 16384 = Heartbeat Monitoring 32768 = Heartbeat Verification	41

"Define access code" wizard

Navigation: Expert → System → Administration → Define access code					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Define access code	8677	String	Read / Write	Max. 16-digit character string comprising numbers, letters and special characters	38
Confirm access code	8685	String	Read / Write	Max. 16-digit character string comprising numbers, letters and special characters	39

"Reset access code" submenu

Navigation: Expert → System → Administration → Reset access code					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Operating time (0652)	2631	Integer	Read	Days (d), hours (h), minutes (m) and seconds (s)	39
Reset access code (0024)	8880	String	Read / Write	Character string comprising numbers, letters and special characters	39

6.3.2 "Sensor" submenu**"Measured values" submenu***"Process variables" submenu*

Navigation: Expert → Sensor → Measured values → Process variables					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Volume flow (1838)	3874	Float	Read	Signed floating-point number	43
Mass flow (1847)	3876	Float	Read	Signed floating-point number	43
Conductivity (1850)	2013	Float	Read	Signed floating-point number	44
Corrected volume flow (1851)	2011	Float	Read	Signed floating-point number	44
Temperature (1852)	2015	Float	Read	Positive floating-point number	44
Corrected conductivity (1853)	3977	Float	Read	Positive floating-point number	45

"Totalizer" submenu

Navigation: Expert → Sensor → Measured values → Totalizer					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Totalizer value 1 to n (0911–1 to n)	1: 2610 2: 2810 3: 3010	Float	Read	Signed floating-point number	45
Totalizer overflow 1 to n (0910–1 to n)	1: 2612 2: 2812 3: 3012	Float	Read	Integer with sign	46

*"Input values" submenu**"Current input 1 to n" submenu*

Navigation: Expert → Sensor → Measured values → Input values → Current input 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Measured values 1 to n (1603–1 to n)	1: 6151 2: 6153 3: 6155	Float	Read	Signed floating-point number	47
Measured current 1 to n (1604–1 to n)	1: 6131 2: 6133 3: 6135	Float	Read	0 to 22.5 mA	47

"Value status input 1 to n" submenu

Navigation: Expert → Sensor → Measured values → Input values → Value status input 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Value status input (1353–1 to n)	1: 2746 2: 4699 3: 4700	Integer	Read	9 = Low 10 = High	48

*"Output values" submenu**"Value current output 1 to n" submenu*

Navigation: Expert → Sensor → Measured values → Output values → Value current output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Output current 1 to n (0361–1 to n)	1: 5931 2: 5933 3: 5935	Float	Read	0 to 22.5 mA	49
Measured current 1 to n (0366–1 to n)	1: 5779 2: 5781 3: 5783	Float	Read	0 to 30 mA	49

"Pulse/frequency/switch output 1 to n" submenu

Navigation: Expert → Sensor → Measured values → Output values → Pulse/frequency/switch output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Output frequency 1 to n (0471–1 to n)	1: 3462 2: 3464 3: 9910	Float	Read	0.0 to 12 500.0 Hz	49
Value per pulse (0455–1 to n)	1: 3034 2: 3036 3: 4714	Float	Read / Write	Signed floating-point number	50
Switch status 1 to n (0461–1 to n)	1: 2485 2: 2486 3: 9917	Integer	Read	1 = Open 2 = Closed	50

"Relay output 1 to n" submenu

Navigation: Expert → Sensor → Measured values → Output values → Relay output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Switch status (0801–1 to n)	1: 3518 2: 3519 3: 9875	Integer	Read	1 = Open 2 = Closed	51
Switch cycles (0815–1 to n)	1: 7625 2: 7627 3: 7629	Integer	Read	Positive integer	51
Max. switch cycles number (0817–1 to n)	1: 21919 2: 21921 3: 21923	Integer	Read	Positive integer	51

"Double pulse output" submenu

Navigation: Expert → Sensor → Measured values → Output values → Double pulse output					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Pulse output (0987)	7041	Float	Read	Positive floating-point number	52

"System units" submenu

Navigation: Expert → Sensor → System units					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Volume flow unit (0553)	2103	Integer	Read / Write	0 = cm ³ /s 1 = cm ³ /min 2 = cm ³ /h 3 = cm ³ /d 4 = dm ³ /s 5 = dm ³ /min 6 = dm ³ /h 7 = dm ³ /d 8 = m ³ /s 9 = m ³ /min 10 = m ³ /h 11 = m ³ /d 12 = ml/s 13 = ml/min 14 = ml/h 15 = ml/d 16 = l/s 17 = l/min 18 = l/h^(*) 19 = l/d 20 = hl/s 21 = hl/min 22 = hl/h 23 = hl/d 24 = Ml/s 25 = Ml/min 26 = Ml/h 27 = Ml/d 32 = af/s 33 = af/min 34 = af/h 35 = af/d 36 = ft ³ /s 37 = ft ³ /min 38 = ft ³ /h 39 = ft ³ /d 40 = fl oz/s (us) 41 = fl oz/min (us) 42 = fl oz/h (us) 43 = fl oz/d (us) 44 = gal/s (us) 45 = gal/min (us) 46 = gal/h (us) 47 = gal/d (us) 48 = Mgal/s (us) 49 = Mgal/min (us) 50 = Mgal/h (us) 51 = Mgal/d (us) 52 = bbl/s (us;liq.) 53 = bbl/min (us;liq.) 54 = bbl/h (us;liq.) 55 = bbl/d (us;liq.) 56 = bbl/s (us;beer) 57 = bbl/min (us;beer) 58 = bbl/h (us;beer) 59 = bbl/d (us;beer) 60 = bbl/s (us;oil) 61 = bbl/min (us;oil) 62 = bbl/h (us;oil) 63 = bbl/d (us;oil) 64 = bbl/s (us;tank) 65 = bbl/min (us;tank) 66 = bbl/h (us;tank) 67 = bbl/d (us;tank) 68 = gal/s (imp) 69 = gal/min (imp) 70 = gal/h (imp)	53

Navigation: Expert → Sensor → System units					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
				71 = gal/d (imp) 72 = Mgal/s (imp) 73 = Mgal/min (imp) 74 = Mgal/h (imp) 75 = Mgal/d (imp) 76 = bbl/s (imp;beer) 77 = bbl/min (imp;beer) 78 = bbl/h (imp;beer) 79 = bbl/d (imp;beer) 80 = bbl/s (imp;oil) 81 = bbl/min (imp;oil) 82 = bbl/h (imp;oil) 83 = bbl/d (imp;oil) 84 = User vol./s 85 = User vol./min 86 = User vol./h 87 = User vol./d 88 = kgal/s (us) 89 = kgal/min (us) 90 = kgal/h (us) 91 = kgal/d (us)	
Volume unit (0563)	2104	Integer	Read / Write	0 = cm ³ 1 = dm ³ 2 = m³(+) 3 = ml 4 = l 5 = hl 6 = Ml Mega 8 = af 9 = ft ³ 10 = fl oz (us) 11 = gal (us) 12 = Mgal (us) 13 = bbl (us;liq.) 14 = bbl (us;beer) 15 = bbl (us;oil) 16 = bbl (us;tank) 17 = gal (imp) 18 = Mgal (imp) 19 = bbl (imp;beer) 20 = bbl (imp;oil) 21 = User vol. 22 = kgal (us)	55
Conductivity unit (0582)	2121	Integer	Read / Write	1 = MS/m 2 = kS/m 3 = S/m 4 = S/cm 5 = mS/m 6 = mS/cm 7 = μS/m 8 = μS/cm 9 = μS/mm 10 = nS/cm	56
Temperature unit (0557)	2109	Integer	Read / Write	0 = °C (+) 1 = K 2 = °F 3 = °R	56

Navigation: Expert → Sensor → System units					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Mass flow unit (0554)	2101	Integer	Read / Write	0 = g/s 1 = g/min 2 = g/h 3 = g/d 4 = kg/s 5 = kg/min 6 = kg/h⁽⁺⁾ 7 = kg/d 8 = t/s 9 = t/min 10 = t/h 11 = t/d 12 = oz/s 13 = oz/min 14 = oz/h 15 = oz/d 16 = lb/s 17 = lb/min 18 = lb/h 19 = lb/d 20 = STon/s 21 = STon/min 22 = STon/h 23 = STon/d 24 = User mass/s 25 = User mass/min 26 = User mass/h 27 = User mass/d	57
Mass unit (0574)	2102	Integer	Read / Write	50 = g 51 = kg⁽⁺⁾ 52 = t 53 = oz 54 = lb 55 = STon 56 = User mass	58
Density unit (0555)	2107	Integer	Read / Write	0 = g/cm ³ 2 = kg/dm ³ 3 = kg/l⁽⁺⁾ 4 = kg/m ³ 5 = SD4°C 6 = SD15°C 7 = SD20°C 8 = SG4°C 9 = SG15°C 10 = SG20°C 11 = lb/ft ³ 12 = lb/gal (us) 13 = lb/bbl (us;liq.) 14 = lb/bbl (us;beer) 15 = lb/bbl (us;oil) 16 = lb/bbl (us;tank) 17 = lb/gal (imp) 18 = lb/bbl (imp;beer) 19 = lb/bbl (imp;oil) 21 = g/m ³	58

Navigation: Expert → Sensor → System units					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Corrected volume flow unit (0558)	2105	Integer	Read / Write	0 = NI/s 1 = NI/min 2 = NI/h⁽⁺⁾ 3 = NI/d 4 = Nm ³ /s 5 = Nm ³ /min 6 = Nm ³ /h 7 = Nm ³ /d 8 = Sm ³ /s 9 = Sm ³ /min 10 = Sm ³ /h 11 = Sm ³ /d 12 = Sft ³ /s 13 = Sft ³ /min 14 = Sft ³ /h 15 = Sft ³ /d 16 = Sgal/s (us) 17 = Sgal/min (us) 18 = Sgal/h (us) 19 = Sgal/d (us) 20 = Sbbbl/s (us;liq.) 21 = Sbbbl/min (us;liq.) 22 = Sbbbl/h (us;liq.) 23 = Sbbbl/d (us;liq.) 24 = Sgal/s (imp) 25 = Sgal/min (imp) 26 = Sgal/h (imp) 27 = Sgal/d (imp) 28 = UserCrVol./s 29 = UserCrVol./min 30 = UserCrVol./h 31 = UserCrVol./d	59
Corrected volume unit (0575)	2106	Integer	Read / Write	100 = NI 101 = Nm³⁽⁺⁾ 102 = Sm ³ 103 = Sft ³ 105 = Sgal (us) 106 = Sbbbl (us;liq.) 107 = Sgal (imp) 108 = UserCrVol.	60
Date/time format (2812)	2150	Integer	Read / Write	0 = dd.mm.yy hh:mm 1 = mm/dd/yy hh:mm am/pm 2 = dd.mm.yy hh:mm am/pm 3 = mm/dd/yy hh:mm	60

"User-specific units" submenu

Navigation: Expert → Sensor → System units → User-specific units					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
User volume text (0567)	2542	String	Read / Write	Max. 10 characters such as letters, numbers or special characters (@, %, /)	61
User volume factor (0568)	2119	Float	Read / Write	Signed floating-point number	62
User volume offset (0569)	2528	Float	Read / Write	Signed floating-point number	62
User mass text (0560)	2531	String	Read / Write	Max. 10 characters such as letters, numbers or special characters (@, %, /)	62
User mass factor (0561)	2115	Float	Read / Write	Signed floating-point number	63
User mass offset (0562)	4339	Float	Read / Write	Signed floating-point number	63
User corrected volume text (0592)	2568	String	Read / Write	Max. 10 characters such as letters, numbers or special characters (@, %, /)	64

Navigation: Expert → Sensor → System units → User-specific units					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
User corrected volume factor (0590)	2573	Float	Read / Write	Signed floating-point number	64
User corrected volume offset (0602)	4662	Float	Read / Write	Signed floating-point number	64

"Process parameters" submenu

Navigation: Expert → Sensor → Process parameters					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Filter options (6710)	2273	Integer	Read / Write	1 = Standard 2 = Standard CIP on 3 = Dynamic 4 = Dynamic CIP on 5 = Binomial filter	65
Flow damping (6661)	2274	Integer	Read / Write	0 to 15	67
Flow override (1839)	5503	Integer	Read / Write	0 = Off 1 = On	67
Conductivity damping (1803)	5508	Float	Read / Write	0 to 999.9 s	68
Temperature damping (1886)	2483	Float	Read / Write	0 to 999.9 s	68
Conductivity measurement (6514)	2268	Integer	Read / Write	0 = Off 1 = On	68
Conductivity temperature coefficient (1891)	2886	Float	Read / Write	Signed floating-point number	69
Reference density (1885)	2536	Float	Read / Write	Positive floating-point number	69

"Low flow cut off" submenu

Navigation: Expert → Sensor → Process parameters → Low flow cut off					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign process variable (1837)	5101	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow	70
On value low flow cutoff (1805)	5138	Float	Read / Write	Positive floating-point number	70
Off value low flow cutoff (1804)	5104	Float	Read / Write	0 to 100.0 %	71
Pressure shock suppression (1806)	5140	Float	Read / Write	0 to 100 s	71

"Empty pipe detection" submenu

Navigation: Expert → Sensor → Process parameters → Empty pipe detection					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Empty pipe detection (1860)	5106	Integer	Read / Write	0 = Off 1 = On	73
Switch point empty pipe detection (6562)	2890	Float	Read / Write	0 to 100 %	73
Response time empty pipe detection (1859)	5108	Float	Read / Write	0 to 100 s	74
New adjustment (6560)	2335	Integer	Read / Write	0 = Cancel 1 = Empty pipe adjust 2 = Full pipe adjust	74

Navigation: Expert → Sensor → Process parameters → Empty pipe detection					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Progress (6571)	2336	Integer	Read	0 = Not ok 6 = Ok 8 = Busy	74
Empty pipe adjust value (6527)	2181	Float	Read	Positive floating-point number	75
Full pipe adjust value (6548)	2832	Float	Read	Positive floating-point number	75
Measured value EPD (6559)	2298	Float	Read	Positive floating-point number	75

"Electrode cleaning circuit" submenu

Navigation: Expert → Sensor → Process parameters → Electrode cleaning circuit					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Electrode cleaning circuit (6528)	2280	Integer	Read / Write	0 = Off 1 = On	76
ECC duration (6555)	2330	Float	Read / Write	0.01 to 30 s	76
ECC recovery time (6556)	2332	Float	Read / Write	1 to 600 s	76
ECC cleaning cycle (6557)	2328	Float	Read / Write	0.5 to 168 h	77
ECC Polarity (6631)	2334	Integer	Read	0 = Positive 1 = Negative	77

"External compensation" submenu

Navigation: Expert → Sensor → External compensation					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Temperature source (6712)	2114	Integer	Read / Write	0 = Internal temperature sensor 2 = Off 11 = Current input 1 * 12 = Current input 2 * 13 = Current input 3 *	78
External temperature (6673)	2125	Float	Read	Floating point number with sign	78
Density source (6615)	2497	Integer	Read / Write	0 = Fixed density 0 = Calculated value 11 = Current input 1 * 12 = Current input 2 * 13 = Current input 3 *	79
External density (6630)	2117	Float	Read	Positive floating-point number	79
Fixed density (6623)	2830	Float	Read / Write	Positive floating-point number	79
Reference temperature (1816)	5136	Float	Read	-273.15 to 99 999 °C	80
Reference density (expansion.) (1892)	6637	Float	Read	Positive floating-point number	80
Linear expansion coefficient (1817)	5132	Float	Read	Signed floating-point number	81
Square expansion coefficient (1818)	5134	Float	Read	Signed floating-point number	82

* Visibility depends on order options or device settings

"Sensor adjustment" submenu

Navigation: Expert → Sensor → Sensor adjustment					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Installation direction (1809)	5501	Integer	Read / Write	0 = Flow in arrow direction 1 = Flow against arrow direction	82
Integration time (6533)	2260	Float	Read	1 to 65 ms	83
Measuring period (6536)	2852	Float	Read	0 to 1 000 ms	83

"Process variable adjustment" submenu

Navigation: Expert → Sensor → Sensor adjustment → Process variable adjustment					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Volume flow offset (1831)	5521	Float	Read / Write	Signed floating-point number	84
Volume flow factor (1832)	5519	Float	Read / Write	Positive floating-point number	84
Mass flow offset (1841)	5525	Float	Read / Write	Signed floating-point number	84
Mass flow factor (1846)	5523	Float	Read / Write	Positive floating-point number	85
Conductivity offset (1848)	5529	Float	Read / Write	Signed floating-point number	85
Conductivity factor (1849)	5527	Float	Read / Write	Positive floating-point number	85
Corrected volume flow offset (1866)	2044	Float	Read / Write	Signed floating-point number	86
Corrected volume flow factor (1867)	2076	Float	Read / Write	Positive floating-point number	86
Temperature offset (1868)	2046	Float	Read / Write	Signed floating-point number	86
Temperature factor (1869)	2042	Float	Read / Write	Positive floating-point number	87

"Calibration" submenu

Navigation: Expert → Sensor → Calibration					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Nominal diameter (2807)	2048	String	Read	DNxx / x"	88
Calibration factor (6522)	2313	Float	Read	Positive floating-point number	88
Zero point (6546)	2870	Float	Read	Signed floating-point number	88
Conductivity calibration factor (6718)	19806	Float	Read	0 to 10 000	88

6.3.3 "I/O configuration" submenu

Navigation: Expert → I/O configuration					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
I/O module 1 to n terminal numbers (3902-1 to n)	1: 6541 2: 6542 3: 6543 4: 6544	Integer	Read	0 = Not used 1 = 26-27 (I/O 1) 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	89
I/O module 1 to n information (3906-1 to n)	1: 8659 2: 8660 3: 8661 4: 8662	Integer	Read	1 = Fieldbus 2 = Configurable 3 = Not configurable 254 = Not plugged 255 = Invalid	89

Navigation: Expert → I/O configuration					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
I/O module 1 to n type (3901-1 to n)	1: 6417 2: 6418 3: 6419 4: 6420	Integer	Read / Write	0 = Off 0 = Current output * 0 = Current input * 0 = Status input * 0 = Pulse/frequency/switch output *	90
Apply I/O configuration (3907)	8665	Integer	Read / Write	0 = Yes 1 = No	90
Conversion code (2762)	6427	Integer	Read / Write	Positive integer	91

* Visibility depends on order options or device settings

6.3.4 "Input" submenu

"Current input 1 to n" submenu

Navigation: Expert → Input → Current input 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Terminal number (1611-1 to n)	1: 6548 2: 6549 3: 6550	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	92
Signal mode (1610-1 to n)	1: 6424 2: 6425 3: 6426	Integer	Read / Write	0 = Passive 2 = Active	92
Current span (1605-1 to n)	1: 6147 2: 6148 3: 6149	Integer	Read / Write	0 = 4...20 mA 1 = 4...20 mA US 2 = 4...20 mA NAMUR (+) 3 = 0...20 mA	92
0/4 mA value (1606-1 to n)	1: 6111 2: 6113 3: 6115	Float	Read / Write	Signed floating-point number	93
20 mA value (1607-1 to n)	1: 6119 2: 6121 3: 6123	Float	Read / Write	Signed floating-point number	93
Failure mode (1601-1 to n)	1: 6159 2: 6160 3: 6161	Integer	Read / Write	1 = Last valid value 2 = Alarm 6 = Defined value	93
Failure value (1602-1 to n)	1: 6163 2: 6165 3: 6167	Float	Read / Write	Signed floating-point number	94

"Status input 1 to n" submenu

Navigation: Expert → Input → Status input 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Terminal number (1358-1 to n)	1: 6554 2: 6555 3: 6556	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	94
Assign status input (1352-1 to n)	1: 2506 2: 4687 3: 4688	Integer	Read / Write	0 = Off 1 = Flow override 2 = Reset all totalizers 3 = Reset totalizer 1 4 = Reset totalizer 2 5 = Reset totalizer 3	95

Navigation: Expert → Input → Status input 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Value status input (1353-1 to n)	1: 2746 2: 4699 3: 4700	Integer	Read	9 = Low 10 = High	95
Active level (1351-1 to n)	1: 2530 2: 4690 3: 4691	Integer	Read / Write	9 = Low 10 = High	96
Response time status input (1354-1 to n)	1: 3404 2: 5753 3: 5755	Float	Read / Write	5 to 200 ms	96

6.3.5 "Output" submenu

"Current output 1 to n" submenu

Navigation: Expert → Output → Current output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Terminal number (0379-1 to n)	1: 6545 2: 6546 3: 6547	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	97
Signal mode (0377-1 to n)	1: 6421 2: 6422 3: 6423	Integer	Read / Write	0 = Passive 2 = Active	98
Assign current output 1 to n (0359-1 to n)	1: 5927 2: 5928 3: 5929	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 4 = Conductivity* 7 = Temperature* 9 = Corrected conductivity* 37 = Flow velocity 39 = Electronic temperature	98
Current span (0353-1 to n)	1: 5923 2: 5924 3: 5925	Integer	Read / Write	0 = 4...20 mA 1 = 4...20 mA US 2 = 4...20 mA NAMUR 3 = 0...20 mA 4 = Fixed current	98
Fixed current (0365-1 to n)	1: 5987 2: 5989 3: 5991	Float	Read / Write	0 to 22.5 mA	99
0/4 mA value (0367-1 to n)	1: 6195 2: 6197 3: 6199	Float	Read / Write	Signed floating-point number	100
20 mA value (0372-1 to n)	1: 5915 2: 5917 3: 5919	Float	Read / Write	Signed floating-point number	101
Measuring mode (0351-1 to n)	1: 5899 2: 5900 3: 5901	Integer	Read / Write	0 = Forward flow 2 = Reverse flow compensation 13 = Forward/Reverse flow	102
Damping output 1 to n (0363-1 to n)	1: 5903 2: 5905 3: 5907	Float	Read / Write	0.0 to 999.9 s	106
Response time (0378-1 to n)	1: 5867 2: 5869 3: 5871	Float	Read	Positive floating-point number	107

Navigation: Expert → Output → Current output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Failure mode (0364-1 to n)	1: 5911 2: 5912 3: 5913	Integer	Read / Write	0 = Min. 1 = Max. 4 = Actual value 5 = Last valid value 6 = Defined value	108
Failure current (0352-1 to n)	1: 5979 2: 5981 3: 5983	Float	Read / Write	0 to 22.5 mA	109
Output current 1 to n (0361-1 to n)	1: 5931 2: 5933 3: 5935	Float	Read	3.59 to 22.5 mA	109
Measured current 1 to n (0366-1 to n)	1: 5779 2: 5781 3: 5783	Float	Read	0 to 30 mA	110

* Visibility depends on order options or device settings

"Pulse/frequency/switch output 1 to n" submenu

Navigation: Expert → Output → Pulse/frequency/switch output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Terminal number (0492-1 to n)	1: 6551 2: 6552 3: 6553	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	111
Signal mode (0490-1 to n)	1: 6235 2: 6236 3: 6237	Integer	Read / Write	0 = Passive 2 = Active	112
Operating mode (0469-1 to n)	1: 4479 2: 4480 3: 9907	Integer	Read / Write	0 = Pulse 1 = Switch 53 = Frequency	112
Assign pulse output 1 to n (0460-1 to n)	1: 2461 2: 2462 3: 4685	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow	114
Value per pulse (0455-1 to n)	1: 3034 2: 3036 3: 4714	Float	Read / Write	Signed floating-point number	114
Pulse width (0452-1 to n)	1: 2836 2: 2838 3: 4702	Float	Read / Write	0.05 to 2 000 ms	115
Measuring mode (0457-1 to n)	1: 2394 2: 2395 3: 4683	Integer	Read / Write	0 = Forward flow 1 = Reverse flow 2 = Reverse flow compensation 13 = Forward/Reverse flow	115
Failure mode (0480-1 to n)	1: 2948 2: 2949 3: 4708	Integer	Read / Write	0 = Actual value 1 = No pulses	116
Pulse output 1 to n (0456-1 to n)	1: 3082 2: 3084 3: 4718	Float	Read	Positive floating-point number	117

Navigation: Expert → Output → Pulse/frequency/switch output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign frequency output (0478-1 to n)	1: 2614 2: 2615 3: 9915	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 4 = Conductivity * 7 = Temperature * 9 = Corrected conductivity * 37 = Flow velocity 39 = Electronic temperature	118
Minimum frequency value (0453-1 to n)	1: 3526 2: 3528 3: 5767	Float	Read / Write	0.0 to 10 000.0 Hz	118
Maximum frequency value (0454-1 to n)	1: 2996 2: 2998 3: 4710	Float	Read / Write	0.0 to 10 000.0 Hz	119
Measuring value at minimum frequency (0476-1 to n)	1: 5887 2: 5889 3: 5891	Float	Read / Write	Signed floating-point number	119
Measuring value at maximum frequency (0475-1 to n)	1: 3514 2: 3516 3: 5759	Float	Read / Write	Signed floating-point number	120
Measuring mode (0479-1 to n)	1: 2922 2: 2923 3: 4706	Integer	Read / Write	0 = Forward flow 2 = Reverse flow compensation 13 = Forward/Reverse flow	120
Damping output 1 to n (0477-1 to n)	1: 3522 2: 3524 3: 5763	Float	Read / Write	0 to 999.9 s	121
Response time (0491-1 to n)	1: 5875 2: 5877 3: 5879	Float	Read	Positive floating-point number	122
Failure mode (0451-1 to n)	1: 2367 2: 2368 3: 4681	Integer	Read / Write	0 = Actual value 1 = 0 Hz 2 = Defined value	122
Failure frequency (0474-1 to n)	1: 3510 2: 3512 3: 9908	Float	Read / Write	0.0 to 12 500.0 Hz	123
Output frequency 1 to n (0471-1 to n)	1: 3462 2: 3464 3: 9910	Float	Read	0.0 to 12 500.0 Hz	123
Switch output function (0481-1 to n)	1: 3022 2: 3023 3: 9914	Integer	Read / Write	0 = Off 1 = On 2 = Diagnostic behavior 3 = Flow direction check 4 = Limit 5 = Status	124
Assign diagnostic behavior (0482-1 to n)	1: 3096 2: 3097 3: 9913	Integer	Read / Write	0 = Alarm 1 = Warning 2 = Alarm or warning	124

Navigation: Expert → Output → Pulse/frequency/switch output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign limit (0483-1 to n)	1: 3184 2: 3185 3: 4722	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 4 = Conductivity * 7 = Temperature * 9 = Corrected conductivity * 16 = Totalizer 1 17 = Totalizer 2 18 = Totalizer 3 37 = Flow velocity 39 = Electronic temperature	125
Switch-on value (0466-1 to n)	1: 3242 2: 3244 3: 4728	Float	Read / Write	Signed floating-point number	127
Switch-off value (0464-1 to n)	1: 3234 2: 3236 3: 4724	Float	Read / Write	Signed floating-point number	127
Assign flow direction check (0484-1 to n)	1: 3363 2: 3364 3: 4732	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow	128
Assign status (0485-1 to n)	1: 3374 2: 3375 3: 4734	Integer	Read / Write	0 = Low flow cut off 1 = Empty pipe detection	128
Switch-on delay (0467-1 to n)	1: 6247 2: 6249 3: 6251	Float	Read / Write	0.0 to 100.0 s	128
Switch-off delay (0465-1 to n)	1: 6239 2: 6241 3: 6243	Float	Read / Write	0.0 to 100.0 s	129
Failure mode (0486-1 to n)	1: 3384 2: 3385 3: 9912	Integer	Read / Write	0 = Actual status 1 = Open 2 = Closed	129
Switch status 1 to n (0461-1 to n)	1: 2485 2: 2486 3: 9917	Integer	Read	1 = Open 2 = Closed	129
Invert output signal (0470-1 to n)	1: 2583 2: 2584 3: 9916	Integer	Read / Write	0 = Yes 1 = No	130

* Visibility depends on order options or device settings

"Relay output 1 to n" submenu

Navigation: Expert → Output → Relay output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Terminal number (0812-1 to n)	1: 8278 2: 8279 3: 8280	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	131
Relay output function (0804-1 to n)	1: 2488 2: 2489 3: 9876	Integer	Read / Write	1 = Open 2 = Closed 2 = Diagnostic behavior 3 = Flow direction check 4 = Limit 5 = Digital Output	131

Navigation: Expert → Output → Relay output 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign flow direction check (0808-1 to n)	1: 8251 2: 8252 3: 8253	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow	132
Assign limit (0807-1 to n)	1: 8248 2: 8249 3: 8250	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 4 = Conductivity* 7 = Temperature* 9 = Corrected conductivity* 16 = Totalizer 1 17 = Totalizer 2 18 = Totalizer 3 37 = Flow velocity 39 = Electronic temperature	132
Assign diagnostic behavior (0806-1 to n)	1: 8245 2: 8246 3: 8247	Integer	Read / Write	0 = Alarm 1 = Warning 2 = Alarm or warning	133
Assign status (0805-1 to n)	1: 8272 2: 8273 3: 8274	Integer	Read / Write	0 = Low flow cut off 1 = Partially filled pipe detection	133
Switch-off value (0809-1 to n)	1: 8260 2: 8262 3: 8264	Float	Read / Write	Signed floating-point number	134
Switch-off delay (0813-1 to n)	1: 8254 2: 8256 3: 8258	Float	Read / Write	0.0 to 100.0 s	134
Switch-on value (0810-1 to n)	1: 8233 2: 8235 3: 8237	Float	Read / Write	Signed floating-point number	134
Switch-on delay (0814-1 to n)	1: 8266 2: 8268 3: 8270	Float	Read / Write	0.0 to 100.0 s	135
Failure mode (0811-1 to n)	1: 8242 2: 8243 3: 8244	Integer	Read / Write	0 = Actual status 1 = Open 2 = Closed	135
Switch status (0801-1 to n)	1: 3518 2: 3519 3: 9875	Integer	Read	1 = Open 2 = Closed	136
Actual relay state (0816-1 to n)	1: 7009 2: 7010 3: 7011	Integer	Read / Write	1 = Open 2 = Closed	136

* Visibility depends on order options or device settings

"Double pulse output" submenu

Navigation: Expert → Output → Double pulse output					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Master terminal number (0981)	5838	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	137
Slave terminal number (0990)	5845	Integer	Read	0 = Not used 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	137

Navigation: Expert → Output → Double pulse output					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Signal mode (0991)	5949	Integer	Read / Write	0 = Passive 2 = Active 3 = Passive NAMUR	138
Assign pulse output 1 (0982-1)	5993	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow	138
Value per pulse (0983)	7495	Float	Read / Write	Signed floating-point number	138
Pulse width (0986)	6998	Float	Read / Write	0.5 to 2 000 ms	139
Phase shift (0992)	6089	Integer	Read / Write	0 = 90° 1 = 180°	139
Measuring mode (0984)	6001	Integer	Read / Write	0 = Forward flow 1 = Reverse flow 2 = Reverse flow compensation 13 = Forward/Reverse flow	139
Failure mode (0985)	6009	Integer	Read / Write	0 = Actual value 1 = No pulses	140
Pulse output (0987)	7041	Float	Read	Positive floating-point number	141
Invert output signal (0993)	6101	Integer	Read / Write	0 = Yes 1 = No	141

6.3.6 "Communication" submenu

"Modbus configuration" submenu

Navigation: Expert → Communication → Modbus configuration					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Bus address (7112)	4910	Integer	Read / Write	1 to 247	142
Baudrate (7111)	4912	Integer	Read / Write	0 = 1200 BAUD 1 = 2400 BAUD 2 = 4800 BAUD 3 = 9600 BAUD 4 = 19200 BAUD 5 = 38400 BAUD 6 = 57600 BAUD 7 = 115200 BAUD	143
Data transfer mode (7115)	4913	Integer	Read / Write	0 = RTU 1 = ASCII	143
Parity (7122)	4914	Integer	Read / Write	0 = Even 1 = Odd 2 = None / 2 stop bits 3 = None / 1 stop bit	143
Byte order (7113)	4915	Integer	Read / Write	0 = 0-1-2-3 1 = 3-2-1-0 2 = 2-3-0-1 3 = 1-0-3-2	144
Telegram delay (7146)	4916	Float	Read / Write	0 to 100 ms	144
Failure mode (7116)	4920	Integer	Read / Write	1 = Last valid value 255 = NaN value	144
Bus termination (7155)	5774	Integer	Read	0 = Off 1 = On	145
Fieldbus writing access (7156)	6807	Integer	Read / Write	0 = Read + write 1 = Read only	145

"Modbus information" submenu

Navigation: Expert → Communication → Modbus information					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Device ID (7153)	2547	Integer	Read	4-digit hexadecimal number	146
Device revision (7154)	4481	Integer	Read	4-digit hexadecimal number	146

"Modbus data map" submenu

Navigation: Expert → Communication → Modbus data map					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Scan list register 0 to 15 (7114)	0: 5001 1: 5002 2: 5003 3: 5004 4: 5005 5: 5006 6: 5007 7: 5008 8: 5009 9: 5010 10: 5011 11: 5012 12: 5013 13: 5014 14: 5015 15: 5016	Integer	Read / Write	1 to 65 535	147

"Web server" submenu

Navigation: Expert → Communication → Web server					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Web server language (7221)	4219	Integer	Read / Write	0 = English 1 = Deutsch * 2 = Français * 3 = Español * 4 = Italiano * 5 = Nederlands * 8 = Svenska * 10 = Bahasa Indonesia * 11 = 日本語 (Japanese) * 12 = Portuguesa * 13 = Polski * 14 = русский язык (Russian) * 15 = čeština (Czech) * 16 = 中文 (Chinese) * 17 = ภาษาไทย (Thai) * 18 = Türkçe * 19 = tiếng Việt (Vietnamese) * 20 = 한국어 (Korean) * 21 = العربية (Arabic) *	148
MAC address (7214)	4210	String	Read	Unique 12-digit character string comprising letters and numbers	148
IP address (7209)	4155	String	Read	4 octet: 0 to 255 (in the particular octet)	149
Subnet mask (7211)	4163	String	Read	4 octet: 0 to 255 (in the particular octet)	149
Default gateway (7210)	4171	String	Read	4 octet: 0 to 255 (in the particular octet)	149

Navigation: Expert → Communication → Web server					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Web server functionality (7222)	4220	Integer	Read / Write	0 = Off 1 = On	149
Login page (7273)	5802	Integer	Read / Write	0 = Without header 1 = With header	150

* Visibility depends on order options or device settings

"WLAN settings" submenu

Navigation: Expert → Communication → WLAN settings					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
WLAN (2702)	6178	Integer	Read / Write	0 = Disable 1 = Enable	151
WLAN IP address (2711)	8643	String	Read / Write	4 octet: 0 to 255 (in the particular octet)	151
WLAN subnet mask (2709)	8651	String	Read / Write	4 octet: 0 to 255 (in the particular octet)	151
WLAN MAC address (2703)	8602	String	Read	Unique 12-digit character string comprising letters and numbers	152
Security type (2705)	6206	Integer	Read / Write	0 = Unsecured 1 = WPA2-PSK	152
WLAN passphrase (2706)	8611	String	Read / Write	8 to 32-digit character string comprising numbers, letters and special characters	152
Assign SSID name (2708)	6218	Integer	Read / Write	0 = Device tag 1 = User-defined	153
SSID name (2707)	8627	String	Read / Write	Max. 32-digit character string comprising numbers, letters and special characters	153
Select antenna (2713)	6102	Integer	Read / Write	0 = External antenna 1 = Internal antenna	153
2.4 GHz WLAN channel (2704)	6182	Integer	Read / Write	1 to 11	154
Apply changes (2712)	6558	Integer	Read / Write	0 = Cancel 6 = Ok	154

6.3.7 "Application" submenu

Navigation: Expert → Application					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Reset all totalizers (2806)	2609	Integer	Read / Write	0 = Cancel 1 = Reset + totalize	155

"Totalizer 1 to n" submenu

Navigation: Expert → Application → Totalizer 1 to n					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign process variable (0914-1 to n)	1: 2601 2: 2801 3: 3001	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow	155
Unit totalizer 1 to n (0915-1 to n)	1: 4604 2: 4605 3: 4606	Integer	Read / Write	0 = cm ³ 1 = dm ³ 2 = m ³ 3 = ml 4 = l 5 = hl 6 = Ml Mega 8 = af 9 = ft ³ 10 = fl oz (us) 11 = gal (us) 12 = Mgal (us) 13 = bbl (us;liq.) 14 = bbl (us;beer) 15 = bbl (us;oil) 16 = bbl (us;tank) 17 = gal (imp) 18 = Mgal (imp) 19 = bbl (imp;beer) 20 = bbl (imp;oil) 21 = User vol. 22 = kgal (us) 50 = g 51 = kg 52 = t 53 = oz 54 = lb 55 = STon 56 = User mass 100 = Nl 101 = Nm ³ 102 = Sm ³ 103 = Sft ³ 105 = Sgal (us) 106 = Sbbl (us;liq.) 107 = Sgal (imp) 108 = UserCrVol.	156
Totalizer operation mode (0908-1 to n)	1: 2605 2: 2805 3: 3005	Integer	Read / Write	0 = Net flow total 1 = Forward flow total 2 = Reverse flow total	157
Control Totalizer 1 to n (0912-1 to n)	1: 2608 2: 2808 3: 3008	Integer	Read / Write	0 = Totalize 1 = Reset + totalize 2 = Preset + hold 3 = Reset + hold 4 = Preset + totalize 5 = Hold	157
Preset value 1 to n (0913-1 to n)	1: 2590 2: 2592 3: 2594	Float	Read / Write	Signed floating-point number	158
Failure mode (0901-1 to n)	1: 2606 2: 2806 3: 3006	Integer	Read / Write	0 = Stop 1 = Actual value 2 = Last valid value	158

6.3.8 "Diagnostics" submenu

Navigation: Expert → Diagnostics					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Actual diagnostics (0691)	2732	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	160
Previous diagnostics (0690)	2734	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	161
Operating time from restart (0653)	2624	Integer	Read	Days (d), hours (h), minutes (m) and seconds (s)	162
Operating time (0652)	2631	Integer	Read	Days (d), hours (h), minutes (m) and seconds (s)	162

"Diagnostic list" submenu

Navigation: Expert → Diagnostics → Diagnostic list					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Diagnostics 1 (0692)	2736	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	163
Diagnostics 2 (0693)	2738	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	163
Diagnostics 3 (0694)	2740	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	164
Diagnostics 4 (0695)	2742	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	165
Diagnostics 5 (0696)	2744	Integer	Read	Symbol for diagnostic behavior, diagnostic code and short message.	166

"Event logbook" submenu

Navigation: Expert → Diagnostics → Event logbook					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Filter options (0705)	4596	Integer	Read / Write	0 = Failure (F) 4 = Maintenance required (M) 8 = Function check (C) 12 = Out of specification (S) 16 = Information (I) 255 = All	167

"Device information" submenu

Navigation: Expert → Diagnostics → Device information					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Device tag (0011)	2026	String	Read	Max. 32 characters, such as letters, numbers or special characters (e.g. @, %, /).	170
Serial number (0009)	7003	String	Read	A maximum of 11-digit character string comprising letters and numbers.	170
Firmware version (0010)	7277	String	Read	Character string in the format xx.yy.zz	170
Device name (0020)	7238	String	Read	Promag300/500	171

Navigation: Expert → Diagnostics → Device information					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Order code (0008)	2058	String	Read	Character string composed of letters, numbers and certain punctuation marks (e.g. /).	171
Extended order code 1 (0023)	2212	String	Read	Character string	171
Extended order code 2 (0021)	2222	String	Read	Character string	172
Extended order code 3 (0022)	2232	String	Read	Character string	172
ENP version (0012)	4003	String	Read	Character string	172

"Mainboard module" submenu

Navigation: Expert → Diagnostics → Mainboard module					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Software revision (0072)	7039	Integer	Read	Positive integer	173

"Sensor electronic module (ISEM)" submenu

Navigation: Expert → Diagnostics → Sensor electronic module (ISEM)					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Software revision (0072)	7039	Integer	Read	Positive integer	173

"I/O module 1" submenu

Navigation: Expert → Diagnostics → I/O module 1					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
I/O module 1 terminal numbers (3902-1)	6541	Integer	Read	0 = Not used 1 = 26-27 (I/O 1) 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	174
Software revision (0072)	7039	Integer	Read	Positive integer	174

"I/O module 2" submenu

Navigation: Expert → Diagnostics → I/O module 2					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
I/O module 2 terminal numbers (3902-2)	6542	Integer	Read	0 = Not used 1 = 26-27 (I/O 1) 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	174
Software revision (0072)	7039	Integer	Read	Positive integer	174

"I/O module 3" submenu

Navigation: Expert → Diagnostics → I/O module 3					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
I/O module 3 terminal numbers (3902-3)	6543	Integer	Read	0 = Not used 1 = 26-27 (I/O 1) 2 = 24-25 (I/O 2) 3 = 22-23 (I/O 3)	174
Software revision (0072)	7039	Integer	Read	Positive integer	174

"Display module" submenu

Navigation: Expert → Diagnostics → Display module					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Software revision (0072)	7039	Integer	Read	Positive integer	176

"Min/max values" submenu

Navigation: Expert → Diagnostics → Min/max values					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Reset min/max values (6541)	2269	Integer	Read / Write	0 = Cancel	185

"Main electronic temperature" submenu

Navigation: Expert → Diagnostics → Min/max values → Main electronic temperature					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Minimum value (6547)	2292	Float	Read	Signed floating-point number	185
Maximum value (6545)	2294	Float	Read	Signed floating-point number	185

"Temperature" submenu

Navigation: Expert → Diagnostics → Min/max values → Temperature					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Minimum value (6681)	2339	Float	Read	Signed floating-point number	186
Maximum value (6680)	2337	Float	Read	Signed floating-point number	186

"Data logging" submenu

Navigation: Expert → Diagnostics → Data logging					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign channel 1 (0851)	2445	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 4 = Conductivity * 7 = Temperature * 9 = Corrected conductivity * 37 = Flow velocity 39 = Electronic temperature 121 = Current output 1 122 = Current output 2 * 123 = Current output 3 *	177
Assign channel 2 (0852)	2446	Integer	Read / Write	Picklist, see Assign channel 1 parameter (→  177)	178
Assign channel 3 (0853)	2548	Integer	Read / Write	Picklist, see Assign channel 1 parameter (→  177)	178
Assign channel 4 (0854)	4286	Integer	Read / Write	Picklist, see Assign channel 1 parameter (→  177)	179
Logging interval (0856)	4288	Float	Read / Write	0.1 to 999.0 s	179
Clear logging data (0855)	4287	Integer	Read / Write	0 = Cancel 1 = Clear data	180
Data logging (0860)	5950	Integer	Read / Write	0 = Overwriting 1 = Not overwriting	180
Logging delay (0859)	5938	Integer	Read / Write	0 to 999 h	180
Data logging control (0857)	5930	Integer	Read / Write	0 = None 1 = Stop 2 = Delete + start	181
Data logging status (0858)	5937	Integer	Read	0 = Done 1 = Stopped 2 = Active 3 = Delay active	181
Entire logging duration (0861)	2827	Float	Read	Positive floating-point number	182

* Visibility depends on order options or device settings

"Simulation" submenu

Navigation: Expert → Diagnostics → Simulation					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Assign simulation process variable (1810)	6813	Integer	Read / Write	0 = Off 1 = Volume flow 2 = Mass flow 3 = Corrected volume flow 3 = Conductivity * 4 = Corrected conductivity * 5 = Temperature *	188
Process variable value (1811)	6814	Float	Read / Write	Depends on the process variable selected	189
Status input simulation (1355)	2620	Integer	Read / Write	0 = Off 1 = On	189
Input signal level (1356)	2638	Integer	Read / Write	9 = Low 10 = High	190

Navigation: Expert → Diagnostics → Simulation					
Parameter	Register	Data type	Access	Selection / User entry / User interface	→ 
Current input 1 to n simulation (1608–1 to n)	1: 6127 2: 6128 3: 6129	Integer	Read / Write	0 = Off 1 = On	190
Value current input 1 to n (1609–1 to n)	1: 6139 2: 6141 3: 6143	Float	Read / Write	0 to 22.5 mA	190
Current output 1 to n simulation (0354–1 to n)	1: 5939 2: 5940 3: 5941	Integer	Read / Write	0 = Off 1 = On	191
Value current output 1 to n (0355–1 to n)	1: 5995 2: 5997 3: 5999	Float	Read / Write	3.59 to 22.5 mA	191
Frequency output simulation 1 to n (0472–1 to n)	1: 6203 2: 6204 3: 6205	Integer	Read / Write	0 = Off 1 = On	191
Frequency value 1 to n (0473–1 to n)	1: 6207 2: 6209 3: 6211	Float	Read / Write	0.0 to 12 500.0 Hz	192
Pulse output simulation 1 to n (0458–1 to n)	1: 6215 2: 6216 3: 6217	Integer	Read / Write	0 = Off 1 = Down-counting value 2 = Fixed value	192
Pulse value 1 to n (0459–1 to n)	1: 6219 2: 6220 3: 6221	Integer	Read / Write	0 to 65 535	193
Switch output simulation 1 to n (0462–1 to n)	1: 6223 2: 6224 3: 6225	Integer	Read / Write	0 = Off 1 = On	193
Switch status 1 to n (0463–1 to n)	1: 6227 2: 6228 3: 6229	Integer	Read / Write	1 = Open 2 = Closed	194
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Device alarm simulation (0654)	6812	Integer	Read / Write	0 = Off 1 = On	196
Diagnostic event category (0738)	4261	Integer	Read / Write	0 = Sensor 1 = Electronics 2 = Configuration 3 = Process	197
Diagnostic event simulation (0737)	4259	Integer	Read / Write	■ Off ■ Diagnostic event picklist (depends on the category selected)	197

* Visibility depends on order options or device settings

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